Core Mathematics

A Practical Guide on Mathematics for Physics Learners

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Lecture notes for first-year university students in physics or physical science, covering mathematical tools you the students need most. Calculus, logic, complex numbers, and vectors, with introductions to vector calculus, differential equations, and statistical analysis. Exercises are central to these notes. Practice each topic repeatedly until you can work through problems quickly and accurately.

Preface

Physics is an activity to describe the world, but how? At present, mathematics (+ *broken* English) is the only language we can use. So, *unfortunately*, we need to learn mathematics! (and English…🥺) But it means we are *fortunate*: we have a language to describe the world. In fact, physics have developed together with mathematics. We understand physics better if we study mathematics well!

When you do DIY, you need to use the tools properly, smoothly, and accurately. It is the same here; when we use math as a tool, it must be **logical**, **quick**, and **accurate**. To be logical, you must think each problem carefully; otherwise your discussion might be incomplete and you can't convince others. To be quick and accurate, you must *"drill"* repeatedly, as if top NBA players do shooting practice almost everyday. You should reach the point where basic calculations feel automatic.

This is not a math textbook. **This is rather** a math "drill" book. Rigorous proofs are mostly omitted. Drill problems are the core of this document, through which students are expected to achieve better conceptual understanding.

Target of this document

This document is primarily for first-year undergraduate students in their second semester, who

- completed basic calculus (differentiation and integration),
- are going to major in physics or physical science, and
- do not hesitate to do repetitive practice to achieve better conceptual understanding.

How to use this document

1. **Buy a A4-sized paper notebook.**

   Digital notebooks on tablets are not suitable (less effective).

2. Read the text and solve **Quizzes**.

   Read all the text carefully. The texts are kept short!! Quizzes should be solved while reading, but you don't have to write down the solution process for Quizzes.

3. Solve **Drill Problems**, until you feel "I can solve them *fluently and accurately*".

   This is to improve your fluency ($\neq$ speed) and accuracy. Write down solution process clearly.

4. Solve other **Problems**.

   This is to have better conceptual understanding, but also to develop your writing skills. Write down the solution process clearly *as an English text*, taking care of *logical completeness*.

Optionally, you you are encouraged to prepare a reference book, so that you can consult it when you want to know more. Sho supposes **[Boas]** listed below is the best option.

Problems

🦾 **Drill problems**, marked with 🦾, are simple and easy. They help you practice the basic calculations until you can do them fluently and accurately. If they feel too easy for you, feel free to skip some of them.

★ These are **minimal basic** problems. To pass this course, you should be able to solve all of them. If you struggle with them, please come talk to Sho—don't wait until the exam!

*** These **basic** problems are the same difficulty as the ★-problems. Solve these for extra practice, to check that you really understand the ★-problems and were not just lucky.

** These are **intermediate** problems. Solve these if you are aiming for an A+. Note also that, at this level, you are asked to write answers carefully with clear explanations.

* These are **advanced** problems, beyond the scope of the lecture, for motivated students who are interested in research activities, a top-level engineering career, or demanding analytical-thinking work.

Bibliography

This document is written under the influence of the following references.

- The Math Book by Hal Tasaki (数学：物理を学び楽しむために, in Japanese)

  A free book with full description, but unfortunately in Japanese. Topic selections and the depth of descriptions are based on this reference.

- First-Year Math Practice (in Japanese)

  This document is grounded in a "learn steadily" approach ("じっくり" in Japanese), which originates from Prof. Gocho and Prof. Kiyono, who were instructors of Sho in his freshman. Problems and rigorous math descriptions are taken from their works.

- **[Boas]** Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed., Wiley, 2006.

- **[AWH]** George B. Arfken, Hans J. Weber, and Frank E. Harris, *Mathematical Methods for Physicists*, 7th ed., Academic Press, 2023.

  These two books are widely used in physics departments around the world. [Boas] seems more friendly to beginners, while [AWH] seems more detailed and complete. Sho recommends you to study [Boas] *in parallel with* this document, using it as a reference, and to consult [AWH] when you want more details or more exercises.

Other references are:

[1] Bureau International des Poids et Mesures, "The International System of Units (SI)," BIPM. [Online]. Available: https://doi.org/10.59161/AUEZ1291

Symbols

SI Units

Historically, different civilizations used different units. Today, science and engineering worldwide use the **SI units** (Système International d'Unités)—the single international standard. It is built from seven **SI base units**:

Table 1: SI base units and their symbols for dimension.

quantity	symbol and name		symbol for dimension
time	s	(second)	T
length	m	(meter)	L
mass	kg	(kilogram)	M
electric current	A	(ampere)	I
temperature	K	(kelvin)	Θ
amount of substance	mol	(mole)	N
luminous intensity	cd	(candela)	J

See [1] for details. Recall that units are case-sensitive.

Mathematical Notations

Chapter 1		Chapter 3	
$\mathbb{C}$	complex numbers	$\wedge$	logical "and"
$\mathbb{R}$	real numbers	$\vee$	logical "or"
$\mathbb{Q}$	rational numbers	$\neg$	logical "not"
$\mathbb{Z}$	integers	$\Longrightarrow$	implication
$\mathbb{N}^0$	natural numbers (0, 1, 2, ...)	$\Longleftrightarrow$	logical equivalence
$\mathbb{N}^+$	natural numbers (1, 2, ...)	$\forall$	for all
$x \in A$	a member of	$\exists$	there exists
$A \cup B$	union	$:=$	is defined as
$A \cap B$	intersection		
$A \setminus B$	set difference		
$f'(x)$	derivative $\left(\text{also } \dfrac{\mathrm{d}f(x)}{\mathrm{d}x}\right)$		
$f^{(n)}(x)$	n-th derivative $\left(\text{also } \dfrac{\mathrm{d}^n f(x)}{\mathrm{d}x^n}\right)$		

Greek symbols

A	α	alpha	H	η	eta	N	ν	nu	T	τ	tau
B	β	beta	Θ	θ	theta	Ξ	ξ	xi	Y	υ	upsilon
Γ	γ	gamma	I	ι	iota	O	o	omicron	Φ	$\varphi\,(\phi)$	phi
Δ	δ	delta	K	κ	kappa	Π	π	pi	X	χ	chi
E	ε	epsilon	Λ	λ	lambda	P	ρ	rho	Ψ	ψ	psi
Z	ζ	zeta	M	μ	mu	Σ	σ	sigma	Ω	ω	omega

Gray-outed ones are almost never used, but others are almost always used. You need to write them so that **others can distinguish each from others**, but Sho thinks Japanese- or Chinese-speakers are OK with it, as we everyday read and write 日 vs 曰 vs 白 or 土 vs 士 vs 工 properly!

Quiz

Q1. Read out the following words.

(1) α-particle **(2)** β-decay **(3)** Γ-function

(4) ε-δ definition **(5)** angle θ and φ **(6)** μ- and τ-leptons

(7) $\psi(x)$ and $\varphi(x)$ **(8)** ζ and ξ **(9)** metric $\eta_{\mu\nu}$

(10) X-ray and γ-ray **(11)** Ω-baryon **(12)** $\varepsilon_{\mu\nu\rho\sigma}$-tensor

Problems

🦾**0.1** Write the following letters so that people can distinguish each from others.

(1) a, α **(2)** b, β **(3)** c, C **(4)** e, E, ε

(5) ζ, ξ **(6)** $\Theta, \theta, Q, \sigma, 6$ **(7)** $g, q, 9$ **(8)** $i, l, 1, I$

(9) k, K, κ **(10)** m, μ **(11)** p, ρ **(12)** s, S

(13) t, τ, T **(14)** u, υ, ν **(15)** w, W, ω **(16)** x, X, χ

Lecture Plan

The content is designed for 150-minute × 14-week lectures, as it is originally for a lecture *Mathematics for Fundamental Physics* (基礎物理數學) in National Sun Yat-sen University (國立中山大學).

1. Derivative
2. Units. Significant Digits.
3. Logic. Theorems. a^x and $\log_a x$.
4. Logic. Theorems. a^x and $\log_a x$.
5. Vectors are arrows.
6. $a \cdot b$ and $a \times b$
7. Matrices (rotation, reflection, scaling)
8. Complex numbers (polar form, Euler's formula, multi-valuedness)
9. Complex vectors, Hermitian inner product, bra-ket notation
10. Vector calculus (grad, div, rot; example of point charge)
11. ODE basics
12. ODE basics
13. Probability and error analysis
14. Probability and error analysis

♣TODO♣a bit on (abstract) vector space to prep for QM?

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Chapter 1

Derivative (Review)

1.1 The first step

Birds sing, fish swim, flowers bloom, stars twinkle, and university students calculate derivatives. Let us begin with basic **derivatives**. Try the next quiz—and note how long it takes.

Quiz

Q1. Calculate the first derivatives of the following functions, **measuring how many minutes it takes.**

(1) $(x+1)^3$ **(2)** $\tan x$ **(3)** $2\cos^2 x$ **(4)** $2\cos x^2$

(5) $x^3 \cos x$ **(6)** $\dfrac{3\sin x}{x}$ **(7)** $x^{3/2}$ **(8)** $\sqrt{\sin x}$

- No mistake, less than two minutes? Amazing! It is as fast as Sho!
- No mistake, less than four minutes? Great, it is exactly as Sho anticipates. Please continue your effort!
- No mistake but takes more than four? That's okay, do not worry. You are just not *fluent*; a little more practice will help you. Try more problems in the Derivative Boot Camp.[#1]
- If you make any mistakes in these eight calculations, then it is a serious issue—just as serious as forgetting how to do the calculations at all. In university, you will perform similar calculations more than 100 times. You need to be both fast and accurate.

At university, students often underestimate the importance of basic calculations. In physics, simple calculations appear constantly, so your speed and accuracy directly affect how well you follow lectures, how efficiently you study, and ultimately your grade. Both can be improved with practice.

This course will help you strengthen these basic calculations—some of which you already know from high school—while also introducing new topics that are essential for physics. You will solve many problems and drills, just as an athlete repeats the same move hundreds of times to master it.

This document (this lecture course) is designed for first-year students in their second semester, **who have already studied derivatives well** in their first-semester calculus. As we do not cover the details of derivatives here, if you are not confident with derivatives, please first study the Derivative Boot Camp.

[#1]Visit `https://misho104.github.io/LecturePublic` and find `gp1_boot1_deriv_true.pdf`.

1.2 Typical Pitfalls

1.2.1 Notations

Consider a function $f(x)$. As we vary the value of x, x is called a **variable**. For a function $f(x)$, there are several equivalent ways to write its derivative:

$$f'(x) = \frac{\mathrm{d}f}{\mathrm{d}x}(x) = \frac{\mathrm{d}f(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}f(x) \quad \text{[All means the same thing]}. \tag{1.1}$$

Similarly, the value of $f'(x)$ at a specific point $x = 3$ can be written as

$$f'(3) = f'(x)\Big|_{x=3} = \frac{\mathrm{d}f}{\mathrm{d}x}\Big|_{x=3} = \frac{\mathrm{d}f}{\mathrm{d}x}(3) = \frac{\mathrm{d}}{\mathrm{d}x}f(3), \tag{1.2}$$

and they are all equivalent. For example, if $f(x) = 3x^2 + 6$, then $f'(x) = 6x$ and $f'(3) = 18$.

Most of students get confused when a **constant** a appears. If a is *declared as a constant*, we can define a function such as $g(x) = ax^2 + 2a$, for which

$$g'(x) = 2ax. \tag{1.3}$$

We can evaluate $g'(x)$ at $x = a$. The result is $g'(a) = 2a^2$, written as

$$g'(a) = g'(x)\Big|_{x=a} = \frac{\mathrm{d}g}{\mathrm{d}x}(a) = \frac{\mathrm{d}}{\mathrm{d}x}g(a) = 2a^2. \tag{1.4}$$

Be careful: The notation $\frac{\mathrm{d}}{\mathrm{d}x}g(a)$ does **not** mean $\frac{\mathrm{d}}{\mathrm{d}x}\big[g(a)\big] = \frac{\mathrm{d}}{\mathrm{d}x}(a^3 + 2a) = 0$.

Fail safe note: If you can't see Eq. (1.3), try setting $a = 3$. Then, you notice $g(x)$ is the same function as $f(x)$, and $g'(x)$ should equal $f'(x)$.

Higher-order derivatives is written as

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right) = \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = f''(x) = f^{(2)}(x), \quad \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right)\right) = \frac{\mathrm{d}^3 f}{\mathrm{d}x^3} = f'''(x) = f^{(3)}(x), \tag{1.5}$$

etc. Please be careful on the position of "2" and "3".

Quiz

Q2. Let a be a constant and $g(x) = ax^2 + 2a$. Calculate

(1) $g(1)$ **(2)** $g(a)$ **(3)** $g(0)$ **(4)** $g(7)$

(5) $g'(1)$ **(6)** $g'(a)$ **(7)** $g''(2)$ **(8)** $\frac{\mathrm{d}}{\mathrm{d}x}g(7)$

Q3. Let $f(x) = x^3 + 2x^2 + 5x + 1$. Calculate

(1) $f^{(2)}(1)$ **(2)** $f'''(1)$ **(3)** $\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}(2)$ **(4)** $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right)\right)$

One more bad news. *Physicists are often lazy* and write $f(x)$ as f. If $f(x) = 2x^2 + 1$, we may write $f' = 4x$ and $f'' = 4$. For $g(t) = 2t^2 - 1$, you may write $g' = 4t$ and $g'' = 4$. Therefore, when you see a function f, you must **identify its variable from the context**.

In physics, a variable can be a function of another variable. The kinetic energy $K(v) = mv^2/2$ is a good example. Its variable v is a function of time t —we write this fact by $v = v(t)$— and thus

$$K(v) = \frac{1}{2}mv^2 \quad \text{but also} \quad K(t) = K(v(t)) = \frac{1}{2}mv(t)^2. \tag{1.6}$$

Now, what does K' mean? —It is ambiguous and we must avoid such notation. Even though, if you see it in a textbook, you need to guess the author's intention from the context.

Advanced note: This rewrite, $K(t)$ and $K(v)$, is common in physics. However, mathematicians do not like this practice because they consider K a unique object: a function should always mean the same rule applied to its input. If they see $K(v) = mv^2/2$, they would say $\big[K$ is an object that converts its input into $m/2 \times (\text{input})^2$, and thus if you feed t into K, you must get $K(t) = mt^2/2\big]$. This is of course not what we mean. For example, imagine $v(t) = \alpha t$ with α a constant. Then

$$\text{physicists:} \quad K(v) = \frac{1}{2}mv^2, \quad K(t) = \frac{1}{2}m\alpha^2t^2; \tag{1.7}$$

but mathematicians would say, $\big[$since it returns $m\alpha^2/2 \times (\text{input})^2$, it is a different object$\big]$ and hence use a different name for it, such as $\tilde{K}$:

$$\text{mathematicians:} \quad K\big(v(t)\big) = \frac{1}{2}m \times v(t)^2 = \tilde{K}(t) = \frac{1}{2}m\alpha^2t^2. \tag{1.8}$$

Both conventions are reasonable, and eventually you will get used to both.

1.2.2 Radian and trigonometric functions

At university, angles are almost always measured in **radians**:

$$360 \text{ degree} \quad (\text{or: } 360°) \quad = \quad 2\pi \text{ radian} \quad (\text{or: } 2\pi \text{ rad}) \tag{1.9}$$

Furthermore, we usually omit "radian" (because we are lazy!). So,

$$\text{a right angle is } \pi/2. \quad \text{The sum of the interior angles of a triangle is } \pi. \tag{1.10}$$

You may write $180° = \pi$, $\pi/180 = 1°$, and so on, but the small circle ° must be written *clearly*.

Quiz

Q4. Express the following angles in radians, and radians in angles.

(1) $180°$	**(2)** $45°$	**(3)** $60°$	**(4)** 90	**(5)** $-30°$
(6) $1°$	**(7)** 1	**(8)** -0.3	**(9)** $x°$	**(10)** x

Why do we insist on radians? The answer comes from the derivative formula

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x. \tag{1.11}$$

Because we have chosen radians as the standard, $(\sin x)'$ becomes this simple.

Advanced note: The simpleness of Eq. (1.11) originates in the fact $\sin(0.01 \text{ rad}) \approx 0.01$. If we used degrees, we would have $\sin(0.01°) \approx 0.01 \times 0.017453$ and everything is messed up with this number 0.017453.

There are a few remarks in the notation of **trigonometric functions**:

$$\begin{aligned} &\sin^2 x \neq \sin x^2. \quad \text{Namely,} \quad (\sin x)^2 = \sin^2 x &&\neq \sin x^2 = \sin(x^2). \\ &\tan^{-1} x \neq \frac{1}{\tan x}. \quad \text{Namely,} \quad \tan^{-1} x = \arctan x &&\neq (\tan x)^{-1} = \frac{1}{\tan x} = \cot x. \end{aligned} \tag{1.12}$$

The following expressions are not incorrect but confusing;

$$\sin^{-2} x, \quad \sin^{1/2} x, \quad \sin(x)^2, \quad \sin(x+1)^2, \quad \ldots$$

We should avoid ambiguity, so please *never* use confusing these notations. Sho thinks we should use $\sin^k x$ only for $k = 2, 3, 4, \ldots$, and use $\arcsin x$ instead of $\sin^{-1} x$.

1.2.3 Several interpretations of derivatives

When you, physics learners, discuss $f'(t)$, you should have the following three interpretations:

- Regarding t as the time, $f'(t)$ is the **rate of change** of $f(t)$ per unit time. If $f'(t) > 0$, the function f is increasing at the time t, while $f'(t) < 0$ means it is decreasing.
- Consider a graph of $f(t)$. Then, $f'(t)$ is the slope of the **tangent line** to the graph at the point t.
- Mathematically, $f'(t)$ is defined by (Check that they are equivalent.)

$$f'(t) := \lim_{\Delta t \to 0} \frac{f(t+\Delta t) - f(t)}{\Delta t} = \lim_{h \to 0} \frac{f(t+h) - f(t-h)}{2h} = \lim_{s \to t} \frac{f(s) - f(t)}{s - t}. \tag{1.13}$$

If you are unsure of them, please consult your first-year Calculus textbooks for further information.

Quiz

Q5. What is the definition of $f'(x)$? Explain.

In physics, it is important to memorize and understand the **definition** of each concept. We will talk about "definitions" in Section 3.5.

1.3 Taylor expansion

Reviewing the definition of $f'(x)$, we have

$$f'(x) := \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}, \quad f'(a) := \lim_{\delta \to 0} \frac{f(a+\delta) - f(a)}{\delta}. \tag{1.14}$$

The second equation is interpreted as follows:

$$\text{If } \delta \approx 0, \quad f'(a) \approx \frac{f(a+\delta) - f(a)}{\delta}, \quad \text{i.e.,} \quad f(a+\delta) \approx f(a) + \delta f'(a) \tag{1.15}$$

and this interpretation is useful in the following example:

Example 1.1

Find the approximate value of the following expressions without using calculators. Then, check your answer with a calculator.

(1) $\sqrt{1.002}$ **(2)** $(1.002)^{10}$ **(3)** $\sqrt{4.004}$

Solution

(1) Apply Eq. (1.15) for $f(x) = \sqrt{x}$, $a = 1$, and $\delta = 0.002$. Then,

$$f'(x) = \frac{1}{2\sqrt{x}}, \quad f'(a) = f'(1) = \frac{1}{2}, \quad f(a+\delta) \approx f(a) + \delta f'(a) = 1 + \frac{\delta}{2} = \underline{1.001}.$$

(2) Doing the same thing for $g(x) = x^{10}$ with $a = 1$ and $\delta = 0.002$,

$$g'(x) = 10x^9, \quad g'(a) = g'(1) = 10, \quad g(a+\delta) \approx g(a) + \delta g'(a) = 1^{10} + 10\delta = \underline{1.02}.$$

(3) Doing the same thing for $h(x) = \sqrt{x}$, $a = 4$, and $\delta = 0.004$, we have

$$h'(4) = \frac{1}{4}, \quad h'(4.004) \approx h(4) + 0.004 \times \frac{1}{4} = \underline{2.001}.$$

Advanced note: "Solutions" contain not only the answer but also how you reached the answer. You are, of course, asked to write such explanations when you solve problems.

Namely, with this technique, you can find **the value of $f(x)$ around a point $x = a$.** This technique, Eq. (1.15), is a special case of Taylor's theorem, which is discussed in ♣TODO♣???.

Quiz

Q6. Find the approximate value of the following expressions without using calculators. Then, check your answer with a calculator.

(1) $\sqrt{1.001}$ **(2)** $(1.0001)^{30}$ **(3)** $\frac{1}{1.001}$

To summarize, we have the following statement:

Statement 1.2 (Taylor expansion: basic)

For a physicists-friendly function $f(x)$,

$$f(a+\delta) \approx f(a) + \delta f'(a), \tag{1.16}$$

or as an equivalent expression,

$$f(x) \approx f(x_0) + (x - x_0) f'(x_0). \tag{1.17}$$

Quiz

Q7. Check these two equations are equivalent. You will be ready to use either of them in the future.

Q8. Review how to derive Eq. (1.16) and summarize it in your own words.

Problems

1.1 Calculate the following expression.

(1) $\sin 30°$ **(2)** $\cos 45°$ **(3)** $\cos 120°$ **(4)** $\sin 150°$ **(5)** $\tan 180°$
(6) $\cos 0$ **(7)** $\sin 2\pi$ **(8)** $\cos \pi$ **(9)** $\sin(\pi/3)$ **(10)** $\cos(-\pi/4)$
(11) $\tan(2\pi/3)$ **(12)** $\sin(\pi/6)$ **(13)** $\tan(5\pi/6)$ **(14)** $\cos(-\pi/6)$ **(15)** $\tan(-\pi/2)$

1.2 For the following $f(x)$, calculate $f'(x)$, $f''(x)$, $f^{(3)}(x)$, $f^{(4)}(x)$, and $f^{(5)}(x)$.

(1) x^3 **(2)** $\sin 2x$ **(3)** e^{2x} **(4)** $\ln 2x$

1.3 Find the approximate value for the following expressions without calculators. Then, check your answer with a calculator.

(1) $(1.002)^7$ **(2)** $(2.002)^7$ **(3)** $(0.998)^4$ **(4)** $(10.01)^3$ **(5)** $1 \div 0.998$
(6) $\sqrt{0.998}$ **(7)** $\sqrt{4.001}$ **(8)** $(0.999)^{1/2}$ **(9)** $(0.999)^{-1/2}$ **(10)** $(0.999)^{-3/2}$
(11) $\dfrac{1}{(2.02)^2}$ **(12)** $\tan 0.001$ **(13)** $\cos 0.002$ **(14)** $\sin 0.003$ **(15)** $\ln 1.001$

****1.4** Find the approximate value for the following expressions without calculators. Then, check your answer with a calculator.

(1) $\sqrt[3]{1.001}$ **(2)** $\sqrt[3]{8.012}$ **(3)** $1.001^{0.5}$ **(4)** $1.001^{-0.9}$ **(5)** $\sin(3.14)$
(6) $\cos(1.57)$ **(7)** $\sin 1°$ **(8)** $\cos 61°$ **(9)** $\ln 0.999$ **(10)** $\log_{10}(10.01)$
(11) $\sqrt{1.001}\sin 0.001$

Fail safe note: If you are not sure about $\sqrt[3]{1.001}$, $(1.001)^{-0.9}$, etc., study Chapter 4 first.

****1.5** For the following $f(x)$, calculate $f^{(n)}(x)$ for all positive integers $n = 1, 2, 3, \ldots$.

(1) x^{10} **(2)** $\sin 2x$ **(3)** e^{2x} **(4)** $\ln x^2$ **(5)** $\sqrt{x}$

***1.6** Write down the definition of the second derivative $f''(x)$. Repeat the discussion of Eq. (1.15) to find the expansion $f'(a+\varepsilon) \approx f'(a) + \varepsilon f'(a) + (\varepsilon^2/2) f''(a)$.

Advanced note: The two equations in Eq. (1.14) have similar but different meanings: the first one defines a new function $f'(x)$, while the second one defines a number that is eventually equal to $f'(x)|_{x=a}$. Anyway, we don't care the difference.

1.4 Mathematical Notation

As *physicists are lazy*, we usually use the following notation:

$x \in \mathbb{C}$	means "x is a complex number".	$x \in \mathbb{Z}$	means "x is an **integer**".[#2]
$x \in \mathbb{R}$	means "x is a real number".	$x \in \mathbb{N}^+$	means "x is a positive integer".
$x \in \mathbb{Q}$	means "x is a **rational number**".	$x \in \mathbb{N}^0$	means "x is a non-negative integer".

[#2] Z comes from "Zahlen" (Zahl).

The symbol "$\in$" means *"is a member of"*, and $\mathbb{N}^+$ means a **set** of positive integers, $\{1, 2, 3, 4, \ldots\}$. So,

$\boxed{x \in \mathbb{N}^+}$ and $\boxed{x \in \{1, 2, 3, 4, 5, \ldots\}}$ means the same thing: "x is a positive integer".

Be careful: In mathematics, "natural numbers" usually mean $\mathbb{N}^0$, including 0. Meanwhile, physicists and other people tend to think 0 is not a natural number.

Quiz

Q9. What is the difference between $\mathbb{N}^+$ and $\mathbb{N}^0$?

Q10. What is the difference between $\mathbb{N}^0$ and $\mathbb{Z}$?

Q11. What is "rational numbers"? What is the difference between $\mathbb{Z}$ and $\mathbb{Q}$?

We write "x is a positive number" by $\boxed{x \in \mathbb{R}, x > 0}$ but also by $\boxed{x > 0}$. Similarly,

$\boxed{x \in \mathbb{R}, x \leq 0}$ or $\boxed{x \leq 0}$ means "x is a non-positive number".

Be careful: Namely, when we say something is "positive" or "negative", we implicitly assume it is a real number.

Advanced note: This is because we cannot compare complex numbers with 0. The operator $>$ is defined only for real numbers. When we write $a > b$, we implicitly consider that a and b are real numbers.

Problems

****1.7** The **intersection** of two sets A and B is a set made by elements both in A and B; we write it by $A \cap B$. The **union** of A and B is a set made by elements in A, B, or both; we write it by $A \cup B$. The **set difference**, $A \setminus B$, is a set made by elements in A but not in B.

For example, let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and $C = \{2, 4\}$. Then,

$$A \cap B = \{2, 3\}, \quad A \cup B = \{1, 2, 3, 4\}, \quad A \setminus C = \{1, 3\}, \quad B \setminus C = \{3\}, \quad C \setminus B = \{\},$$

where $\{\}$ (or $\varnothing$) means the **empty set**, the set without any elements.

(1) Write the following sets.

(a) $A \cap C$ **(b)** $B \cap C$ **(c)** $A \cup C$ **(d)** $B \cup C$ **(e)** $A \cup B \cup C$

(f) $B \setminus A$ **(g)** $C \setminus A$ **(h)** $C \setminus B$ **(i)** $(B \setminus A) \cup A$ **(j)** $\mathbb{N}^0 \setminus \mathbb{N}^+$

(2) Express the following statements only with $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{N}^+$, $\mathbb{N}^0$, $\cap$, $\cup$, $\setminus$, and "$x \in$".

(a) x is a rational number. **(b)** x is an irrational number.

(c) x is a rational number but not an integer. **(d)** x is a non-positive integer.

(e) x is a negative integer. **(f)** x is a non-zero integer.

Chapter 2

Units and Significant figures

A **physical quantity** is usually not just a number, but also has a **unit** and an **uncertainty** (error). For example, "$(72 \pm 2)\,\mathrm{kg}$" has a number "72" and a unit "kg", which compose the **central value** "72 kg". It also has uncertainty because every measurement has an uncertainty. In experimental physics, uncertainty is usually more important than the central value, and thus uncertainty handling is a fundamental skill of physicists.

In this chapter, we first review physical quantities and how to handle units. Then, we discuss **significant figures**, a simple and convenient method to express the uncertainty. Further discussions on uncertainty analysis are given in Chapter ♣TODO♣chapter.

2.1 Physical quantity

The expression "$(72 \pm 2)\,\mathrm{kg}$" means the central value is 72 kg and the **absolute uncertainty** is 2 kg. Accordingly, its **relative uncertainty** is given by $(2\,\mathrm{kg})/(72\,\mathrm{kg}) = 0.028$. Let's see other examples.

	unit	central value	absolute uncertainty	relative uncertainty
$10\,\mathrm{m} \pm 1\,\mathrm{cm}$	m (meter)	10 m	$1\,\mathrm{cm} = 0.01\,\mathrm{m}$	0.001 (or 0.1%)
$1.6\,\mathrm{A} \pm 0.04\,\mathrm{A}$	A (ampere)	1.6 A	0.04 A	0.025 (or 2.5%)
$(5 \pm 1) \times 10^{-3}\,\mathrm{C}$	C (coulomb)	0.005 C	0.001 C	0.2 (or 20%)
$(50 \pm 1)\,\mathrm{m/s}$	m/s	50 m/s	1 m/s	0.02 (or 2%)

Remark: Upper- and lowercase letters are distinguished. "M" **does not** mean "meter". The unit "A" is "ampere", not "Ampere". The unit "coulomb" is "C", not "c".

Quiz

Q1. For each of the following values, find its unit, central value, absolute uncertainty, and relative uncertainty.

(1) $50\,\mathrm{kg} \pm 500\,\mathrm{g}$ **(2)** $(0.05 \pm 0.001)\,\mathrm{A}$

(3) $(1.6 \pm 0.001) \times 10^{-19}\,\mathrm{C}$ **(4)** $72\,\mathrm{km/h} \pm 1\,\mathrm{m/s}$

Q2. The next passage has six (6) errors in the use of uppercase and lowercase letters. Find them out.

In the SI system, temperature is expressed in k (kelvin), a unit named after Lord Kelvin. The unit of force is N (newton), named after the british scientist Isaac newton. The units A (Ampere) and c (coulomb) are named after French scientists. In contrast, Kg (kilogram) is not named after a person. , but comes from Greek.

As an undergraduate student, you need to follow the following rules:

Statement 2.1 (Rules for Physical Quantities: Basic)

1. Numbers are always with units, even in calculations.
2. Include units in a symbol.
3. For physical quantities, use decimals (e.g., 1.6 A). Do not use fractions like (8/5) A.
4. Use significant figures to express the measurement precision.

These rules are explained below.

Be careful: We use *decimals for physical quantities* and *fractions for mathematical concepts*. Since this lecture is mainly on math and math concepts, you will use decimals ♣TODO♣only in this section.

2.2 Units

When we educate kids, we write "my height is $\boxed{}$ cm", or "my height is h cm", where $h = 172$ is just a number. But this is not nice! We want to convert the units freely and write equations such as $1.72\,\mathrm{m} = 172\,\mathrm{cm} = 0.00172\,\mathrm{km}$. So, **we always include units in symbols** $h = 172\,\mathrm{cm}$. Then

$$\text{we can write: } h = 172\,\mathrm{cm} = 1.72\,\mathrm{m} = 0.00172\,\mathrm{km} = 1.82\times 10^{-16}\,\text{light-year}.$$

Similarly, if $m = 110\,\mathrm{g}$ and $\mathrm{g} = 9.8\,\mathrm{m/s^2}$,

$$\begin{aligned} \text{we may write: } & w = mg = 110\,\mathrm{g} \times 9.8\,\mathrm{m/s^2} = 1.1\,\mathrm{kg\cdot m/s^2} \\ \text{but not: } & w = mg = 0.11 \times 9.8 = 1.1\,\mathrm{kg\cdot m/s^2} \end{aligned}$$

This second equation is incorrect because m is not equal to 0.11; m is equal to 0.11 kg or 110 g.

Remark: Usually, physicists use upright fonts for units and *italic fonts* for quantities. For example, m, T, and C are quantities, which can be mass, temperature, capacitance, etc. Meanwhile, m, T, and C are units: meter, tesla, and coulomb, respectively.

Quiz

Q3. **(1)** If $mr\omega^2 = 10.0\,\mathrm{kg\cdot m/s^2}$, $m = 5.0\,\mathrm{kg}$, and $r = 1.0\,\mathrm{m}$, what is ω?

(2) If $v_0 t + (1/2)at^2 = 5.0\,\mathrm{m}$ with $a = -4.0\,\mathrm{m/s^2}$ and $v_0 = 7.0\,\mathrm{m/s}$, what is t?

(3) At time $t = 0$, a particle is at $(x, y) = (2.0\,\mathrm{m}, 0)$. It moves with a constant velocity $(v_x, v_y) = (3.0, 4.0)\,\mathrm{m/s}$. What is its position at $t = 1.0\,\mathrm{s}$?

Every physical concept has its own unit. For example, speed has m/s, acceleration has $\mathrm{m/s^2}$, and energy has $\mathrm{kg\cdot m/s^2} = \mathrm{N\cdot m} = \mathrm{J}$. The next table lists the quantities you have learned. Notice that angle (rad) has no dimension. It is called a **dimensionless** quantity.

concept	dimension	typical unit	
speed	$\mathsf{L\,T^{-1}}$	m/s	
acceleration	$\mathsf{L\,T^{-2}}$	$\mathrm{m/s^2}$	
linear momentum	$\mathsf{M\,L\,T^{-1}}$	kg·m/s	
force	$\mathsf{M\,L\,T^{-2}}$	$\mathrm{kg\cdot m/s^2}$	(= N)
energy / work	$\mathsf{M\,L^2\,T^{-2}}$	$\mathrm{kg\cdot m^2/s^2}$	(= J)
frequency	$\mathsf{T^{-1}}$	1/s	(= Hz)
angle	1 (dimensionless)	rad	

Advanced note: To understand why angle has no dimension, you may consider the definition of the radian: it is defined as the ratio of the arc length to the radius, so the units cancel (meter / meter = 1).

Here, please do not write: $\boxed{\text{The force is } \mathrm{kg\cdot m^2/s^2}\text{. This is also called newton (N).}}$, because *we cannot mix units and concepts*. A correct (and easy) way is to use English words:

$\boxed{\text{the unit of the force is } \mathrm{kg\cdot m/s^2}}$ or $\boxed{\text{the force has the unit } \mathrm{kg\cdot m/s^2}.}$

Similarly, the following statement is incorrect because it mixes concepts and units:

$\boxed{\text{Since } m = \mathrm{kg} \text{ and } g = \mathrm{m/s^2},\ mg = \mathrm{kg\cdot m/s^2} = \mathrm{N}.}$

Instead, you need to write

$\boxed{\text{Since } m \text{ has a unit of kg and } g \text{ has a unit of } \mathrm{m/s^2},\ mg \text{ has a unit of } \mathrm{kg\cdot m/s^2} = \mathrm{N}.}$

Notice the last equation: units can be equal to other units, so we may write $\boxed{\mathrm{N} = \mathrm{J/m} = \mathrm{kg\cdot m/s^2}}$ $\boxed{\mathrm{m} = \mathrm{J/N}}$ $\boxed{\mathrm{s} = \sqrt{\mathrm{kg\cdot m/N}}}$. These are scientifically correct equations.

Remark: We sometimes use informal notations, such as $\boxed{[F] = \mathrm{N}}$ or $\boxed{F \rightsquigarrow \mathrm{N}}$ to express "the unit of F is N (newton)". Still, using an equal symbol (=) is **not allowed**. Namely, $\boxed{F = \mathrm{N}}$ is always incorrect.

Quiz

Q4. What are the units of the following concepts? Write English sentences to explain it.

(1) speed **(2)** velocity **(3)** force **(4)** energy **(5)** power

Advanced note: In very formal situations, we use **symbols for dimensions** (see Table 1 on page 4):

Since $\dim(m) = \mathsf{M}$ and $\dim(g) = \mathsf{L\,T^{-2}}$, $\dim(mg) = \mathsf{M\,L\,T^{-2}}$,

with the operator "dim" [1]. However, it seems too complicated for most situations.

Problems

★**2.1** Express the following units only with SI base units, e.g., $N = kg{\cdot}m/s^2$.

(1) W (watt) **(2)** Pa (pascal) **(3)** J (joule) **(4)** C (coulomb)

*****2.2** Find a few more examples of dimensionless quantities in physics.

****2.3** Use "dim" notation to express the dimensions of the following quantities. For example, if v is speed, then $\dim(v) = \mathsf{L}\,\mathsf{T}^{-1}$.

(1) speed v **(2)** force F
(3) angular momentum L **(4)** electric charge Q
(5) resistance R **(6)** specific heat capacity c
(7) Planck constant h **(8)** fine-structure constant α

2.3 Significant figures

In researches, we need to treat uncertainties in the method given in ♣TODO♣chap. However, most of lectures use **significant figures** to express the accuracy of a value, so that students becomes familiar with the concept of uncertainties. For example,

3.14	means the last digit "4" is uncertain. It can be 3.1***6***, 3.1***5***, 3.1***1***, 3.1***0***, ...
3.141	means "3.14" is for sure but it can be 3.14***3***, 3.14***2***, 3.14***0***, 3.1***39***, ...
3.1415	might be 3.141***6***, 3.1415***3***, etc., but the writer is sure *it is not* 3.145 or 3.135.

We write the "uncertain digit" in ***a different style*** for clarity. You should notice the difference between:

1.000***0***	meaning the last digit "0" is uncertain so it may be 1.0002, 1.0001, 0.9998, ...
1.0***0***	meaning it can be 1.03 or 0.99, but the writer is sure *it is not* 1.2 or 0.8.

It is usually convenient to use **scientific notation**:

0.00012***3***	is OK but we prefer	$1.2\boldsymbol{3}\times10^{-4}$.	Both mean the value can be $1.2\boldsymbol{2}\times10^{-4}$ etc.
0.0010***0***	is OK but we prefer	$1.0\boldsymbol{0}\times10^{-3}$.	
2997***9***	is OK but we prefer	$2.997\boldsymbol{9}\times10^{4}$.	

Fail safe note: $10^3 = 1000$ and $10^{-3} = 1/10^3 = 1 \div 1000$. Go to ♣TODO♣chap for details!

We need to *avoid ambiguous notations*. Namely,

2040	is ambiguous and not nice. We are not sure the author means 2.04×10^3 or 2.040×10^3.
120	is not nice. We are not sure the author means It may mean 1.20×10^2 or 1.2×10^2.
42000	is not nice because it has four ways to interpret the author's intention.

Quiz

Q5. What are the four interpretations of 42000? Write them in scientific notation.

We use **rounding**-to-the-nearest (四捨五入) when necessary. For example, if you need to convert to three significant figures, it will be

$$1.234 \to 1.23, \quad 1.235 \to 1.24, \quad 31.98 \to 32.0, \quad 1234 \to 1.23\times10^3, \quad \text{and so on.}$$

Quiz

Q6. Round the following numbers to three significant figures.

(1) 41.11 **(2)** 98.76 **(3)** 100.12 **(4)** 15.449

(5) 2.2360 **(6)** 0.0123456 **(7)** 9876 **(8)** 9.999×10^4

Remark: Programmers use `2.99e8` (or `2.99E8`) to mean 2.99×10^8, or `1.6e-12` (or `1.6E-12`) to mean 1.6×10^{-12}. However, we **should not** use them in handwriting.

Problems

2.4 Write the following numbers in scientific notation. However, some of them are ambiguous, so answer "ambiguous" if so.

152 340 1000 9999 43210 0.0300 0.00213 31.0 31.00 31

2.5 The expressions 0.123, 1.23, and 123 are numbers with ***three*** *significant figures*. Similarly, 1234 and 1.234×10^{-3} are numbers with ***four*** *significant figures*. How about the following expressions?

(1) 3.14 **(2)** 0.11 **(3)** 1.1×10^{-2} **(4)** 1.1×10

(5) 1.0008 **(6)** 1.0020 **(7)** 1.0000 **(8)** 4.00×10^{-10}

(9) 0.0008 **(10)** 0.00080 **(11)** 12345 **(12)** 67890

2.4 Calculation with Significant figures

We need to do calculations of numbers with uncertainties, such as $1.23+4.56$ or $1.23/4.56$, but how? Here we discuss a simple method for

- addition $(x+y)$ and subtraction $(x-y)$
- multiplication $(x\times y)$ and division $(x\div y$ or $x/y)$

Other calculations, such as $\sqrt{x}$, e^x or $\sin(x)$, need the professional method given in ♣TODO♣chap.

Multiplication and Division

As a rule, if x has m significant figures and y has n significant figures, then $x\times m$ or $x\div y$ should be rounded to have $\min(m,n)$ significant figures. ...But the rule is tough to understand. Probably you should see the example, and learn through practice.

Example 2.2

Calculate the following expressions, using calculators.

(1) 8.912×2.4 **(2)** $5.0 \div 3.14$ **(3)** 2.01×49.8 **(4)** 1.210^3

Solution

Here we use a shorthand notation "3-SF" to mean "three significant figures".

(1) Since 8.912 has 4-SF and 2.4 has 2-SF, we round the result to (the smaller) 2-SF:

$$8.912 \times 2.4 = 21.3888 \to \underline{21}.$$

(2) Since 5.0 has 2-SF and 3.14 has 3-SF, we round the result to 2-SF:

$$5.0 \div 3.14 = 1.592\ldots \to \underline{1.6}.$$

(3) Both 2.01 and 49.8 have 3-SF, so we keep 3-SF, but don't forget to avoid ambiguity!

$$2.01 \times 49.8 = 100.098 \to 100 \to \underline{1.00 \times 10^2}.$$

(4) $1.210^3 = 1.210 \times 1.210 \times 1.210$, so we round the result to 4-SF:

$$1.210^3 = 1.771561 \to \underline{1.772}.$$

If you want, we can justify this treatment by carrying out a hand calculation of the long multiplication.

$$
\begin{array}{cccccc}
 & & & 1. & 2 & \mathit{0} \\
\times & & & 1. & 3 & \mathit{1} \\
\hline
 & & & \mathit{1} & \mathit{2} & \mathit{0} \\
 & & 3 & 6 & \mathit{0} & \\
 & 1 & 2 & \mathit{0} & & \\
\hline
 & 1 & 5 & \mathit{7} & \mathit{2} & \mathit{0}
\end{array}
\qquad
\begin{array}{ccccccc}
 & & & 8. & 9 & 1 & \mathit{2} \\
\times & & & & & 2. & \mathit{4} \\
\hline
 & & \mathit{3} & \mathit{5} & \mathit{6} & \mathit{4} & \mathit{8} \\
 & 1 & 7 & 8 & 2 & \mathit{4} & \\
\hline
 & 2 & \mathit{1} & \mathit{3} & \mathit{8} & \mathit{8} & \mathit{8}
\end{array}
\qquad
\begin{array}{ccccccc}
 & & & & 2. & 0 & \mathit{1} \\
\times & & & & 4 & 9. & \mathit{8} \\
\hline
 & & & \mathit{1} & \mathit{6} & \mathit{0} & \mathit{8} \\
 & & 1 & 8 & 0 & \mathit{9} & \\
 & & 8 & 0 & \mathit{4} & & \\
\hline
 & 1 & 0 & \mathit{0.} & \mathit{0} & \mathit{9} & \mathit{8}
\end{array}
$$

Observe which digits are uncertain, and how they "pollute" the results in each step. In $1.2\mathit{0} \times 1.3\mathit{1}$, the last $\mathit{1}$ pollutes the part of $\mathit{1} + 6$ and we get an uncertain number $\mathit{7}$. So, the result is $1.5\mathit{7}$.

Addition and subtraction

Different rules are applied for addition and subtraction. You can understand the rules easily if you do long addition and observe which digits are polluted.

Example 2.3

Calculate the following expressions.

(1) $12.3 + 4.56$ **(2)** $123 + 4.56$ **(3)** $0.50 - 0.032$ **(4)** $1.20 \times 10^3 - 27$

Solution

$$
\begin{array}{ccccc}
 & 1 & 2. & \mathit{3} & \\
+ & & 4. & 5 & \mathit{6} \\
\hline
 & 1 & 6. & \mathit{8} & \mathit{6}
\end{array}
\qquad
\begin{array}{cccccc}
 & 1 & 2 & \mathit{3} & & \\
+ & & & 4. & 5 & \mathit{6} \\
\hline
 & 1 & 2 & \mathit{7.} & \mathit{5} & \mathit{6}
\end{array}
\qquad
\begin{array}{ccccc}
 & 0. & 5 & \mathit{0} & \\
- & 0. & 0 & 3 & \mathit{2} \\
\hline
 & 0. & 4 & \mathit{6} & \mathit{8}
\end{array}
\qquad
\begin{array}{ccccc}
 & 1 & 2 & \mathit{0} & \mathit{0} \\
- & & & 2 & \mathit{7} \\
\hline
 & 1 & 1 & \mathit{7} & \mathit{3}
\end{array}
$$

and we round the results. So, the answers are $16.\mathit{9}$, $12\mathit{8}$, $0.4\mathit{7}$, and $1.1\mathit{7} \times 10^3$.

Problems

2.6 Calculate the following, taking care of significant figures. You may use calculators.

(1) 1.23×4.5 **(2)** 1.50×2.0 **(3)** 2.00×5.00 **(4)** 4.0×2.5

(5) $1.23 + 4.5$ **(6)** $1.50 + 2.0$ **(7)** $2.00 + 8.00$ **(8)** $100.0 + 0.123$

(9) $1.23 \div 4.5$ **(10)** $6.0 \div 3.00$ **(11)** $9.00 \div 4.5$ **(12)** $1.50 \div 3.0$

(13) $1.23 - 4.5$ **(14)** $3.10 - 0.10$ **(15)** $3.10 - 0.1$ **(16)** $3.10 - 3.0$

(17) $131 + 69$ **(18)** $131 + 6.9$ **(19)** $131 + 0.69$ **(20)** $1.3 \times 10^2 + 69$

(21) $(1.2 \times 10^5) \times 9.4$ **(22)** $1.2 \div (5.2 \times 10^5)$

(23) $(3.33 \times 10^5) \times (6.3 \times 10^4)$ **(24)** $(3.33 \times 10^5) \times (6.3 \times 10^{-4})$

(25) $(3.33 \times 10^5) \div (6.3 \times 10^4)$ **(26)** $(3.33 \times 10^5) \div (6.3 \times 10^{-4})$

(27) $1.00 \times 10^4 + 2.5 \times 10^3$ **(28)** $3.00 \times 10^3 - 4.50 \times 10^2$

(29) $2.99 \times 10^2 + 0.814$ **(30)** $1.23 \times 10^{-4} + 4.5 \times 10^{-6}$

****2.7** Calculate the following, taking care of significant figures. You may use calculators.

(1) $1.50 \times 2.000 \times 3.50$ **(2)** $8.00 \div 4.00 \div 0.25$ **(3)** 5.0^3 **(4)** 2.10^3

(5) $1.0 + 2.0 + 3.0 \times 4.0$ **(6)** $17 - 8.0 - 3.0$ **(7)** $3.0^3 + 1.2$ **(8)** $5.0^3 - 5$

(9) $(1.2 + 3.45) \times 2.1$ **(10)** $(8.0 - 1.25) \div 2.0$ **(11)** $1.2 \times 3.4 + 5.6$

(12) $18.0 - 2.5 \times 1.2$ **(13)** $1.20 \times 3.40 + 5.60$ **(14)** $\left(5.0 \times 10^2 + 3.4\right) \times 1.2$

2.8 Calculate the following with calculators, taking care of significant figures and units.

(1) $3.1\,\mathrm{m/s} \times 1.0\,\mathrm{hour}$ **(2)** $3.1\,\mathrm{m} \div 25\,\mathrm{cm}$

(3) $1.5\,\mathrm{kg} \times 9.8\,\mathrm{m/s^2}$ **(4)** $1.50\,\mathrm{km} + 195\,\mathrm{m}$

(5) $\left(5.2\,\mathrm{kg/m^3}\right) \times (4.0\,\mathrm{\mu m})^3$ **(6)** $\left(5.2\,\mathrm{kg/m^3}\right) \times \left(4.0\,\mathrm{\mu m^3}\right)$

(7) $1.2\,\mathrm{\mu m} \div 2.00\,\mathrm{nm}$ **(8)** $\left(1.7 \times 10^{-27}\,\mathrm{kg}\right) \times \left(3.00 \times 10^8\,\mathrm{m/s}\right)^2$

(9) $2.000\,\mathrm{kg} \times \left(3.0\,\mathrm{m/s^2}\right)$ **(10)** $2.000\,\mathrm{kg} \times (3.0\,\mathrm{m/s})^2$

(11) $\left(1.60 \times 10^{-19}\,\mathrm{C}\right) \div 2.0\,\mathrm{\mu s}$ **(12)** $\left(1.60 \times 10^{-19}\,\mathrm{C}\right) \times 5.0\,\mathrm{kV}$

(13) $\left(6.63 \times 10^{-34}\,\mathrm{J \cdot s}\right) \times \left(3.0 \times 10^{12}\,\mathrm{Hz}\right)$ **(14)** $1.50\,\mathrm{MJ} \div 20.0\,\mathrm{m}$

Advanced note: Uncertainties can often be determined subjectively, but central values are also without specific definitions; it is often the averaged value of the measurements, but one may assume some probabilistic distribution and use its median, mean, or mode as the central value.

Chapter 3

Logic

Solve Problem 1.7 (on page 20) again.

(1) If $mr\omega^2 = 10.0\,\text{kg}\cdot\text{m/s}^2$, $m = 5.0\,\text{kg}$, and $r = 1.0\,\text{m}$, what is ω?
(2) If $v_0 t + (1/2)at^2 = 5.0\,\text{m}$ with $a = -4.0\,\text{m/s}^2$ and $v_0 = 7.0\,\text{m/s}$, what is t?
(3) At time $t = 0$, a particle is at $(x, y) = (2.0\,\text{m}, 0)$. It moves with a constant velocity $(v_x, v_y) = (3.0, 4.0)\,\text{m/s}$. What is its position at $t = 1.0\,\text{s}$?

The answers are

(1) $\omega = \pm 1.4\,\text{s}^{-1}$ **(2)** $t = 1.0\,\text{s}, 2.5\,\text{s}$ **(3)** $x = 5.0\,\text{m}, y = 4.0\,\text{m}$

but **what do they mean?**

> **Quiz**
>
> **Q1.** Guess the correct meaning of each of above expressions.
>
> **(1)** ω is both $+1.4\,\text{s}^{-1}$ and $-1.4\,\text{s}^{-1}$ v.s. ω is either $+1.4\,\text{s}^{-1}$ or $-1.4\,\text{s}^{-1}$
>
> **(2)** t is both $1.0\,\text{s}$ and $2.5\,\text{s}$ v.s. t is either $1.0\,\text{s}$ or $2.5\,\text{s}$
>
> **(3)** x is $5.0\,\text{m}$ and y is $4.0\,\text{m}$ v.s. Either x is $5.0\,\text{m}$, or y is $4.0\,\text{m}$.

Elementary educations do not discuss these differences because kids do not know **logical thinking**. Now, you are a grown-up university student. You need to think about it.

3.1 Basic Logics

Since elementary school, you have written many equalities and inequalities, such as

$$3 + 5 = 8, \quad 6 + 2 = 10, \quad 1 + 3 > -3, \quad 1 + 3 \neq 0, \quad 3^3 = 9, \quad 1 > 2, \quad \text{and} \quad \sin(\pi) \neq 0.$$

> **Quiz**
>
> **Q2.** The above statements are either **true** (T) or **false** (F). For each of them, state whether it is true or false.
>
> **Q3.** Confirm that the following statements are all true:
>
> **(1)** $3 + 5 = 8$ is true. **(2)** $6 + 2 = 10$ is false.
>
> **(3)** $3 \neq 3$ is false. **(4)** False is false.
>
> **(5)** "False is false" is true. **(6)** "False is true" is false.

We can only write true things, such as "$3 + 5 = 8$", "$3^2 = 9$ is true", or "$6 + 2 = 10$ is false". In other words, **when you write something, you must confirm that it is true.** Sometimes, you are asked to confirm a statement, or **prove** a statement. The process is called **proof** of the statement.

Numbers can be manipulated by operators such as $+$, $\div$. Similarly, true (T) and false (F) can be manipulated by **and**, **or**, and **not** operators.

and	(conjunction)	"A and B" ($A \wedge B$) is true if both A and B are true, and false otherwise.
or	(disjunction)	"A or B" ($A \vee B$) is true if at least one of A and B is true. False if both are false.
not	(negation)	"not A" ($\neg A$) is true if A is false, and false if A is true.

We can write these property as equations:

$$\begin{aligned} &(\mathrm{T} \wedge \mathrm{T}) = \mathrm{T}, && (\mathrm{T} \wedge \mathrm{F}) = (\mathrm{F} \wedge \mathrm{T}) = (\mathrm{F} \wedge \mathrm{F}) = \mathrm{F}, \\ &(\mathrm{F} \vee \mathrm{F}) = \mathrm{F}, && (\mathrm{T} \vee \mathrm{T}) = (\mathrm{T} \vee \mathrm{F}) = (\mathrm{F} \vee \mathrm{T}) = \mathrm{T}, \\ &(\neg\, \mathrm{T}) = \mathrm{F}, && (\neg\, \mathrm{F}) = \mathrm{T}, \end{aligned}$$

but the **truth table**, shown to the right, is more useful.

A	B	A and B	A or B	not A
T	T	T	T	F
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

Remark: Note that "$1 + 1 = 2$ or $2 + 2 = 4$" is true, which says the word "or" is a bit different from everyday English. In daily conversation, *"Sho will eat ramen or sushi tonight"* usually means "but not both". However, in mathematics, if Sho eats both ramen and sushi for a dinner[#3], the statement *"Sho had sushi or ramen tonight"* is true.

Quiz

Q4. State whether it is true or false for the following statements.

(1) $1 + 2 = 3$ and $3 + 4 < 5$. **(2)** $10 \div 3 > 0$ or $10 \div 2 > 0$.

(3) $\neg(3 > 5)$. **(4)** $(6 + 3 > 0) \wedge \neg(5 - 2 > 0)$.

Fail safe note: Don't be confused by daily English.
"$x = 1$ and $x = -1$ are the two solutions of $x^2 = 1$." (correct daily English)
"$x^2 = 0$ is satisfied for $x = 1$ or $x = -1$." (correct mathematical English)
are both correct, but if you say "$x = 1$ and $x = -1$", then it means an impossible equality $x = 1 = -1$.

3.2 Implication $\Longrightarrow$

We can discuss the following statements: are they true or false?

Example 3.1

- If $x > 5$, then $x > 1$. ... This is true. (Number larger than 5 are larger than 1.)
- If $x < 9$, then $x < 2$. ... This is false, because we have a counterexample $x = 5$.
- If $x^2 = 1$, then $x \leq 1$. ... This is true.
- If $x^2 = 1$, then $x \leq 0$. ... This is false, because we have a counterexample $x = 1$.
- If $x^2 = 1$, then $x = -1$. ... This is false, because we have a counterexample $x = 1$.

[#3] Sho sometimes does it.

In general, it is easy to claim that a statement is false, i.e., to *refute* or *disprove* a statement. You just have to find one **counterexample**. Meanwhile, it is more difficult to claim that a statement is true: you need to write a rigorous proof.

Now, let's try to find some counterexamples.

Quiz

Q5. The following statements are false. Find a counter example for each.

(1) If $x^2 > 0$, then $x > 0$. **(2)** If $x^2 - x = 0$, then $x = 0$.

(3) If $x^2 = y^2$, then $x = y$. **(4)** If $x + y < 1$, then $x < 1$ and $y < 1$.

We use the symbol "$\Longrightarrow$" to express "if ... then ..." statement.

$A \Longrightarrow B$ means If A is true, then B is true. or equivalently, A implies B.

and this is called **implication**. For example, the statements in the above quiz can be written as

$$x^2 > 0 \Rightarrow x > 0, \quad x^2 - x = 0 \Rightarrow x = 0, \quad x^2 = y^2 \Rightarrow x = y, \quad x + y < 1 \Rightarrow x < 1 \wedge y < 1.$$

Quiz

Q6. Similarly rewrite the statements in Example 3.1 with using the symbol "$\Longrightarrow$".

Advanced note: In formal logic, $A \Rightarrow B$ is defined by $(\neg A) \vee B$, which means "if A is true, B must also be true; but if A is false, we do not care about B". So, if A is always false, $A \Rightarrow B$ is always true, no matter what B is. For example, $(x \in \mathbb{R} \wedge x^2 < -1) \Rightarrow x = 999$ is a true statement. Through this definition, it is clear that $A \Rightarrow B$ is equivalent to $\neg B \Rightarrow \neg A$, which we will see below.

3.3 Equivalent Statements—What does "solve" mean?

Since elementary school, you have solved many equations, but what does "solving an equation" mean?

Quiz

Q7. **(1)** Choose true statements.

(a) $2x - 1 = 0 \Longrightarrow x = 1/2.$ **(b)** $x^2 = 4 \Longrightarrow x = 2.$

(c) $2x - 1 = 0 \Longrightarrow x > -10.$ **(d)** $x^2 = 4 \Longrightarrow x = 0.$

(2) Do you think these statements can be considered as "solving an equation"?

(b) and **(d)** are false statements (find counterexamples!), so we cannot write them. **(a)** and **(c)** are true statements, and **(a)** looks "solving an equation", but we do not consider **(c)** is "solving an equation". The difference between **(a)** and **(c)** is that

$$2x - 1 = 0 \Longleftarrow x = 1/2 \quad \text{is true, but} \quad 2x - 1 = 0 \Longleftarrow x > -10 \quad \text{is false.}$$

So, for **(a)**, both $\Longrightarrow$ and $\Longleftarrow$ are correct: we use $\Longleftrightarrow$ to express such cases.

Definition 3.2 (Equivalent)

If both $A \Longrightarrow B$ and $A \Longleftarrow B$ are true, we call "A and B are **equivalent**", and write $A \Longleftrightarrow B$.

$A \Longrightarrow B$ is read by B if A in English.
$A \Longleftrightarrow B$ is read by B if and only if A or B **iff** A (namely, **if**-and-only-**if**).

Now you need to do a bit of practice. It is actually tough, unfortunately, but this drill is important for logical thinking.

Problems

3.1 Fill in the blanks with $\Longrightarrow$, $\Longleftarrow$, or $\Longleftrightarrow$, where x is a real number.

(1) $x = 2 \ \boxed{} \ x > 0$ **(2)** $x > 1 \ \boxed{} \ x > 0$ **(3)** $x = 1 \ \boxed{} \ x > 0$
(4) $x = -3 \ \boxed{} \ x^2 = 9$ **(5)** $x = 0 \ \boxed{} \ x^2 = 0$ **(6)** $x < 0 \ \boxed{} \ x^2 > 0$
(7) $x = 3 \ \boxed{} \ x \neq 5$ **(8)** $3x = 3 \ \boxed{} \ x = 1$ **(9)** $x^3 = x^2 \ \boxed{} \ x = 1$

If the statement "$A \Rightarrow B$" is true, then

- B is called a **necessary condition** for A, because B is necessary for A; if not B, then not A.
- A is called a **sufficient condition** for B, because if A is true, B is "sufficiently" true.

So, if A and B are equivalent, B (resp. A) is called *necessary-and-sufficient condition* for A (resp. B). For example, because $x = 1 \Longrightarrow x^2 = 1$,

- $x = 1$ is sufficient to satisfy $x^2 = 1$. ($x = 1$ is a sufficient condition for $x^2 = 1$.)
- $x^2 = 1$ is necessary to satisfy $x = 1$. ($x^2 = 1$ is a necessary condition for $x = 1$.)

Quiz

Q8. Fill in the blanks with "a sufficient condition" or "a necessary condition".

(1) $x = -1$ is $\boxed{}$ for $x^2 = 1$. **(2)** $x^2 = 4$ is $\boxed{}$ for $x = 2$.
(3) $x > 1$ is $\boxed{}$ for $x > 0$. **(4)** $x^2 > 0$ is $\boxed{}$ for $x < 0$.
(5) $x < 0$ is $\boxed{}$ for $x^2 > 0$. **(6)** $x^2 > 0$ is $\boxed{}$ for $x > 0$.

Now we have the answer for the question: What does "solving an equation" mean?

Definition 3.3 (Solving an equation)

"Solving an equation" means "finding an *equivalent* equation in the form of $x = \boxed{}$".

Consider the equations in Problem 2.8. As $2x - 1 = 0 \Leftrightarrow x = 1/2$ and $x^2 = 4 \Leftrightarrow x = 2 \vee x = -2$ are true statements, we say $x = 1/2$ and $x = \pm 2$ are the solutions of the equations, respectively.

When you solve an equation, you have to check that the solution is **necessary and sufficient**.

- If you don't check necessity, you may have an "incomplete solution".
- If you don't check sufficiency, you may have an "extraneous solution".

Quiz

Q9. **(1)** Solve $\sqrt{x+2} = x$. You might reach an *extraneous solution*, which you need to *exclude* it.

(2) Solve $|2x| = |x+1|$. Obviously $x = 1$ satisfies this equation, but it is an *incomplete solution* and you need to find more solutions.

Before discussing advanced topics on logics, you should do some drills.

Problems

3.2 State whether each statement is true (T) or false (F). For example, $2 > 5$ is "F".

(1) $(3 > 1) \wedge (2 < 5)$ **(2)** $3 < 1 \vee 2 > 5$ **(3)** $\neg(3 > 1)$

(4) $\neg(2 > 5)$ **(5)** $(\neg(3 < 1)) \wedge (2 < 5)$ **(6)** $\neg(3 < 1 \wedge 2 < 5)$

(7) $(4 = 4) \wedge \neg(2 > 5)$ **(8)** $(3 < 1) \vee \neg(3 = 1)$ **(9)** $(\neg(3 < 1)) \vee \neg(3 > 1)$

3.3 State whether each statement is true or false. If false, give a counter example.

(1) If $x = 2$, then $x^2 = 4$. **(2)** If $x^2 = 4$, then $x = 2$.

(3) $x^2 = 0 \Longrightarrow x = 0$ **(4)** $x > 2 \Longrightarrow x > 0$

(5) $x > 0 \Longrightarrow x > 2$ **(6)** $x = 3 \Longrightarrow |x| = 3$

(7) $(x = 1) \vee (x = -1) \Longrightarrow x^2 = 1$ **(8)** $x^2 = 1 \Longrightarrow (x = 1) \vee (x = -1)$

★3.4 Solve each equation, assuming x is a real number.

(1) $\sqrt{x} = 3$ **(2)** $\sqrt{x-1} = 2$ **(3)** $|x| = 5$ **(4)** $|x| = |-3|$

(5) $\sqrt{x^2} = 3$ **(6)** $|x^2| = 5$ **(7)** $|x-2| = 3$ **(8)** $|2x+1| = 7$

****3.5** Solve each equation, assuming x is a real number.

(1) $\sqrt{x-3} = 2$ **(2)** $\sqrt{x-3} = x-3$ **(3)** $\sqrt{x-3} = 3-x$

(4) $\sqrt{(x-3)^2} = 2$ **(5)** $\sqrt{(x-3)^2} = x-1$ **(6)** $\sqrt{(x-3)^2} = x-3$

(7) $\sqrt{(x-3)^2} = 3-x$ **(8)** $\sin x = 1/2$ **(9)** $\tan x = 0$

★3.6 The following "solutions" have logical errors. Identify errors and find the right answer.

(1) A student solves $x^2 = 9$ and writes $x = 3$.

(2) He solves $x^2 - 3x = 0$ by dividing both sides by x, obtaining $x - 3 = 0$, so $x = 3$.

(3) He solves $x^6 = x^4$ by dividing both sides by x^4, obtaining $x^2 = 1$, so $x = \pm 1$.

Next, fill in the blanks with $\Longrightarrow$, $\Longleftarrow$, or $\Longleftrightarrow$.

(4) $x^2 = 9 \ \boxed{}\ x = 3$ **(5)** $x^2 - 3x = 0 \ \boxed{}\ x - 3 = 0 \ \boxed{}\ x = 3$

(6) $x^6 = x^4 \ \boxed{}\ x^2 = 1 \ \boxed{}\ x = \pm 1$

★**3.7** Let x and y be real numbers. Fill in the blanks with $\Longrightarrow$, $\Longleftarrow$, or $\Longleftrightarrow$. If your answer is $\Longrightarrow$ or $\Longleftarrow$, explain the reason by giving relevant counter examples.

(1) $x = 2 \; \boxed{} \; x^2 = 4$ **(2)** $x^2 = 4 \; \boxed{} \; x = 2 \lor x = -2$

(3) $x > 0 \; \boxed{} \; x^2 > 0$ **(4)** $|x| = 2 \; \boxed{} \; (x = 2 \lor x = -2)$

(5) $x + 1 = 0 \; \boxed{} \; x = -1$ **(6)** $x^2 = x \; \boxed{} \; x(x-1) = 0 \; \boxed{} \; x = 0$

(7) $\sqrt{x} = 123 \; \boxed{} \; x = 123^2$ **(8)** $\sqrt{x^2} = 0 \; \boxed{} \; x = 0$

(9) $\sqrt{x^2} = 3 \; \boxed{} \; x = 3$ **(10)** $\sqrt{x^2} = 3 \land x > 0 \; \boxed{} \; x = 3$

(11) $\sqrt{x^2} = \sqrt{y^2} \; \boxed{} \; x = y$ **(12)** $x + y > 0 \; \boxed{} \; x > 0 \land y > 0$

(13) $xy = 0 \; \boxed{} \; x = y = 0$ **(14)** $x^2 + y^2 = 0 \; \boxed{} \; x = y = 0$

****3.8** Let a be a real constant and x be a real number. For (1)–(4), fill in the blanks with $\Longrightarrow$, $\Longleftarrow$, or $\Longleftrightarrow$. If your answer is $\Longrightarrow$ or $\Longleftarrow$, explain the reason by counter examples.

(1) $\sqrt{x} = 4 \; \boxed{} \; x = 16$ **(2)** $\sqrt{x^2} = 4 \; \boxed{} \; x = 4$

(3) $\sqrt{x} = a^2 \; \boxed{} \; x = a^4$ **(4)** $\sqrt{x^2} = a^2 \; \boxed{} \; x = a^2$

(5) Solve the equation $\sqrt{x^2} = a$, noting that a can be negative.

****3.9** **(1)** Recall that $A \Rightarrow B$ is defined by $(\neg A) \lor B$. Also, recall that $A \Leftrightarrow B$ is defined by $A \Rightarrow B \land A \Leftarrow B$. Write a truth table (see page 27) for $A \Rightarrow B$, $A \Leftarrow B$, and $A \Leftrightarrow B$.

(2) Explain the reason we can understand $A \Leftrightarrow B$ as $A = B$.

(3) Write a truth table for the following expressions:

$$A \land B, \quad \neg(A \land B), \quad (\neg A) \lor (\neg B), \quad \neg(A \lor B), \quad (\neg A) \land (\neg B)$$

Explain that this truth table is considered as a *proof* of **de Morgan's theorem**

$$\neg(A \lor B) = (\neg A) \land (\neg B), \qquad \neg(A \land B) = (\neg A) \lor (\neg B). \tag{3.1}$$

(4) Prove the following, which we will discuss in the next section.

$$(A \Rightarrow B) = \big((\neg B) \Rightarrow (\neg A)\big), \qquad (A \Rightarrow B) = \neg(A \land \neg B). \tag{3.2}$$

***3.10** This lecture does not cover **quantifiers** "for-all $\forall$" and "exists $\exists$". Learn them by yourselves. In particular, prove the following:

(1) $\exists x, P(x) \Longleftrightarrow \neg\big(\forall x, \neg P(x)\big)$ **(2)** $\neg\big(\exists x, P(x)\big) \Longleftrightarrow \forall x, \neg P(x)$

(3) $\forall x, P(x) \Longleftrightarrow \neg\big(\exists x, \neg P(x)\big)$ **(4)** $\neg\big(\forall x, P(x)\big) \Longleftrightarrow \exists x, \neg P(x)$

***3.11** For each statement, write its negation, such as "not ($x \geq 0$ for all real x)", in a natural form. Then, state whether the original statement and the negated one are true or false.

(1) $x^2 \geq 0$ for all real x. **(2)** There exists a real x such that $x^2 = -1$.

(3) For all real x, $x > 0 \Rightarrow x^2 > 0$. **(4)** $|x| = x$ for all real x.

3.4 A few more notes about logic and proof

There is a useful theorem for "not" operator, called **de Morgan's theorem**.

Theorem 3.4 (De Morgan's theorem)

1. "not (A and B)" is equivalent to "(not A) or (not B)", i.e., $\neg(A \wedge B) = (\neg A) \vee (\neg B)$.
2. "not (A or B)" is equivalent to "(not A) and (not B)", i.e., $\neg(A \vee B) = (\neg A) \wedge (\neg B)$.

The proof is given as Problem 3.9.

Example 3.5

For each of the following statements, write their negation in a natural form.

(1) $x = 3$ **(2)** $y < 2$ **(3)** $x = 3$ or $y < 2$ **(4)** $x = 3$ and $y < 2$

Solution

The negation of **(1)** is $x \neq 3$ and the negation of **(2)** is $y \geq 2$. Since de Morgan's theorem says not(A or B) = (not A and not B), the negation of **(3)** is $x \neq 3$ and $y \geq 2$. Similarly, with de Morgan's theorem, the negation of **(4)** is $x \neq 3$ or $y \geq 2$.

Quiz

Q10. For each of the following, write its negation.

(1) $x > 0$ or $x < 1$. **(2)** $x \neq 4$ and $x \neq 5$. **(3)** $0 < x < 1$.

(4) $x > 0$ or $y > 0$. **(5)** $x = 0$ and $y = 0$. **(6)** $x \neq 0$ and $y \neq 0$.

In general, $A \Rightarrow B$ and $B \Rightarrow A$ are different. However, if $A \Rightarrow B$, then $\neg B \Rightarrow \neg A$ (not $B \Rightarrow$ not A) is always correct. It is called the **contrapositive** of $A \Rightarrow B$. Let's see an example.

Example 3.6

If $0 < x < 2$, then x^2 is always less than 4. We can write it by $(0 < x < 2) \Rightarrow \left(x^2 < 4\right)$. Let A be the statement "$0 < x < 2$" and B be the statement "$x^2 < 4$". "Not A" is "$x \leq 0$ or $x \geq 2$". Meanwhile, "not B" is "$x^2 \geq 4$".

- Its **converse** ($A \Leftarrow B$) is $\left(x^2 < 4\right) \Rightarrow (0 < x < 2)$, which is false (counterexample: $x = -1$).
- Its **inverse** (not $A \Rightarrow$ not B) is ($x \leq 0$ or $x \geq 2$) $\Rightarrow \left(x^2 \geq 4\right)$, which is false.
- Its **contrapositive** (not $A \Leftarrow$ not B) is $\left(x^2 \geq 4\right) \Rightarrow$ ($x \leq 0$ or $x \geq 2$), which is true.

Because the contrapositive of a true statement is always true, we can prove a statement by proving its contrapositive. This method is called **proof by contrapositive**.

Quiz

Q11. For each of the following statements, write its converse, inverse, and contrapositive. Then state whether each of them is true or false.

(1) If $x > 3$, then $x > 0$. **(2)** If $x = 2$, then $x^2 = 4$.

(3) If $|x| = 0$, then $x = 0$. **(4)** If $x^2 = 4$, then $x \neq 3$.

Q12. Prove the following statements by *proof by contrapositive*. Namely, first write the contrapositive, then prove it.

(1) If $x^2 > 1$, then $|x| > 1$. **(2)** If $x^3 + x^2 + x < 0$, then $x < 0$.

(3) If $xy \neq 0$, then $x \neq 0$. **(4)** If n^2 is not integer, then n is not integer.

Another powerful method is **proof by contradiction** (deductio ad absurdum). Imagine you prove "if A, then B". You assume "A" and "not B" are both true, and derive a contradiction. If you reach a contradiction, it means "A and not B" is false, i.e., "not (A and not B)" is true, which means $A \Rightarrow B$.

Quiz

Q13. Prove the following statements by *proof by contradiction*.

(1) If x^2 is an even integer, then x cannot be an odd integer.

(2) If a, b, and c are integers and abc is even, at least one of a, b, or c is even.

Remark: The validity of "proof by contrapositive" and "proof by contradiction" was already discussed on page 31, Eq. (3.2).

Advanced note: You learned another useful method for proof, **mathematical induction**, in high school. In university, you may learn more advanced methods, such as **infinite descent** and **transfinite induction**.

3.5 Facts, Definitions, Assumptions, and Conclusions

One of the main reasons students find university physics difficult is that they treat all statements in the same way. In physics, every statement has its own role:

- experimental fact
- definition (in mathematics: definition)
- assumption / approximation (in mathematics: axiom or assumption)
- derived statements (in mathematics: theorem, corollary, or proposition)

You should always know which role each statement plays.

Remark: Mathematics does not have experimental facts because it is not based on experiments; it is purely based on definitions.

Example 3.7 (Types of statements in a calculation)

A particle moves as $x(t) = kt^2 + x_0$, where k and x_0 are constants. Find the averaged velocity between $t = 0$ and $t = t_0$.

Solution

We should first clarify assumptions and definitions:

- Assumptions (explicitly written): $x(t) = kt^2 + x_0$. k and x_0 are constants.
- Assumptions (implicit): k, x_0, and t_0 are real numbers, and t means time.
- Definition: averaged velocity is *defined* by $v_{\text{avg}} := \dfrac{x(t_0) - x(0)}{t_0 - 0}$.

Now, we can calculate $\dfrac{x(t_0) - x(0)}{t_0 - 0} = \dfrac{\left(kt_0^2 + x_0\right) - x_0}{t_0} = kt_0$ to give a derived statement:

- Derived statement (conclusion): the averaged velocity between $t = 0$ and t_0 is $v_{\text{avg}} = kt_0$.

Quiz

Q14. Solve the following problem, but with clarifying the nature of each of your statements (as is done in the above example).

A particle moves as $x(t) = kt^2 + x_0$, where k and x_0 are constants. Find the instantaneous velocity at $t = t_1$.

Consider an **experimental fact**, such as "gravity is always attractive". We have to accept it. We need to remember it and understand what it says. "Protons have charge $+|e|$ and electrons have charge $-|e|$" is another experimental fact, but we can understand it as the definition of the elementary charge $|e|$.

Definitions often appear as mathematical equations, such as "instantaneous velocity is defined by $v(t) := \mathrm{d}x/\mathrm{d}t$". Another example is "uniform circular motion is a motion with constant speed along a circle"; here, the word "uniform circular motion" is defined.

Meanwhile, **assumption** is what we can impose freely. For example, you can decide whether you ignore air resistance or not. However, every conclusion is valid only under its assumptions. If you impose bad assumptions, your conclusion will be meaningless. For example, if you want to analyze the free fall of a sheet of paper but you ignore air resistance, your result should be invalid and meaningless. You must be very careful when you make assumptions.

Remark: We often use "$A := B$" to say "define A by B". For example, $v(t) := \mathrm{d}x/\mathrm{d}t$ is the definition of instantaneous velocity, and $T := 1/f$ is the definition of period. Meanwhile, "$A = B$" just means "A is equal to B".

Some physicists use $A \equiv B$ to say "A is defined by B", but Sho does not recommend it because mathematicians use the symbol $\equiv$ for other purposes.

Problems

★**3.12** Prove the following statements by contrapositive or contradiction.

(1) $x^2 + y^2 = 0 \Longrightarrow x = 0$.

(2) $ab = 0 \Longrightarrow a = 0$ or $b = 0$.

(3) $ab = 0 \Longleftarrow a = 0$ or $b = 0$.

★**3.13** Find a logical mistake: Let $a = b = 1$. Then $a^2 = ab$, so $a^2 - b^2 = ab - b^2$. So, $(a - b)(a + b) = b(a - b)$, which gives $a + b = b$, so $2 = 1$.

★**3.14** With de Morgan's theorem, rewrite the following statements in a form without "not".

(1) not ($x > 0$ and $y > 0$) **(2)** not ($x = 0$ or $y = 0$)

(3) not (if $x > 1$, then $x^2 > 1$) **(4)** not ($x > 0$ iff $x^2 > 0$)

*****3.15** With de Morgan's theorem, rewrite the following statements in a natural form.

(1) not ((not A) or (not B)) **(2)** not ((not A) and (not B))

(3) $\neg((\neg A) \wedge (\neg B))$ **(4)** $\neg(A \vee \neg B)$

*****3.16** For each of the following statements, write its converse, inverse, and contrapositive. State T or F for the original statement, its converse, inverse, and contrapositive.

(1) If $x^2 - 5x + 6 = 0$, then $x = 2$ or $x = 3$.

(2) If $x > 0$ and $y > 0$, then $x + y > 0$.

(3) If x and y are both even integers, then xy is divisible by 4.

*****3.17** A student claims: "I proved $\neg B \Rightarrow \neg A$, so I have proved $B \Rightarrow A$." Is this correct? Explain, and state what the student has actually proved.

*****3.18** The following two statements look similar but are logically different. For each, state whether it is T or F and explain.

(1) For all real x: if $x > 0$, then $x^2 > 0$.

(2) For all real x: if $x^2 > 0$, then $x > 0$.

*****3.19** True or false? Give a reason.

(1) The contrapositive of a false statement is also false.

(2) If $A \Rightarrow B$ is true and $B \Rightarrow A$ is false, then A and B are not equivalent.

(3) If A is false, then $A \Rightarrow B$ is always true, for any B.

***3.20** **(1)** What is a **proposition**?

(2) Prove the following statements on propositions P, Q, and R.

(a) $(P \Rightarrow Q \wedge Q \Rightarrow R) \Longrightarrow P \Rightarrow R$. (**syllogism**)

(b) $(P \Rightarrow Q) \wedge (P \Rightarrow R) \Longleftrightarrow P \Rightarrow (Q \wedge R)$.

Chapter 4

Power, Exponential, and Logarithm

4.1 Powers

Let's begin with a review: can you recall all the rules for powers...?

Quiz

Q1. Calculate without a calculator.

(1) $2^4,\quad 3^5,\quad 4^2,\quad 10^3,\quad 1^{25},\quad 2^{10}$

(2) $5^0,\quad 3^0,\quad 2^0,\quad 1^0$

(3) $2^{-1},\quad 12^{-1},\quad 3^{-3},\quad 2^{-4},\quad 5^{-2},\quad 2^{-10}$

(4) $\left(\frac{1}{2}\right)^{-1},\quad \left(\frac{1}{3}\right)^{-2},\quad \left(\frac{1}{12}\right)^{-1},\quad \left(\frac{1}{5}\right)^{-2},\quad \left(\frac{1}{2}\right)^{-10},\quad \left(\frac{1}{3}\right)^{0},\quad \left(\frac{1}{10}\right)^{0}$

(5) $\left(\frac{1}{2}\right)^{4},\quad \left(\frac{2}{3}\right)^{3},\quad \left(\frac{5}{2}\right)^{2},\quad \left(\frac{2}{5}\right)^{-2},\quad \left(\frac{7}{2}\right)^{2},\quad \left(\frac{7}{2}\right)^{-2},\quad \left(\frac{1}{3}\right)^{2},\quad \left(\frac{1}{3}\right)^{-2}.$

(6) $4^{1/2},\quad 81^{1/2},\quad 3^{1/2},\quad 27^{1/3},\quad \left(\frac{1}{4}\right)^{1/2},\quad \left(\frac{1}{7}\right)^{1/2},\quad \left(\frac{8}{9}\right)^{1/2},\quad \left(\frac{15}{3}\right)^{1/2}$

(7) $4^{-1/2},\quad 81^{-1/2},\quad 3^{3/2},\quad 27^{2/3},\quad 27^{-2/3},\quad \left(\frac{1}{4}\right)^{3/2},\quad \left(\frac{1}{7}\right)^{-1/2},\quad \left(\frac{1}{2}\right)^{-5/2}$

Q2. Fill in the blanks, where $A > 0$, $B > 0$, and x, y are real numbers.

(1) $A^3 \cdot A^3 = A^{\Box}$ **(2)** $(A^3)^3 = A^{\Box}$ **(3)** $(A \cdot B)^4 = \Box \cdot \Box$

(4) $A^x \cdot A^y = A^{\Box}$ **(5)** $(A^x)^y = A^{\Box}$ **(6)** $(A \cdot B)^x = A^x \cdot \Box$

(7) $A^{\Box} = \dfrac{1}{A^x}$ **(8)** $A^x = \dfrac{1}{\Box}$ **(9)** $\dfrac{A^x}{A^y} = A^{\Box}$

(10) $A^0 = \Box$ **(11)** $A^{\Box} = A$ **(12)** $A^{\Box} = \sqrt{A}$

(13) $A^{\Box} = \sqrt[3]{A}$ **(14)** $\dfrac{A^x}{B^x} = \Box$ **(15)** $\left(\dfrac{A}{B}\right)^{-1} = \Box$

Q3. With $A > 0$, $B > 0$, $x \in \mathbb{R}$, and $y \in \mathbb{R}$, the following statements are all false. Find a counterexample for each.

(1) $A^x + A^y = A^{x+y}$ **(2)** $(A+B)^x = A^x + B^x$ **(3)** $(A \cdot B)^x = A \cdot B^x$

(4) $(A^x)^y = A^{(x^y)}$ **(5)** $(A^x)^y = A^{x+y}$

Do not worry if you cannot recall all the rules! Learn this section and come back here, and then you will be able to answer all the questions!

We should first recall $\sqrt{a}$... what is that?

Definition 4.1 (n-th root of non-negative numbers)

For $a \geq 0$, we define $\sqrt[n]{a}$ for $n = 2, 3, 4, \ldots$ by the **positive** solution x of the equation $x^n = a$ and call it the ***n*-th root** of a. In particular, write we $\sqrt{a}$ instead of $\sqrt[2]{a}$.

Since we have assumed $a \geq 0$, the equation $x^n = a$ always has a unique positive solution. It guarantees that $\sqrt[n]{a}$ is **always and uniquely** determined; we usually say "$\sqrt[n]{a}$ is **well-defined**" to describe this situation. Also, notice that $\sqrt[n]{a} \geq 0$ *by definition*.

Example 4.2 (n-th root of a positive number)

$\sqrt[3]{8}$ is the positive solution of $x^3 = 8$. So, $\sqrt[3]{8} = 2$. Similarly, $\sqrt[4]{1.4641}$ is the positive solution of $x^4 = 1.4641$, so $\sqrt[4]{1.4641} = 1.1$. Notice $\sqrt[n]{0}$ is always 0 for any $n = 2, 3, 4, \ldots$

Be careful: We **do not** consider $\sqrt[n]{x}$ for $x < 0$.

Advanced note: It is actually not impossible to consider such cases, but Sho recommends not to use such notations because they involve mathematical subtleties. You will just be confused.

We are ready to define the **power**, a^x. Here, a is called the **base** and x is called the **exponent**.

Definition 4.3 (Power for positive base, real exponent)

For positive **base** $a > 0$ and real **exponent** $x \in \mathbb{R}$, we define the **power** a^x by the following steps.

- $a^0 := 1$, $a^1 := a$.
- For $x = 2, 3, 4, \ldots$, we define $a^x := a \cdot a^{x-1}$ (*recursive definition*).

We have covered the case $x \in \mathbb{N}^0$. Next, we consider positive rational numbers, $x \in \mathbb{Q}, x > 0$. Recall that a positive rational number x can be expressed as $x = n/m$ with $n \in \mathbb{N}^0, m \in \mathbb{N}^+$.

- If $x > 0$ and $x \in \mathbb{Q}$ but $x \notin \mathbb{N}$, then we define $a^x = a^{n/m}$ ($n, m \in \mathbb{N}^+, m \geq 2$) by

$$a^{1/m} := \sqrt[m]{a}, \qquad a^{n/m} := \left(a^{1/m}\right)^n = \left(\sqrt[m]{a}\right)^n.$$

- If $x > 0$ and $x \notin \mathbb{Q}$, we can consider a sequence of rational numbers $x_k \in \mathbb{Q}$ such that $x_k \to x$. Then we define $a^x := \lim_{k\to\infty} a^{x_k}$.

Now we have defined a^x for all $x \geq 0$. Finally,

- If $x < 0$, we define $a^x := 1/\left(a^{|x|}\right)$.

and then we have defined a^x for all $x \in \mathbb{R}$, if $a > 0$.

Example 4.4 (First two items: integer-power of a positive number)

- $2^0 := 1$, because of the first item in the above definition.
- The second item says $2^5 := 2 \times 2^4$, $2^4 = 2 \times 2^3$, $2^3 = 2 \times 2^2$, and $2^2 = 2 \times 2^1 = 4$. So, *recursively*, $2^3 = 2 \times 4 = 8$, $2^4 = 2 \times 8 = 16$, and $2^5 = 32$.
- Similarly, $1.1^2 = 1.1 \times 1.1 = 1.21$ and $1.1^3 = 1.1 \times 1.21 = 1.331$.

Example 4.5 (Third and fourth items: positive-power of a positive number)

For rational number x, we use the third item of the above definition to consider a^x. For example, $27^{1/3} = \sqrt[3]{27} = 3$ and $1.4641^{1/4} = \sqrt[4]{1.4641} = 1.1$. Then, we can get $27^{2/3} = \left(27^{1/3}\right)^2 = 3^2 = 9$, $1.4641^{0.25} = \sqrt[4]{1.4641} = 1.1$, and $1.4641^{0.75} = \left(1.4641^{0.25}\right)^3 = 1.1^3 = 1.331$.

If x is not rational, we need to use the fourth item. For example, if $x = \sqrt{2} = 1.4142...$, we consider a sequence $1, 14/10, 141/100, 1414/1000, \ldots$ which converges to $\sqrt{2}$. Then,

- $2^{\sqrt{2}}$ is the limit of $2^1, 2^{14/10}, 2^{141/100}, 2^{1414/1000}, \ldots \longrightarrow 2^{\sqrt{2}} = 2.665....$
- $3^\pi = 3^{3.14159\ldots}$ is the limit of $3^3, 3^{31/10}, 3^{314/100}, 3^{3141/1000}, \ldots \longrightarrow 3^\pi = 31.544....$

Quiz

Q4. Using calculators, calculate the following numbers.

(1) 3^3 and $3^{3.1}$. **(2)** $\sqrt[10]{3}$ and $\left(\sqrt[10]{3}\right)^{31}$

(3) $\sqrt[100]{3}$ and $\left(\sqrt[100]{3}\right)^{314}$ **(4)** $3^{3.14}$ and $3^{3.141592653579}$

Example 4.6 (negative-power of a positive number)

For a^x with $x < 0$, we use the last item in the above definition.

$$1.1^{-2} = \frac{1}{1.1^2} = \frac{1}{1.21} \qquad 1.4641^{-0.75} = \frac{1}{1.4641^{0.75}} = \frac{1}{1.331} \qquad 3^{-\pi} = \frac{1}{3^\pi} = \frac{1}{31.544\ldots}$$

We have discussed a^x for positive base a. Powers a^x for $a \le 0$ require very careful consideration and we postpone it to the later chapters; Table 2 gives a summary of a^x for various a and x. Two simplest cases are discussed here.

Definition 4.7 (Power for negative base)

For a negative base $a < 0$, we only consider a^n for integer n.

$$a^0 := 1, \qquad a^n := a \cdot a^{n-1} \text{ for } n > 0, \qquad a^n := 1/a^{|n|} \text{ for } n < 0. \tag{4.1}$$

This is equivalent to $a^n := (-1)^n |a|^n$, where we define $(-1)^n$ is $+1$ for even n and -1 for odd n.

Definition 4.8 (Power for zero base)

We define $0^x := 0$ for $x > 0$, and $\sqrt[n]{0} := 0$ for $n \in \mathbb{N}^+$. (We do not consider 0^x for $x \le 0$.)

Quiz

Q5. Some of the following expressions are not defined. Find all the undefined ones.

$$\pi^{-3} \quad \pi^{-1/\pi} \quad (-2)^0 \quad (-2)^{-3} \quad (-2)^{3/8} \quad -2^{-\pi} \quad \sqrt{-4}$$

$$\sqrt[3]{3.5} \quad \sqrt[3.5]{3} \quad \sqrt[-4]{3} \quad \sqrt[3]{0} \quad 0^{2.5} \quad 0^{-2.5} \quad 0.1^{0.1}$$

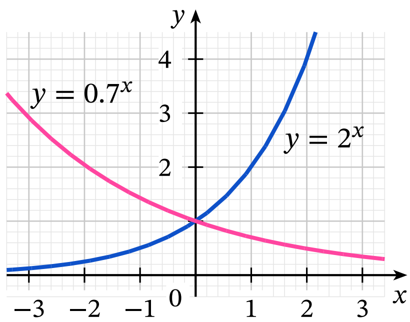
Table 2: Summary of the definition of a^x for various a and x.

$a > 0$	$x \in \mathbb{R}$	$(\text{positive})^{(\text{real})}$	Definition 4.3
	$x \in \mathbb{C} \setminus \mathbb{R}$	$(\text{positive})^{(\text{non-real})}$	exp-complex-properties
$a < 0$	$x \in \mathbb{Z}$	$(\text{negative})^{(\text{integer})}$	Definition 4.7: $a^x = \lvert a\rvert^x \times (-1)^x$
	$x \notin \mathbb{Z}$	$(\text{negative})^{(\text{others})}$	♣TODO♣chapter
$a = 0$	$x > 0$	$0^{(\text{positive})}$	Definition 4.8: $0^x = 0$
	$x = 0$	0^0	*not considered* (but sometimes $0^0 := 1$)
	not $x \geq 0$	$0^{(\text{other})}$	*not considered*

In the rest of this chapter, we will only consider powers of positive base, i.e., a^x for $a > 0$.

4.2 Exponential functions and Logarithmic functions

Powers with positive bases, such as 2^x and 0.7^x, are considered as an **exponential function** defined for $x \in \mathbb{R}$. The figure to the right shows the graphs of $y = 2^x$ and $y = 0.7^x$.



Obviously, $f(x) = a^x$ is **strictly increasing** if $a > 1$; **strictly decreasing** if $0 < a < 1$. In Section 4.4, we will check this property by calculating $f'(x)$. It is also important that a^x can take any positive real value. We will use these facts to define $\log_a x$ below in this section.

Quiz

Q6. Draw the graphs of $y = 2^x$, $y = 0.5^x$, and $y = 1^x$ by hand, without using calculators or computers.

Theorem 4.9 (Properties of Exponential Functions)

For $a > 0$, $b > 0$, and $x, y \in \mathbb{R}$,

$$a^0 = 1, \quad \frac{1}{a^x} = a^{-x}, \quad \sqrt[n]{a^x} = a^{x/n} \quad (n \in \mathbb{N}, n \geq 2); \tag{4.2}$$

$$a^x a^y = a^{x+y}, \quad \frac{a^x}{a^y} = a^{x-y}, \quad (a^x)^y = a^{xy}, \quad (ab)^x = a^x b^x, \quad \left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}. \tag{4.3}$$

Remark: Usually, a^{x^y} is interpreted as $a^{(x^y)}$, not as $(a^x)^y = a^{xy}$.

Quiz

Q7. Transform the following numbers to the form of $2^{\square}$.

(1) 2	**(2)** 1024	**(3)** $\sqrt{2}$	**(4)** $\sqrt[3]{4}$	**(5)** 0.25	**(6)** $\sqrt[5]{0.25}$
(7) 4^x	**(8)** $\left(\sqrt{2}\right)^x$	**(9)** $4 \cdot 2^x$	**(10)** $2^x/16$	**(11)** $2^x 4^y$	**(12)** $2^x/4^y$

Q8. Without calculators, arrange the following numbers in order from smallest to largest.

$$2,\quad 1024,\quad 2^5,\quad 8^2,\quad \sqrt{2},\quad \sqrt[3]{4},\quad 0.25,\quad 0.5^{0.3},\quad \sqrt[5]{0.25},\quad 0,\quad 1$$

Consider the equation $p = a^x$, where $a > 0$, $a \neq 1$ and $p \in \mathbb{R}$. For a given p, how many solutions x does it have? As $a \neq 1$, a^x is strictly increasing or decreasing and it takes any value $0 < a^x < \infty$. Therefore, $p = a^x$ has a unique real solution x for any $p > 0$. We call the solution $x = \log_a p$.

Definition 4.10 (Logarithm)

For $a > 0$, $a \neq 1$, and $p > 0$, we define the **logarithm** $\log_a p$ as the unique solution x of $a^x = p$.

$$\text{For } a > 0, a \neq 1, \text{and } p > 0, \qquad a^x = p \iff x = \log_a p. \tag{4.4}$$

Notice that $\log_a p$ is undefined if not $(a > 0 \wedge a \neq 1 \wedge p > 0)$, i.e., if $(a < 0 \vee a = 1 \vee p \leq 0)$.

Fail safe note: Go back to Theorem 3.4, de Morgan's theorem, on page 32.

Quiz

Q9. Based only on the above discussion, explain why $k = \log_a(a^k)$ and $a^{\log_a k} = k$.

We here review basic properties of $\log_a x$. Drill problems are prepared later in this section.

Theorem 4.11 (Properties of Logarithmic Functions)

For $a > 0$ but $a \neq 1$, $b > 0$ but $b \neq 1$, $A > 0$, and $B > 0$,

$$\log_a 1 = 0, \quad \log_a a = 1, \quad \log_a A^k = k \log_a A \quad (k \in \mathbb{R}), \tag{4.5}$$

$$\log_a(AB) = \log_a A + \log_a B, \quad \log_a\left(\frac{A}{B}\right) = \log_a A - \log_a B, \tag{4.6}$$

$$\log_a A = \frac{\log_b A}{\log_b a} \quad \text{[changing the base]}. \tag{4.7}$$

4.3 Napier's number and Natural logarithm

There is a special number for the base, called **Napier's number** e:

$$\mathrm{e} = 2.718281828\ldots = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \sum_{k=0}^{\infty}\frac{1}{k!} = 1+1+\frac{1}{2}+\frac{1}{6}+\frac{1}{24}+\frac{1}{120}+\ldots \tag{4.8}$$

It is special because of the following theorem:

Theorem 4.12 (Napier's number)

$$\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^x = \mathrm{e}^x, \quad \frac{\mathrm{d}}{\mathrm{d}x}\ln x = \frac{1}{x}, \tag{4.9}$$

where the **natural logarithm** ($\ln x$) is defined by $\ln x := \log_e x$. Also, we often write e^x as $\exp(x)$.

Quiz

Q10. Draw the graphs of $y = \exp(x)$ and $y = \ln x$, using computers or calculators.

Q11. Calculate the following, where $a > 0$, $b > 0$, and $b \neq 1$.

(1) $\frac{\mathrm{d}}{\mathrm{d}x}\exp(x)$ **(2)** $\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{-3x}$ **(3)** $\frac{\mathrm{d}}{\mathrm{d}x}a^x$ **(4)** $\frac{\mathrm{d}}{\mathrm{d}x}\ln(3x)$ **(5)** $\frac{\mathrm{d}}{\mathrm{d}x}\log_b x$

Fail safe note: Review Problem 3.20 for (3). Review Eq. (4.7) for (5). Recall e is just a number.

Problems

You may skip these two drill problems, as (Sho expects) you **already have learned** them.

4.1 Differentiate them, where a is a positive constant such that $a \neq 1$.

(1) $\exp(5x)$ **(2)** e^{-x^2} **(3)** $3\exp(2x)$ **(4)** $\mathrm{e}^{\sin x}$ **(5)** $x(\mathrm{e}^x)^2$

(6) $\sqrt{\mathrm{e}^x+1}$ **(7)** $\mathrm{e}^x\mathrm{e}^{x^2}$ **(8)** $\ln(2x)$ **(9)** $\ln|3x|$ **(10)** $\ln(x^5+x)$

(11) $\ln|\cos x|$ **(12)** $x\ln x$ **(13)** $\ln(1/x)$ **(14)** $\mathrm{e}^x\ln x$ **(15)** $\ln|2x-1|$

(16) a^{4x} **(17)** a^{x^2} **(18)** $(2a)^{2x}$ **(19)** $\log_a x^5$ **(20)** $\log_a(x^5+x)$

4.2 Simplify each expression, where a is a positive constant such that $a \neq 1$.

(1) $9^{-1/2}$ **(2)** $16^{3/4}$ **(3)** $8^{-2/3}$ **(4)** $\mathrm{e}^{3\ln 2}$

(5) $\mathrm{e}^{\ln 3+\ln 4}$ **(6)** $\mathrm{e}^{\ln(6+7)}$ **(7)** $\ln \mathrm{e}^{6+7}$ **(8)** $\mathrm{e}^3(\mathrm{e}^4)^4$

(9) $\log_4 8$ **(10)** $\log_{10} 0.01$ **(11)** $\log_{0.5} 2$ **(12)** $\log_3(1/27)$

(13) $\log_6 6^{10}$ **(14)** $\ln 6+\ln(1/6)$ **(15)** $2\ln 3-\ln 9$ **(16)** $\ln 12-\ln 4+\ln 2$

(17) 1^{-a} **(18)** $\log_a 1$ **(19)** $\log_a(2a^2)^4$ **(20)** $\log_a(\ln(\mathrm{e}^{5a}))$

Problems

4.3 Some of the following expressions are not defined. Find all the undefined ones.

π^0	π^{-3}	$\pi^{2.5}$	$\pi^{-2.5}$	π^π	$\pi^{-\pi}$	$\pi^{-1/\pi}$
0^0	0^{-3}	$0^{2.5}$	$0^{-2.5}$	0^π	$0^{-\pi}$	$0^{-1/\pi}$
$(-2)^0$	$(-2)^{-3}$	$(-2)^{2.5}$	$(-2)^{-2.5}$	$(-2)^\pi$	$-2^{-\pi}$	$-2^{-1/\pi}$
$\sqrt{4.5}$	$\sqrt{-4}$	$\sqrt[3]{3.5}$	$\sqrt[3.5]{3}$	$\sqrt[-4]{3}$	$\sqrt[4]{-3}$	$\sqrt[4]{1+\sqrt{2}}$
$\sqrt[1.5]{1}$	$\sqrt[4]{-8}$	$\sqrt[-3]{8}$	$\sqrt[0]{8}$	$\sqrt[3]{\pi}$	$\sqrt[\pi]{3}$	$\sqrt[4]{1-\sqrt{2}}$

★**4.4** Solve these equations.

(1) $2^{x+1} = 64$ **(2)** $e^{3x-1} = 4$ **(3)** $e^x = 5$

(4) $\ln|x-1| = 0$ **(5)** $\ln x = 4$ **(6)** $\log_5(2x+3) = 2$

(7) $2^{-2x} = 7$ **(8)** $\log_3(2x+1) = 3$ **(9)** $e^{2x} - 3e^x + 2 = 0$

*****4.5** Draw the graphs of the following functions.

(1) $y = 2^x$ **(2)** $y = 0.5^x$ **(3)** $y = \left(\sqrt[3]{2}\right)^x$ **(4)** $y = 2 \cdot 2^x$ **(5)** $y = \dfrac{2^x}{2}$

Fail safe note: Draw $y = 2^x$ first. Rewrite the other functions in the form of $y = 2^{\square}$.

****4.6** Solve these equations.

(1) $\ln(x^2-3) = \ln(2x)$ **(2)** $\ln x + \ln(x-1) = \ln 6$

(3) $\ln|x^2-3| = \ln|2x|$ **(4)** $\ln|x| + \ln|x-1| = \ln 6$

(5) $\log_2 x + \log_2(x+2) = 3$ **(6)** $\ln(x+1) + \ln(x-1) = \ln 3$

(7) $\log_2 x - \log_2(x-2) = 3$ **(8)** $\ln(x+1) - \ln(x-1) = \ln 4$

(9) $\log_2|x| + \log_2|x+2| = 3$ **(10)** $\ln|x+1| + \ln|x-1| = \ln 4$

****4.7** **Hyperbolic functions** are defined by

$$\cosh x := \frac{e^x + e^{-x}}{2}, \quad \sinh x := \frac{e^x - e^{-x}}{2}, \quad \tanh x := \frac{\sinh x}{\cosh x}, \quad \text{etc.} \tag{4.10}$$

(1) Calculate the derivative of $\cosh x$, $\sinh x$, and $\tanh x$.

(2) Draw the graphs of $\cosh x$, $\sinh x$, and $\tanh x$.

(3) Show the following properties:

(a) $\cosh^2 x - \sinh^2 x = 1$. **(b)** $\cosh x \geq 1$ for all x.

(c) $\cosh x$ is an even function. **(d)** $\sinh x$ and $\tanh x$ are odd functions.

****4.8** Calculate the derivative of $\cosh 2x^2$ and $3x \tanh x^2$.

****4.9** Calculate the derivative of x^x.

***4.10** Calculate the derivative of x^{x^x} and $(x^x)^x$.

4.4 Exponential Growth and Decay

Consider $f_0(t) = \exp(At)$ with $A \in \mathbb{R}$, and imagine $t \in \mathbb{R}$ is the time. You may find

$$f_0(t) = \exp(At) \text{ is **strictly increasing** if } A > 0, \text{ while **strictly decreasing** if } A < 0. \tag{4.11}$$

Therefore, $f_0(t) = \exp(At)$ is called exponential growth if $A > 0$ and exponential decay if $A < 0$.

Quiz

Q12. Consider the above function $f_0(t) = \exp(At)$ with $A, t \in \mathbb{R}$.

(1) What does "strictly increasing" mean?

(2) Calculate $f_0'(t)$ and confirm the two statements in Eq. (4.11).

(3) Show that $f_0(0) = 1$ and $f_0'(t) = Af_0(t)$.

Notice $e^{At} = 1$ at $t = 0$. If the number at $t = 0$ is N (we assume $N > 0$), we should use Ne^{At}:

Statement 4.13 (Exponential growth and decay)

Consider the function $f(t) = N\exp(At)$ with $N > 0$ and $A, t \in \mathbb{R}$.

- If $A > 0$, it is strictly increasing; we call $f(t)$ **exponential growth**.
- If $A < 0$, it is strictly decreasing; we call $f(t)$ **exponential decay**.

The function $f(t)$ is characterized by

$$f(0) = N, \quad f'(t) = Af(t). \tag{4.12}$$

(We will come back to Eq. (4.12) in ♣TODO♣ode-section.)

Exponential decay is usually expressed by using a positive number $\Gamma > 0$, called the **decay rate**:

$$f(t) = N\exp(-\Gamma t) = N\exp\left(-\frac{t}{\tau}\right) \tag{4.13}$$

with $\tau := 1/\Gamma$. Since τ has the same *dimension*[#4] as the time t, we call τ the **lifetime**. We also define the **half-life** $T_{1/2}$ so that, at the time $T_{1/2}$, the number becomes half ($N/2$) of the number N at $t = 0$.

Quiz

Q13. **(1)** Show that $f'(t) = -\Gamma f(t)$.

(2) Show that $T_{1/2} = \tau \ln 2$. Namely, show $f(t) = N/2$ at the time $t = \tau \ln 2$.

(3) Assume $f(0) = 40$. Find $f(T_{1/2})$, $f(2T_{1/2})$, and $f(3T_{1/2})$.

(4) Assume $f(t_0) = 40$. Find the time t at which $f(t) = 20$ and $f(t) = 10$.

(5) Show that $f(t + T_{1/2}) = f(t)/2$ for any t.

For example, if $f(1) = 12$, then $f(1 + T_{1/2}) = 6$, and $f(1 + 2T_{1/2}) = 3$. Each time one half-life passes, the quantity is reduced by half.

[#4] ⟶ Chapter 2

Problems

***4.11 An RC circuit has charge $Q(t) = Q_0 e^{-t/\tau}$, where $\tau = RC$.

(1) Verify that $Q(\tau) = Q_0/\mathrm{e}$. Then, find the half-life.

(2) Assume $R = 2.0\,\mathrm{k\Omega}$ and $C = 50\,\mu\mathrm{F}$. Calculate τ. Then, find the time t at which Q dropped to 25% of Q_0.

Chapter 5

Vectors as arrows

Physicists understand a vector in three ways:

1. an arrow in n-dimensional space ($n \in \mathbb{N}^+$, but usually $n = 3$),
2. a list of n numbers arranged vertically ($n \in \mathbb{N}^+$), and
3. an element of a vector space (such as a Hilbert space).

In this chapter, we will focus on the first two interpretations. The last interpretation, more abstract and mathematical, will be discussed in ♣TODO♣chap:linear-algebra.

Remark: This document discusses vectors mainly in terms of mathematics; more detailed *physical* discussion about vectors can be found in Sho's Vector Boot Camp.[#5]

Be careful: In university, we **never** use the horizontal notation $\vec{v} = (x, y, z)$ to describe vectors. Please **always** use the vertical form $\vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ to be prepared for Chapter 7.

Remark: We use the arrow notation $\vec{v}$ because this is a first-year lecture. Usually, physicists use boldface ($\boldsymbol{a}, \boldsymbol{x}, \boldsymbol{\beta}$, etc.) to denote vectors.

Advanced note: The discussion in this chapter is only valid for *finite-dimensional* vectors because here we will define vectors as arrows in a "space". As the "space", readers are expected to imagine *the 3d space* of our Universe, a 2d-sheet of paper in our Universe, or something like those, and then the **dimension** of the space (defined in Theorem 5.16) will be limited to a finite integer.

5.1 What is a Vector?

You have seen vectors, such as $(5, -1)$, $(1, 0, 2)$, or $(-1, 3)$, or in the vertical form $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, etc. However, it represents a *mathematical* nature of vectors. As a **physicist**, you have to forget about such expressions in this section (we will return to it in Section 5.7). Instead, we will first *define* vectors in a more physical way.

Definition 5.1 (Vector for physicists)

Consider a space. A **vector** is a straight arrow drawn in the space, or in general, a quantity that has both **magnitude** (= length) and **direction** in the space.

We here do not investigate what "the space" is, but you may well imagine a lecture room as the space.

[#5] Visit `https://misho104.github.io/LecturePublic` and find `gp2_boot2_vector.pdf`.

Notation 5.2 (Vectors and Scalars)

We denote vectors with an arrow over the symbol, such as $\vec{v}$, $\vec{F}$, $\vec{a}$, $\vec{p}$. Their magnitudes are written as $|\vec{v}|$, $|\vec{F}|$, $|\vec{a}|$, $|\vec{p}|$, respectively. Meanwhile, a quantity that has no direction is called a **scalar**. Symbols without arrows, such as v, F, a, p, are scalars.

Be careful: v and $\vec{v}$ are **completely different objects**. They are totally unrelated, just as A and a are unrelated. However, in physics, we are sometimes *lazy* enough to write v to mean $|\vec{v}|$, the magnitude of $\vec{v}$. Please not be confused.

Quiz

Q1. Choose scalar quantities. Choose vector quantities.

(1) $\vec{a}$	**(2)** p	**(3)** $\lvert\vec{b}\rvert$	**(4)** $\lvert g\rvert$	**(5)** magnitude of $\vec{v}$
(6) mass	**(7)** velocity	**(8)** speed	**(9)** $\vec{x} + \lvert\vec{x}\rvert$	**(10)** direction of $\vec{v}$

Q2. **(1)** What do we call $|\vec{v}|$? Also, clearly write down its definition.

(2) What do we call $|v|$? Also, clearly write down its definition.

Fail safe note: For $x \in \mathbb{R}$, we define $|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$ and call it "the **absolute value** of x".

There is a special vector "**zero vector**", which has magnitude 0 and no direction.

Definition 5.3 (Zero vector)

There is a vector with magnitude 0 and no direction. We call it **the zero vector** and denote $\vec{0}$.

Any other vectors have a direction and **positive** magnitude. Namely,

$$|\vec{0}| = 0. \qquad |\vec{v}| = 0 \iff \vec{v} = \vec{0}. \qquad |\vec{v}| > 0 \iff \vec{v} \neq \vec{0}. \tag{5.1}$$

Quiz

Q3. Explain why $\vec{0} \neq 0$. Explain why $|\vec{0}| = 0$.

Advanced note: The uniqueness of $\vec{0}$ is easy to prove, once we clarify the definition of "=". Namely, "$\vec{a} = \vec{b}$" means "$\vec{a}$ and $\vec{b}$ have the same magnitude and direction". (Try to complete the proof.)

5.2 Addition and Scalar multiplication

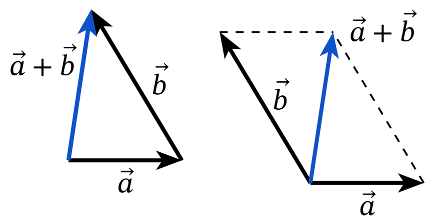
Like + and ÷ for numbers and ∧ and ∨ for true/false, we have two operations for vectors:

- **addition**: two vectors $\vec{a}$ and $\vec{b}$ can be added; we write the result by $\vec{a} + \vec{b}$.
- **scalar multiplication**: a vector $\vec{a}$ is multiplied by a scalar k; we write the result by $k\vec{a}$.

Definition 5.4 (Vector addition)

If $\vec{a}$ and $\vec{b}$ are vectors drawn in the same space, we can define $\vec{a}+\vec{b}$ by the vector obtained by placing the tail of $\vec{b}$ at the head of $\vec{a}$.

The triangle in the figure to the right represents this definition: the tail of $\vec{b}$ is put at the head of $\vec{a}$ to get the blue arrow $\vec{a}+\vec{b}$. You may also use the parallelogram method shown to the right: consider a parallelogram made by $\vec{a}$ and $\vec{b}$ sharing the same tail. Then its diagonal, the blue arrow, is $\vec{a}+\vec{b}$.



Notice that the two horizontal arrows in the figure are both $\vec{a}$, although their locations are different. Similarly, both blue arrows show the vector $\vec{a}+\vec{b}$. Vectors are the same if and only if they have the same magnitude and direction; location does not matter.

Definition 5.5 (Scalar multiplication)

For a vector $\vec{v}$ and a real number (scalar) k, we define $k\vec{v}$ as follows:

- if $k>0$, $k\vec{v}$ has the magnitude $|k||\vec{v}|$ and has the same direction as $\vec{v}$.
- if $k<0$, $k\vec{v}$ has the magnitude $|k||\vec{v}|$ and is anti-parallel to $\vec{v}$.
- if $k=0$, $k\vec{v}=\vec{0}$.

Remark: The word "**anti-parallel**" means "in the opposite direction". Meanwhile, **we should avoid** the ambiguous word "**parallel**" for vectors, as it may mean either the same direction or the opposite direction; use the phrase "has the same direction" instead. For more vocabulary to describe directions, please check the Vector Boot Camp.

Advanced note: Watch out we assume $k\in\mathbb{R}$ in Definition 5.5. It restricts the discussion in this chapter to *real vectors*. In Section 6.5, we will discuss *complex vectors* by modifying this definition.

We should carefully digest these definitions. Let's see an example, and try the next quiz.

Example 5.6

(1) Explain the meaning of $|k\vec{v}|$ and $|k||\vec{v}|$.

(2) Prove $|k\vec{v}|=|k||\vec{v}|$, where $\vec{v}$ is a vector and $k\in\mathbb{R}$.

(3) Based on the above definitions, explain what $-\vec{v}$ is.

Solution

(1) $|k\vec{v}|$ means the magnitude of a vector $k\vec{v}$, which is a scalar multiplication of $\vec{v}$ by k. Meanwhile, $|k||\vec{v}|$ means the absolute value of k times the magnitude of $\vec{v}$.

(2) If $k\neq 0$, then, according to the above definition, $k\vec{v}$ has the magnitude $|k||\vec{v}|$ and it means $|k\vec{v}|=|k||\vec{v}|$. If $k=0$, then $k\vec{v}=\vec{0}$ and thus LHS $=|k\vec{v}|=0$, while RHS $=0|\vec{v}|=0$. ∎

(3) $-\vec{v}$ is a shorthand notation of $(-1)\vec{v}$, the scalar multiplication of $\vec{v}$ by -1. So, $-\vec{v}$ has the same magnitude as $\vec{v}$ but is anti-parallel to $\vec{v}$.

Quiz

Q4. Find out how the following vectors are defined based on the above definitions of addition and scalar multiplication.

(1) $\vec{a}+\vec{a}$ **(2)** $2\vec{a}$ **(3)** $2\vec{a}+3\vec{b}$ **(4)** $-\vec{a}$ **(5)** $\vec{a}-\vec{b}$ **(6)** $-\vec{a}-2\vec{b}$

Addition and scalar multiplication have the following properties:

Theorem 5.7

For vectors $\vec{a}$ and $\vec{b}$ in the same space and $p, q \in \mathbb{R}$,

(A) $\vec{a}+\vec{b}=\vec{b}+\vec{a}$,

(B) $(\vec{a}+\vec{b})+\vec{c}=\vec{a}+(\vec{b}+\vec{c})$,

(C) $\vec{a}+\vec{0}=\vec{a}$,

(D) $\vec{a}+(-\vec{a})=\vec{0}$,

(E) $p\vec{a}+p\vec{b}=p(\vec{a}+\vec{b})$,

(F) $p\vec{a}+q\vec{a}=(p+q)\vec{a}$,

(G) $(pq)\vec{a}=p(q\vec{a})$,

(H) $1\vec{a}=\vec{a}$.

These properties seem obvious, but in fact, they play a fundamental role in ♣TODO♣....

Problems

*****5.1** Consider a vector $\vec{a} \neq \vec{0}$ and a constant $k \in \mathbb{R}$. Consider three vectors $\vec{a}$, $k\vec{a}$, and $k^2\vec{a}$.

(1) Which are in the same direction? Which are anti-parallel to each other?

(2) Compare their magnitudes; which are the longest and the shortest?

*****5.2** Consider a vector $\vec{a}$ and a constant k.

(1) Write a vector that has magnitude $|\vec{a}|$ and is anti-parallel to $\vec{a}$.

(2) Write a vector that has magnitude $3|\vec{a}|$ and is in the same direction as $\vec{a}$.

(3) Write a vector that has magnitude $3k|\vec{a}|$ and is in the same direction as $\vec{a}$.

Vectors with magnitude 1 are called **unit vectors**.

(4) Write a vector that has magnitude 1 and is in the same direction as $\vec{a}$. (Namely, write a unit vector that has the same direction as $\vec{a}$.)

(5) Write a unit vector that is anti-parallel to $\vec{a}$.

*****5.3** Prove the next "**triangle inequality**" geometrically, i.e., only with the above definitions.

$$\text{For any two vectors } \vec{a} \text{ and } \vec{b} \text{ in the same space, } |\vec{a}+\vec{b}| \leq |\vec{a}|+|\vec{b}|. \tag{5.2}$$

****5.4** Starting from Definition 5.4 and Definition 5.5, prove the properties in Theorem 5.7.

****5.5** Prove that, for any two vectors $\vec{a}$ and $\vec{b}$ in the same space, $\big||\vec{a}|-|\vec{b}|\big| \leq |\vec{a}-\vec{b}|$.

Most of the following discussion is only for *real vectors* and do not apply for *complex vectors*.

5.3 Inner Product

Imagine two arrows. Probably you can think the angle θ between the arrows. The angle leads you to the following **geometric** definition of the inner product.

Definition 5.8 (Inner product (geometrical definition))

For vectors $\vec{a}$ and $\vec{b}$ drawn in the same space, the **inner product** is defined by

$$\vec{a} \cdot \vec{b} \coloneqq |\vec{a}|\,|\vec{b}| \cos\theta, \tag{5.3}$$

where θ is the angle between $\vec{a}$ and $\vec{b}$; if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then $\vec{a} \cdot \vec{b} \coloneqq 0$.

Quiz

Q5. Assume the length of $\vec{a}$ is 3 and the length of $\vec{b}$ is 2.

(1) If $\vec{a} \perp \vec{b}$, then what is $\vec{a} \cdot \vec{b}$?

(2) If $\vec{a}$ and $\vec{b}$ are anti-parallel, what is $\vec{a} \cdot \vec{b}$?

(3) What is the maximum value of $\vec{a} \cdot \vec{b}$? When is it achieved?

(4) If $\vec{a} \cdot \vec{b} = -3$, what can you say about the angle between $\vec{a}$ and $\vec{b}$?

Fail safe note: $\vec{a} \perp \vec{b}$ means that $\vec{a}$ and $\vec{b}$ are **perpendicular** to each other, or **normal** to each other; in other words, the angle θ between them is $\pi/2 = 90°$.

This inner product has the following properties:

Theorem 5.9 (Real-vector inner product)

For vectors $\vec{a}$ and $\vec{b}$ drawn in the same space and a constant $k \in \mathbb{R}$,

(A) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$,

(B) $\vec{a} \cdot \vec{a} \geq 0$ for any vector $\vec{a}$,

(C) $\vec{a} \cdot \vec{a} > 0$ for any vector $\vec{a} \neq \vec{0}$,

(D) $\vec{a} \cdot \vec{a} = 0 \iff \vec{a} = \vec{0}$,

(E) $(\vec{a} + \vec{b}) \cdot \vec{c} = \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}$,

(F) $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$,

(G) $(k\vec{a}) \cdot \vec{b} = k(\vec{a} \cdot \vec{b})$,

(H) $\vec{a} \cdot (k\vec{b}) = k(\vec{a} \cdot \vec{b})$.

Be careful: These properties are valid only for “real vectors”.

We will skip their proof. Instead, we focus on these four very important properties. You need to memorize them securely.

Theorem 5.10 (Properties of real-vector inner product)

$$\text{For a vector } \vec{a}, \quad |\vec{a}| = \sqrt{\vec{a}\cdot\vec{a}}. \tag{5.4}$$

$$\text{For non-zero vectors } \vec{a} \text{ and } \vec{b}, \quad \vec{a}\cdot\vec{b} = 0 \iff \vec{a} \perp \vec{b}. \tag{5.5}$$

$$\text{For vectors } \vec{a} \text{ and } \vec{b}, \quad -|\vec{a}||\vec{b}| \le \vec{a}\cdot\vec{b} \le |\vec{a}||\vec{b}| \quad (\textbf{Cauchy-Schwarz inequality}). \tag{5.6}$$

$$\text{For vectors } \vec{a} \text{ and } \vec{b}, \quad |\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + |\vec{b}|^2 \tag{5.7}$$

Be careful: These properties are valid only for “real vectors”.

Quiz

Q6. Prove the following theorems directly from Definition 5.8.

(1) $\vec{a} \perp \vec{b} \Longrightarrow \vec{a}\cdot\vec{b} = 0$.

(2) $\vec{a}\cdot\vec{b} = 0 \Longrightarrow (\vec{a} \perp \vec{b}) \vee (\vec{a} = \vec{0}) \vee (\vec{b} = \vec{0})$.

(3) $-|\vec{a}|\,|\vec{b}| \le \vec{a}\cdot\vec{b} \le |\vec{a}|\,|\vec{b}|$.

(4) $|\vec{a}\cdot\vec{b}| \le |\vec{a}|\,|\vec{b}|$. [Hint: Recall that $|x| \le 3$ means $-3 \le x \le 3$.]

(5) $|\vec{a}|^2 = \vec{a}\cdot\vec{a}$.

(6) $|\vec{a}| = \sqrt{\vec{a}\cdot\vec{a}}$. [Hint: Most students make mistakes in this question.]

Q7. Prove the following equation, using Definition 5.8, Theorem 5.9, and the equations in the previous quiz.

(1) $(\vec{a}+\vec{b})\cdot\vec{a} = |\vec{a}|^2 + \vec{a}\cdot\vec{b}$. **(2)** $|\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + |\vec{b}|^2$.

Problems

***5.6 Two vectors $\vec{a}$ and $\vec{b}$ satisfy $|\vec{a}| = 2$, $|\vec{b}| = 3$, and $\vec{a}\cdot\vec{b} = 3$. Let x, y be real numbers.

(1) Find the angle between $\vec{a}$ and $\vec{b}$.

(2) Calculate $|\vec{a}+\vec{b}|$.

(3) Calculate the magnitude of $4\vec{a}+3\vec{b}$ and $x\vec{a}+y\vec{b}$.

(4) Explain why $|x\vec{a}+y\vec{b}|^2$ is **not** in general equal to x^2+y^2.

Now let $\vec{e}_1$ and $\vec{e}_2$ satisfy $|\vec{e}_1| = |\vec{e}_2| = 1$ and $\vec{e}_1\cdot\vec{e}_2 = 0$. Let p, q be real numbers.

(5) Find the angle between $\vec{e}_1$ and $\vec{e}_2$.

(6) Explain why $(p\vec{e}_1 + q\vec{e}_2)\cdot(x\vec{e}_1 + y\vec{e}_2) = px + qy$.

(7) Explain why $|x\vec{e}_1 + y\vec{e}_2| = \sqrt{x^2+y^2}$.

**5.7 Three vectors $\vec{A}, \vec{B}, \vec{C}$ satisfy $|\vec{A}| = 2$, $|\vec{B}| = 3$, $\vec{A}\cdot\vec{B} = 3\sqrt{2}$, and $\vec{A}\cdot\vec{C} = -1$.

(1) Find the angle between $\vec{A}$ and $\vec{B}$.

(2) Calculate $|\vec{A}+\vec{B}|^2$, $|\vec{A}-\vec{B}|^2$, and $|2\vec{A}+4\vec{B}|^2$.

(3) Calculate $(\vec{A}-2\vec{B})\cdot(2\vec{A}+\vec{B}+\vec{C}) + 2\vec{B}\cdot\vec{C}$.

(4) Find k such that $|\vec{A}+k\vec{B}| = \sqrt{10}$.

(5) Find c such that $\vec{B}+c\vec{C}$ is perpendicular to $\vec{A}$.

5.4 Cross Product (only for 3d real-vectors)

For two arrows drawn in three-dimensional space, we can define the **cross product**.

Definition 5.11 (Cross product)

For $\vec{a}$ and $\vec{b}$ drawn in a *three-dimensional* space, the **cross product** $\vec{a}\times\vec{b}$ is defined as follows:

- It is a vector with magnitude $|\vec{a}\times\vec{b}| = |\vec{a}|\,|\vec{b}|\sin\theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$.
- If $|\vec{a}\times\vec{b}| \neq 0$, we determine its direction so that $\left(\vec{a}\times\vec{b}\right)\perp\vec{a}$ and $\left(\vec{a}\times\vec{b}\right)\perp\vec{b}$.

Here, we always have two possible directions. We impose another condition to make it unique:

- If you hold the *right* hand so that your thumb points in the direction of $\vec{a}$ and your index finger in the direction of $\vec{b}$, your middle finger points in the direction of $\vec{a}\times\vec{b}$.
  (This is often quoted that "the ordered triple $(\vec{a}, \vec{b}, \vec{a}\times\vec{b})$ obeys the **right-hand rule**".)

We will not discuss much about the cross product, but only the following properties:

Theorem 5.12

For vectors $\vec{a}, \vec{b}, \vec{c}$ and real number k,

(A) $\vec{a}\times\vec{a} = \vec{0}$.

(B) $\vec{a}\times\vec{b} = -\vec{b}\times\vec{a}$.

(C) $0 \le |\vec{a}\times\vec{b}| \le |\vec{a}|\,|\vec{b}|$.

(D) $(k\vec{a})\times\vec{b} = \vec{a}\times(k\vec{b}) = k(\vec{a}\times\vec{b})$.

(E) $\vec{a}\times(\vec{b}+\vec{c}) = \vec{a}\times\vec{b}+\vec{a}\times\vec{c}$.

(F) $\vec{a}\times\vec{b} = \vec{0}$ if $\vec{a}$ and $\vec{b}$ are parallel or anti-parallel.

(G) $\vec{a}\cdot\left(\vec{a}\times\vec{b}\right) = \vec{b}\cdot\left(\vec{a}\times\vec{b}\right) = 0$.

(H) $\vec{a}\cdot(\vec{b}\times\vec{c}) = \vec{b}\cdot(\vec{c}\times\vec{a}) = \vec{c}\cdot(\vec{a}\times\vec{b})$.

The equation **(B)** is the most important. For **(A)**, notice $\vec{a}\times\vec{a}$ is not "zero".

Advanced note: Geometrical interpretation of the cross product is sometimes useful:

- $|\vec{a}\times\vec{b}|$ is the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$.
- $(\vec{a}\times\vec{b})\cdot\vec{c}$ is the (signed) volume of the parallelepiped formed by $\vec{a}$, $\vec{b}$, and $\vec{c}$.

Problems

★5.8 Prove **(A)**, **(B)**, **(C)**, **(F)**, and **(G)** of Theorem 5.12.

****5.9** In physics, we **always** use the **right-handed Cartesian coordinate system** to describe our three-dimensional space, which is characterized by three unit vectors $\vec{e}_x$, $\vec{e}_y$, $\vec{e}_z$ defined so that they are perpendicular to each other and $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ obeys the right-hand rule. Calculate their inner products and cross products, such as $\vec{e}_x \cdot \vec{e}_y$ and $\vec{e}_x \times \vec{e}_z$

***5.10** Prove **(B)** of Theorem 5.9 and **(E)** of Theorem 5.12 geometrically (i.e., based on the definitions given in this chapter). Sho has his own proof but not very confident. Can you find a better proof?

5.5 Position Vectors

We have defined vectors as arrows. Arrows are not positions, so vectors are not positions. However, *once we define the origin* O in the space, we can use vectors to represent each position in the space.

Definition 5.13 (Position vector)

Consider a space and fix a point O as the origin. To each point P, we associate the vector $\overrightarrow{\mathrm{OP}}$ and call it the **position vector** of P. We often write $\vec{p} = \overrightarrow{\mathrm{OP}}$, $\vec{q} = \overrightarrow{\mathrm{OQ}}$, and so on.

Consider $\vec{p} = \overrightarrow{\mathrm{OP}}$, $\vec{q} = \overrightarrow{\mathrm{OQ}}$, and $\vec{r} = \overrightarrow{\mathrm{OR}}$. They are obviously dependent on the choice of the origin O. Also, the point described by $\vec{p} + \vec{q}$ will be different if we chose a different point as the origin. However, we can check that

- the vector $\overrightarrow{\mathrm{PQ}}$ is given by $\vec{q} - \vec{p}$
- the length of the segment PQ is given by $|\vec{q} - \vec{p}|$.
- the middle point M of the segment PQ has the position vector $\vec{m} = (\vec{p} + \vec{q})/2$.

These properties are independent of the choice of O. Namely, whatever choice we did for origin, $(\vec{p} + \vec{q})/2$ represents the midpoint of PQ. For further discussion, please check the Vector Boot Camp.

5.6 Intermission: Vector or Scalar or Not

Let us summarize the operations on vectors. With k a real number and $\vec{a}, \vec{b}$ vectors:

magnitude	$\lvert\vec{a}\rvert$	$\to$ scalar
scalar multiplication	$k\vec{a}$	$\to$ vector
addition	$\vec{a}+\vec{b}$	$\to$ vector
inner product	$\vec{a}\cdot\vec{b}$	$\to$ scalar
cross product	$\vec{a}\times\vec{b}$	$\to$ vector (only in 3d)

Then, how about them? Try to ensure that you understand the meaning of each operation.

Problems

5.11 For each expression, answer **V** if it is a vector, **S** if scalar, and **N** if invalid (not defined). Here, $\vec{a}, \vec{b}, \ldots$ are three-dimensional vectors and $a, b, \ldots$ are scalars (real numbers).

(1) $\vec{a}$ (2) $\vec{a}^2$ (3) $(\vec{a})^2$ (4) $|\vec{a}|^2$ (5) $|a|^2$
(6) 0 (7) $\vec{0}+0$ (8) $0+\vec{0}$ (9) $0\cdot\vec{0}$ (10) $0\,\vec{0}$
(11) $\dfrac{1}{\vec{a}}$ (12) $\dfrac{\vec{a}}{\vec{a}}$ (13) $\dfrac{\vec{a}}{\vec{b}}$ (14) $\dfrac{1}{|\vec{a}|}$ (15) $\dfrac{1}{|\vec{a}|^2}$
(16) $\dfrac{1}{(\vec{a})^2}$ (17) $\dfrac{\vec{a}}{(\vec{a}\cdot\vec{b})}$ (18) $\dfrac{\vec{a}}{(\vec{a}\cdot\vec{b})^2}$ (19) $\dfrac{\vec{a}}{|\vec{a}-\vec{b}|}$ (20) $\dfrac{\vec{a}}{(\vec{a}\cdot\vec{b})}$
(21) $\sqrt{\vec{a}^2}$ (22) $\sqrt{\vec{a}}$ (23) $\sqrt{|\vec{a}|}$ (24) $\sqrt{|a|}$ (25) $\sqrt{a^2}$
(26) $a+1$ (27) $a+\vec{a}$ (28) $\vec{a}\times\vec{a}$ (29) $a\cdot\vec{a}$ (30) $a\vec{a}$
(31) $\vec{a}+1$ (32) $\vec{a}+\vec{a}$ (33) $a\times\vec{a}$ (34) $\vec{a}\cdot\vec{a}$ (35) $\vec{a}\vec{a}$
(36) $\vec{a}^{-2}$ (37) $|\vec{a}|^{-2}$ (38) $|\vec{a}|^{-2}$ (39) $|\vec{a}|^{-1}\,\vec{a}$ (40) $|\vec{a}|^{-1/2}\,\vec{a}$
(41) $\vec{p}\cdot\vec{q}+\vec{p}\times\vec{q}$ (42) $\vec{p}\times(\vec{q}\times\vec{r})$ (43) $\vec{p}\cdot(\vec{q}\times\vec{r})$ (44) $p(\vec{q}\times\vec{r})$
(45) $(\vec{p}\times\vec{q})^2$ (46) $\dfrac{1}{|\vec{x}-\vec{r}|^{3/2}}$ (47) $\dfrac{\vec{x}-\vec{r}}{|\vec{x}-\vec{r}|^{3/2}}$ (48) $\dfrac{\vec{a}\times(\vec{x}-\vec{r})}{|\vec{x}-\vec{r}|^{3/2}}$
(49) $\dfrac{d\vec{y}}{dx}$ (50) $\dfrac{dy}{d\vec{x}}$

5.7 Axes and Components

So far, we have considered vectors as arrows drawn in a space. Vectors are characterized only by the magnitude and direction. Now, we are going to *recall* the component-wise notation such as $\vec{a}=\binom{1}{2}$. We define "orthonormal basis vectors" and then see we can *describe an arrow by a list of numbers*.

Definition 5.14 (Orthonormal basis vectors)

If n vectors $\vec{e}_1, \ldots, \vec{e}_n$ satisfy the following properties, we call them **an orthonormal basis**:

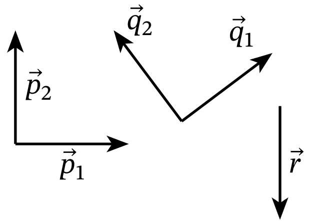
- All of them are unit vectors, i.e., $|\vec{e}_i| = 1$ for all i.
- Any of them are perpendicular, i.e., $i \neq j \implies \vec{e}_i \cdot \vec{e}_j = 0$.
- We cannot add any more vectors without violating the above two rules.

Precisely speaking, we call the set $\{\vec{e}_1, \ldots, \vec{e}_n\}$ "*an* orthonormal basis". Then, if we fix *one* orthonormal basis to use, we call its members "orthonormal basis vectors".

Fail safe note: A woman, two women. A nucleus, two nuclei. A basis, two bases. A matrix, two matrices.

Example 5.15 (Basis vectors for arrows on this sheet)

Let's limit our "space" to this 2d sheet and consider the five vectors drawn to the right. Assume their length are all "1".



If we choose $\vec{p}_1$ and $\vec{p}_2$, they form *an* orthonormal basis $\{\vec{p}_1, \vec{p}_2\}$ because they are unit vectors, perpendicular to each other, and we cannot add any more. Similarly, $\{\vec{q}_1, \vec{q}_2\}$ is *another* orthonormal basis.

Meanwhile, $\{\vec{p}_1, \vec{q}_2\}$ and $\{\vec{p}_1, \vec{p}_2, \vec{q}_1, \vec{q}_2\}$ are not orthonormal bases because the members are not orthogonal. $\{\vec{p}_1\}$ is not an orthonormal basis because we can add one more vector, such as $\vec{r}$, without breaking the conditions; after adding $\vec{r}$, we have an orthonormal basis $\{\vec{p}_1, \vec{r}\}$.

Quiz

Q8. Consider the figure in the above example. Which sets are orthogonal bases?

$$\{\vec{p}_1, -\vec{p}_2\}, \quad \{\vec{p}_1, \vec{r}\}, \quad \{\vec{p}_2, \vec{r}\}, \quad \{\vec{p}_1, \vec{p}_2, \vec{r}\}, \quad \{\vec{q}_1, 2\vec{q}_2\}, \quad \{-\vec{q}_1, -\vec{q}_2\}.$$

This example shows we can find many orthogonal bases, but **the number** of the members is always two. This number is called the **dimension** of the space, and this is why we call "this *2d* sheet" in the above example.

In general, a space has *infinitely many* orthonormal bases and we can choose *an* orthonormal basis at our convenience; we will come back to this point in Section 7.2. However, the number of the orthonormal basis vectors, the **dimension**, is fixed by the space we considered.

Theorem 5.16 (Properties of orthonormal basis vectors)

- Any orthonormal bases of a space have the same number of vectors. We call the number **the dimension** of the space.
- If $\{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n\}$ is an orthonormal basis, any arrow $\vec{v}$ in the space can be expressed as

$$\vec{v} = c_1\vec{e}_1 + c_2\vec{e}_2 + \dots + c_n\vec{e}_n = \sum_{k=1}^{n} c_k\vec{e}_k, \quad \text{where} \quad c_k \in \mathbb{R} \tag{5.8}$$

and this expression is *unique*, i.e., if $\vec{v}$ is expressed by

$$\begin{aligned} \vec{v} &= c_1\vec{e}_1 + c_2\vec{e}_2 + \dots + c_n\vec{e}_n \\ &= d_1\vec{e}_1 + d_2\vec{e}_2 + \dots + d_n\vec{e}_n, \end{aligned}$$

then all the coefficients are equal: $c_k = d_k$.

Quiz

Q9. Show that the numbers c_k in Eq. (5.8) are actually determined by $c_k = \vec{e}_k \cdot \vec{v}$.

Advanced note: This theorem seems not difficult to prove because we only think finite-dimensional spaces, but there could be caveats that Sho did not notice. A more rigorous construction is in ♣TODO♣abs-vec.

If you choose an orthonormal basis, then it automatically defines the **axes** of the space:

- For a 2d space, we call the directions of the basis vectors as x-axis and y-axis, respectively. Sho usually writes the basis vectors by $\vec{e}_x$ and $\vec{e}_y$, but other textbooks may write as $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$.
- For a 3d space, we call the directions of the basis vectors as x-axis, y-axis, and z-axis, respectively. The basis vectors are expressed by $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ or $(\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}})$. Here, physicists **always** choose the axes so that $\vec{e}_x, \vec{e}_y, \vec{e}_z)$ obeys the **right-hand rule** (see Section 5.4).

They are called **2d Cartesian coordinate system** and **3d right-handed Cartesian coordinate system**, respectively. In general, a coordinate system defined by an orthonormal basis is called Cartesian coordinate system.

Now we are ready to express vectors in their **components** because we have reached Eq. (5.8):

Definition 5.17 (Components of a vector)

If we fix an orthonormal basis and label the basis vectors by $\vec{e}_1, \dots, \vec{e}_n$, then Theorem 5.16 says any vector $\vec{v}$ can be written as $\vec{v} = c_1\vec{e}_1 + \dots + c_n\vec{e}_n$ with uniquely determined $c_k \coloneqq \vec{e}_k \cdot \vec{v} \in \mathbb{R}$. We call c_k **the k-th component** of $\vec{v}$ and express $\vec{v}$ with the components as

$$\vec{v} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} \vec{e}_1 \cdot \vec{v} \\ \vec{e}_2 \cdot \vec{v} \\ \vdots \\ \vec{e}_n \cdot \vec{v} \end{pmatrix}. \tag{5.9}$$

Vectors in n-dimensional spaces are called ***n*-dimensional vectors**. Since an orthonormal basis in a n-dimensional space has n basis vectors, a n-dimensional vectors are expressed with n real numbers $c_1, \dots, c_n$.

Advanced note: We have to make sure this representation is *well-defined*; we do not want to have two different expressions for one vector, or two different vectors having the same expressions.

Because c_k is uniquely determined, the component-wise notation is unique for a vector. Meanwhile, if $\vec{a}$ and $\vec{b}$ are different but have the same component-wise notation, it means $\vec{a} - \vec{b} =: \vec{\Delta}$ has the same notation as $\vec{0}$ (why?). It means $\vec{\Delta}/|\vec{\Delta}|$ is a unit vector orthogonal to all of $\vec{e}_k$, which contradicts that $\{\vec{e}_1, \dots, \vec{e}_n\}$ is the basis (why?). Accordingly, different vectors must have different component-wise notation.

Example 5.18

Prove the following equations in a 2d space.

(1) $\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} a+p \\ b+q \end{pmatrix}$ **(2)** $\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} p \\ q \end{pmatrix} = ap + bq$ **(3)** $\left|\begin{pmatrix} a \\ b \end{pmatrix}\right| = \sqrt{a^2 + b^2}$

Solution

(1) Since $\begin{pmatrix} a \\ b \end{pmatrix}$ means $a\vec{e}_x + b\vec{e}_y$ and $\begin{pmatrix} p \\ q \end{pmatrix}$ means $p\vec{e}_x + q\vec{e}_y$ under some orthonormal basis $(\vec{e}_x, \vec{e}_y)$,

$$\text{LHS} = \left(a\vec{e}_x + b\vec{e}_y\right) + \left(p\vec{e}_x + q\vec{e}_y\right) = (a+p)\vec{e}_x + (b+q)\vec{e}_y = \text{RHS},$$

where we used the equations in Theorem 5.7. ∎

(2) Similarly, using the equations in Theorem 5.9 and Definition 5.14,

$$\begin{aligned}\text{LHS} &= \left(a\vec{e}_x + b\vec{e}_y\right) \cdot \left(p\vec{e}_x + q\vec{e}_y\right) \\ &= ap(\vec{e}_x \cdot \vec{e}_x) + bp(\vec{e}_y \cdot \vec{e}_x) + aq(\vec{e}_x \cdot \vec{e}_y) + bq(\vec{e}_y \cdot \vec{e}_y) = ap + bq. \blacksquare\end{aligned}$$

(3) Thanks to **(2)**, $(\text{LHS})^2 = \begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = a^2 + b^2$. Since $(\text{LHS}) \geq 0$, $\text{LHS} = \sqrt{a^2 + b^2} = \text{RHS}$. ∎

Fail safe note: In **(3)**, you must write $\boxed{\text{LHS} \geq 0}$. Without it, you can only claim $\text{LHS} = \pm\sqrt{a^2 + b^2}$.

Notice that we *proved* these equations based on Theorem 5.7 etc. These equations are not *definitions* or *assumptions*, but *derived statements*[#6] with proofs. There are a few more statements to be proved:

Quiz

Q10. Consider a 2d space.

(1) Prove $\vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\vec{e}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and $\vec{e}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. [Hint: Recall $c_k = \vec{e}_k \cdot \vec{v}$.]

(2) Prove $k\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$, where $k \in \mathbb{R}$.

Problems

******5.12** Consider a 3d space. An orthonormal basis $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ is taken according to the right-hand rule. Let $\vec{a} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ and $\vec{p} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$.

(1) Prove $\vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, $\vec{e}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\vec{e}_y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\vec{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

(2) Prove $k\vec{a} + l\vec{b} = \begin{pmatrix} ka + lp \\ kb + lq \\ kc + lr \end{pmatrix}$.

(3) Prove $\vec{a} \cdot \vec{p} = ap + bq + cr$ and $|\vec{a}| = \sqrt{a^2 + b^2 + c^2}$.

(4) Express $\vec{a} \times \vec{p}$ with using a, b, c, p, q, r.

[#6] → Section 3.5

**5.13 Consider a n-dimensional space ($n \in \mathbb{N}^+$) and an orthonormal basis $(\vec{e}_1, \dots, \vec{e}_n)$ of it. Prove the following.

- $\vec{0} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$ and $(i\text{-th component of } \vec{e}_j) = \begin{cases} 1 \text{ if } i = j, \\ 0 \text{ if } i \neq j \end{cases}.$
- Consider $p, q \in \mathbb{R}$ and n-dimensional vectors $\vec{a}$ and $\vec{b}$. Let the k-th component of $\vec{a}$ be a_k and the k-th component of $\vec{b}$ be b_k. Then, the k-th component of $p\vec{a} + q\vec{b}$ is equal to $pa_k + qb_k$. Also, $\vec{a} \cdot \vec{b} = \sum_{k=1}^{n} a_k b_k$ and $|\vec{a}| = \sqrt{\sum_{k=1}^{n} a_k^2}$.

Let us summarize the above discussion.

Theorem 5.19 (Component-wise interpretation of real-vector arithmetic)

Consider n-dimensional real vectors, $\vec{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$, and $k \in \mathbb{R}$. Then, addition, scalar multiplication by $k \in \mathbb{R}$, and inner product are given by, respectively,

$$\vec{a} + \vec{b} = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix}, \qquad k\vec{a} = \begin{pmatrix} ka_1 \\ \vdots \\ ka_n \end{pmatrix}, \tag{5.10}$$

$$\vec{a} \cdot \vec{b} = \sum_{k=1}^{n} a_k b_k = a_1 b_1 + a_2 b_2 + \dots + a_n b_n, \tag{5.11}$$

Be careful: Again, these formulas are *consequences* of the definitions we gave earlier. They are **not** new definitions.

Since you must be familiar with this component-wise notation and calculations based on it, we do not discuss it further. Consult your high-school math textbook or Sho's Vector Boot Camp if you are not confident in such calculations.

Vectors are closely related to matrices, the main topic of Chapter 7. We will there discuss both real and complex matrices, and thus complex vectors will be introduced there. Since complex vectors are not arrows, we need to *define* complex vectors in a way different from what we did in this chapter.

To prepare for the discussion on complex matrices in Chapter 7, we first review **complex numbers** in the next chapter (Chapter 6).

Problems

*5.14 Complete the proof in "Advanced Note" after Definition 5.17.

*5.15 When we define a Cartesian coordinate system for a 3d space, we require that $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ satisfies the right-hand rule. However, similar rules are not required for 2d spaces. Why?

Chapter 6

Complex Numbers

You have learned complex numbers in high school but it is worth reviewing them in a university style. As you learned in Section 3.5, try to distinguish between definitions and derived consequences.

> **Remark:** Geometrical discussions, such as **complex planes**, are not included in this document, mainly because you are expected to be familiar with it.

6.1 Complex Numbers

Definition 6.1 (Complex numbers)

A **complex number** is given by the form

$$z = a + b\mathrm{i}, \quad \text{where } a, b \in \mathbb{R} \quad \text{and} \quad \mathrm{i} \text{ is a symbol satisfying } \mathrm{i}^2 = -1. \tag{6.1}$$

We call the newly-introduced symbol i the **imaginary unit**. Also, a is called the **real part** of z and b is called the **imaginary part** of z, written by

$$\mathrm{Re}(z) \coloneqq a, \quad \mathrm{Im}(z) \coloneqq b.$$

The set of all complex numbers is written by $\mathbb{C}$.

> **Be careful:** $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ are both real. If $z = 3 + 5\mathrm{i}$, then $\mathrm{Im}(z) = 5$, not $5\mathrm{i}$.

> **Remark:** We use i to denote the imaginary unit. You may see other notations such as i, j, j, or $\sqrt{-1}$. However, Sho recommends you **not** to write $\sqrt{-1}$ since it may make you confused. Recall, in Chapter 4, we *avoided* definitions of $\sqrt{x}$ for $x < 0$.

> **Advanced note:** Notice that i is just a symbol and it has only one property $\mathrm{i} \cdot \mathrm{i} = -1$. You may well ask "$-\mathrm{i}$ also satisfies $(-\mathrm{i})^2 = -1$; which should we choose as i?" but it sounds a weird question since "$-\mathrm{i}$" was not *a priori*: it was not there before we introduced i, so we can discuss $-\mathrm{i}$ only after we fix i.

Hereafter, when we write $\boxed{z = a + b\mathrm{i} \in \mathbb{C}}$, we implicitly assume a and b are real numbers.

Definition 6.2

Consider $z = a + b\mathrm{i} \in \mathbb{C}$.

- z is a real number if $\mathrm{Im}\, z = b = 0$. Namely, if $b = 0$, then $z = a + 0\mathrm{i} = a$.
- z is called **pure-imaginary** if $\mathrm{Re}\, z = a = 0$ but $z \neq 0$, namely, if $z = b\mathrm{i}$ with $b \neq 0$.

Also, for another complex number $c + d\mathrm{i}$, we define $\quad a + b\mathrm{i} = c + d\mathrm{i} \iff a = c \land b = d.$

It is debatable if 0 is pure-imaginary or not, so you need to be careful when you discuss 0 (but usually 0 is not included in pure-imaginary numbers.) Some people write the set of all pure-imaginary number by $\boxed{\mathrm{i}\mathbb{R}}$, but then you need to pay attention to this issue.

For complex numbers $z_1 = a_1 + b_1\mathrm{i}$ and $z_2 = a_2 + b_2\mathrm{i}$, addition and multiplication are defined by

$$z_1 + z_2 = (a_1 + a_2) + (b_1 + b_2)\mathrm{i}, \qquad \begin{aligned} z_1 z_2 &= (a_1 + b_1\mathrm{i})(a_2 + b_2\mathrm{i}) \\ &= \left(a_1a_2 + b_1b_2\mathrm{i}^2\right) + (a_1b_2 + a_2b_1)\mathrm{i} \\ &= (a_1a_2 - b_1b_2) + (a_1b_2 + a_2b_1)\mathrm{i}. \end{aligned}$$

and powers of complex numbers z^n for integer n are defined as usual:

- $\alpha^1 = \alpha, \quad \alpha^2 = \alpha\alpha, \quad \alpha^3 = \alpha\alpha\alpha, \ldots$
- For $\alpha \neq 0, \quad \alpha^0 = 1, \quad \alpha^{-1} = \dfrac{1}{\alpha}, \quad \alpha^{-2} = \dfrac{1}{\alpha^2}, \quad \ldots$

Remark: Here we define z^n for $n \in \mathbb{Z}$.

Furthermore, two more important operations are defined:

Definition 6.3 (Complex conjugate and Absolute value)

For $z = a + b\mathrm{i} \in \mathbb{C}$, we define

- its **complex conjugate** by $\overline{z} := a - b\mathrm{i}$,
- its **absolute value** by $|z| := \left(\overline{z}\, z\right)^{1/2} = \left(a^2 + b^2\right)^{1/2}$.

Quiz

Q1. Prove that, if $z = a + b\mathrm{i}$, then $\overline{z}z$ is a real and non-negative number.

Q2. Explain why $|z| \geq 0$. [Hint: See the definition carefully. What does $x^{1/2}$ mean?]

Q3. For each z below, find $\mathrm{Re}(z)$, $\mathrm{Im}(z)$, $\overline{z}$, and $|z|$.

$$z = 2 + 7\mathrm{i}, \quad z = -3 - \mathrm{i}, \quad z = 5, \quad z = -4\mathrm{i}, \quad z = 0.$$

Q4. Find $x, y \in \mathbb{R}$ such that $(x + 1) + (2y - 3)\mathrm{i} = 4 - \mathrm{i}$.

The fact $z\overline{z} \in \mathbb{R}$ is used to calculate fractions of complex numbers:

$$\frac{w}{z} = \frac{w\overline{z}}{z\overline{z}} = \frac{w\overline{z}}{|z|^2}. \quad \text{For example, } \frac{11 + 4\mathrm{i}}{1 + 2\mathrm{i}} = \frac{(11 + 4\mathrm{i}) \times (1 - 2\mathrm{i})}{(1 + 2\mathrm{i}) \times (1 - 2\mathrm{i})} = \frac{19 - 18\mathrm{i}}{5}. \tag{6.2}$$

Thus, for $z = a + b\mathrm{i} \neq 0$ and $n \in \mathbb{Z}$,

$$z^{-n} = \frac{1}{z^n} = \frac{\overline{z}^n}{z^n\overline{z}^n} = \frac{\overline{z}^n}{\left(z\overline{z}\right)^n} = \frac{\overline{z}^n}{|z|^{2n}} = \frac{(a - b\mathrm{i})^n}{\left(a^2 + b^2\right)^n}. \tag{6.3}$$

Problems

6.1 [Note: This is a timed practice; just write down the answer, without explanation.] Consider the following numbers. Assume $a, b, k \in \mathbb{R}$.

$$-2+\mathrm{i},\quad 1-\mathrm{i},\quad 3+4\mathrm{i},\quad -2(3+4\mathrm{i}),\quad -2,\quad -2\mathrm{i},\quad -9\mathrm{i},\quad 0,\quad \mathrm{i},\quad 4,$$
$$\sqrt{2}+\mathrm{i},\quad -\sqrt{3}+2\mathrm{i},\quad 3+\mathrm{i}\sqrt{5},\quad -1-\mathrm{i}\sqrt{7},\quad -\sqrt{11}+\mathrm{i}\sqrt{14},$$
$$\frac{\mathrm{i}+\sqrt{5}}{\sqrt{2}},\quad \frac{\mathrm{i}-\sqrt{5}}{\sqrt{2}},\quad \frac{\sqrt{3}}{\sqrt{2}}+\frac{\sqrt{3}}{\sqrt{2}}\mathrm{i},\quad -\frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}\mathrm{i},\quad a+2\mathrm{i},\quad a-b\mathrm{i},\quad k\mathrm{i}.$$

(1) What are the real part, imaginary part, and complex conjugate of each? (goal time:)

(2) What are their absolute values? (goal time: 120 seconds)

6.2 Calculate the following.

(1) $(3+2\mathrm{i})+(4-5\mathrm{i})$ **(2)** $(-2+7\mathrm{i})-(5+3\mathrm{i})$ **(3)** $(6-4\mathrm{i})-(-1+2\mathrm{i})$

(4) $(2+3\mathrm{i})(4-\mathrm{i})$ **(5)** $(-1+5\mathrm{i})(2+3\mathrm{i})$ **(6)** $(3-2\mathrm{i})(3+2\mathrm{i})$

(7) $\mathrm{i}(3+\mathrm{i})-\mathrm{i}(1-4\mathrm{i})$ **(8)** $(1-2\mathrm{i})+\overline{4+\mathrm{i}}$ **(9)** $\overline{(3-4\mathrm{i})+\overline{1+2\mathrm{i}}}$

(10) $(1+2\mathrm{i})^4$ **(11)** $(1-2\mathrm{i})^4$ **(12)** $(2+3\mathrm{i})^3(2-3\mathrm{i})^3$

(13) $\dfrac{2+\mathrm{i}}{-\mathrm{i}}$ **(14)** $\dfrac{11+4\mathrm{i}}{1+2\mathrm{i}}$ **(15)** $\dfrac{1-\mathrm{i}}{(1+\mathrm{i})^2}$

**6.3 Find $z \in \mathbb{C}$ that satisfies $z^2 = 1+2\mathrm{i}$.

The following relations hold for these operations. They are somewhat obvious, but with using them you can speed up your calculations.

Theorem 6.4 (Properties of Complex-number arithmetic)

For $\alpha, \beta, \gamma \in \mathbb{C}$ and $n, m \in \mathbb{Z}$,

$$\begin{aligned}&(\alpha+\beta)+\gamma=\alpha+(\beta+\gamma),\quad \alpha+\beta=\beta+\alpha,\quad 0+\alpha=\alpha,\\ &(\alpha\beta)\gamma=\alpha(\beta\gamma),\quad \alpha\beta=\beta\alpha,\quad 1\alpha=\alpha,\quad \alpha(\beta+\gamma)=\alpha\beta+\alpha\gamma.\end{aligned} \tag{6.4}$$

$$\alpha^n\alpha^m=\alpha^{n+m},\quad (\alpha^n)^m=\alpha^{nm},\quad (\alpha\beta)^n=\alpha^n\beta^n,\quad \alpha^{-1}=\frac{\overline{\alpha}}{|\alpha|^2},\quad \left(\frac{\alpha}{\beta}\right)^n=\frac{\alpha^n}{\beta^n}; \tag{6.5}$$

$$\overline{\overline{\alpha}}=\alpha,\quad \overline{\alpha+\beta}=\overline{\alpha}+\overline{\beta},\quad \overline{\alpha\beta}=\overline{\alpha}\,\overline{\beta},\quad \overline{\alpha^n}=(\overline{\alpha})^n,\quad \overline{\left(\frac{\alpha}{\beta}\right)}=\frac{\overline{\alpha}}{\overline{\beta}}; \tag{6.6}$$

$$|\overline{\alpha}|=|\alpha|,\quad |\alpha\beta|=|\alpha||\beta|,\quad \left|\frac{\alpha}{\beta}\right|=\frac{|\alpha|}{|\beta|},\quad |\alpha^n|=|\alpha|^n, \tag{6.7}$$

$$|\alpha|-|\beta|\le|\alpha-\beta|,\quad |\alpha+\beta|\le|\alpha|+|\beta|, \tag{6.8}$$

Theorem 6.5

$$\mathrm{Re}(z) = \frac{z+\bar{z}}{2}, \quad \mathrm{Im}(z) = \frac{z-\bar{z}}{2\mathrm{i}}. \tag{6.9}$$

Example 6.6

Express $\mathrm{Re}(z)^2$, $\mathrm{Re}(z^2)$, and $\mathrm{Re}(z)\,\mathrm{Im}(z)$ without using “Re” and “Im”, i.e., only with z and $\bar{z}$.

Solution

$$\mathrm{Re}(z)^2 = \left(\frac{z+\bar{z}}{2}\right)^2 = \frac{(z+\bar{z})^2}{4}; \quad \mathrm{Re}(z^2) = \frac{z^2+\overline{z^2}}{2} = \frac{z^2+\bar{z}^2}{2}; \quad \mathrm{Re}(z)\,\mathrm{Im}(z) = \frac{z^2-\bar{z}^2}{4\mathrm{i}}.$$

Quiz

Q5. Calculate them.

(1) $\left|\left(\sqrt{2}+\mathrm{i}\sqrt{7}\right)^4\right|$ **(2)** $|11+22\mathrm{i}|^2$ **(3)** $\left|\frac{1+\mathrm{i}}{1-\mathrm{i}}\right|$ **(4)** $(1+\mathrm{i})\overline{(1+\mathrm{i})^2}$

Q6. Express following expressions without “Re” and “Im”, i.e., only with z and $\bar{z}$.

(1) $(\mathrm{Im}(z))^2$ **(2)** $\mathrm{Im}(2z)$ **(3)** $\mathrm{Re}(z) + \mathrm{Im}(z)$

Problems

★6.4 Let $z = 2+\mathrm{i}$ and $w = 1-2\mathrm{i}$. Calculate:

(1) $z+w$ **(2)** $\bar{z}$ **(3)** $\bar{w}$ **(4)** $|z|$ **(5)** $|w|$

(6) $\bar{z}+\bar{w}$ **(7)** $\overline{z+w}$ **(8)** $z\bar{w}$ **(9)** $\bar{z}w$ **(10)** $|zw|$

(11) $\left|\frac{z}{w}\right|$ **(12)** $|z+w|$ **(13)** $|z|+|\bar{w}|$ **(14)** $|zw+z\bar{w}|$ **(15)** $|\bar{z}w+z\bar{w}|$

★6.5 Calculate i^2, i^4, and i^{127}. Also, for $z = (1+\mathrm{i})/\sqrt{2}$, calculate z^{127}.

★6.6 Prove the following statements, where $z, \alpha, \beta \in \mathbb{C}$.

(1) $\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$. **(2)** $|z_1 z_2| = |z_1|\,|z_2|$. **(3)** $\mathrm{Im}(z) = (z-\bar{z})/(2\mathrm{i})$.

****6.7** Prove that, for $a, b, c \in \mathbb{R}$ and $z \in \mathbb{C}$, the solution of $az^2+bz+c = 0$ is given by

$$z = \frac{-b \pm \sqrt{b^2-4ac}}{2a} \quad \text{if } b^2 \geq 4ac; \quad z = \frac{-b \pm \mathrm{i}\sqrt{|b^2-4ac|}}{2a} \quad \text{if } b^2 < 4ac. \tag{6.10}$$

****6.8** Find all complex numbers z satisfying the following equations.

(1) $z^2+16 = 0$ **(2)** $z^2+4z+5 = 0$ **(3)** $z^2+16\mathrm{i} = 0$ **(4)** $z^2+4z+5\mathrm{i} = 0$

****6.9** Let's prove $0 \leq |\alpha+\beta| \leq |\alpha|+|\beta|$ step by step, where $\alpha, \beta \in \mathbb{C}$.

(1) For $z \in \mathbb{C}$, prove the next: $z \neq 0 \implies z\bar{z} > 0$. [Hint: Let $z = a+b\mathrm{i}$.]

(2) For $z \in \mathbb{C}$, prove $|z| \geq \mathrm{Re}(z)$. Then, let $z = \alpha\bar{\beta}$ and simplify $2|z| - 2\,\mathrm{Re}(z)$.

(3) Calculate $(|\alpha|+|\beta|)^2 - |\alpha+\beta|^2$ and prove it is not negative (≥ 0).

(4) Prove $0 \leq |\alpha+\beta| \leq |\alpha|+|\beta|$. Also, find the condition for $|\alpha+\beta| = |\alpha|+|\beta|$.

**6.10 Let $p(z) = a_n z^n + a_{n-1} z^{n-1} + \ldots + a_1 z + a_0$ be a polynomial with *real* coefficients ($a_k \in \mathbb{R}$). Prove that if $z_0 \in \mathbb{C}$ is a solution of $p(z) = 0$, then so is $\overline{z_0}$.

6.2 Polar Form

Theorem 6.7 (Polar form)

A complex number $z = a + b\mathrm{i}$, where $a, b \in \mathbb{R}$, can be expressed in the following **polar form**:

$$z = r\big(\cos\theta + \mathrm{i}\sin\theta\big), \quad \text{where } r = |z| \text{ is the absolute value of } z, \quad \theta \in \mathbb{R} \text{ is called the } \textbf{argument} \text{ of } z. \tag{6.11}$$

For $z \neq 0$, this expression is unique up to the 2π-periodicity of θ.

Proof: Let $z = a + b\mathrm{i}$ with $a, b \in \mathbb{R}$. For $z = 0$, the statement is valid because $r = |z| = 0$. Now we assume $z \neq 0$ and, with $r := |z|$, consider a point P at $(a/r, b/r)$ on xy-plane. Since $(a/r)^2 + (b/r)^2 = 1$, P is on the unit circle around the origin O and thus can be expressed by $(\cos\theta, \sin\theta)$ with θ being the angle from the positive x-axis to $\overrightarrow{\mathrm{OP}}$ and the choice of θ is unique up to the 2π-periodicity. It means $a/r = \cos\theta$ and $b/r = \sin\theta$, i.e., $z = r\cos\theta + \mathrm{i}r\sin\theta$. ∎

Advanced note: In this document, the **trigonometric functions** $\cos\theta$ and $\sin\theta$ are defined using the unit circle in xy-plane and this proof employs that definition. In advanced mathematics, we often take another approach, where we first define e^x for $x \in \mathbb{R}$ by a **power series**. Then we extend e^x to $\mathbb{C}$ and, from this complex function e^z, we define $\cos x$ and $\sin x$ for $x \in \mathbb{R}$ by Euler's formula. See ♣TODO♣Lorem ipsum dolor sit. for further discussion.

Quiz

Q7. Find the complex number expressed by the following polar form:

(1) $r = \sqrt{2}, \theta = \pi/4$ **(2)** $r = 2, \theta = -\pi/6$ **(3)** $r = 4, \theta = \pi/2$

(4) $r = \sqrt{2}, \theta = 9\pi/4$ **(5)** $r = 2, \theta = -5\pi/6$ **(6)** $r = 4, \theta = -\pi$

Similar to $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$, we may consider a function $\mathrm{Arg}(z)$ to return its argument. Here, however, we need to care the **multivaluedness** in θ. For example, **(1)** and **(4)** in the above quiz express the same number $z = 1 + \mathrm{i}$, so there can be multiple values of θ for $z = 1 + \mathrm{i}$. So we need to choose one; a convenient choice is to define $-\pi < \mathrm{Arg}(z) \leq \pi$:

Definition 6.8

For $z \in \mathbb{C}$, $z \neq 0$,

$$\mathrm{Arg}(z) := \text{the argument of } z, \text{chosen between } -\pi \text{ (exclusive) and } \pi \text{ (inclusive).} \tag{6.12}$$

This strategy is called "choosing the **principle value** of the argument of z so that $-\pi < \mathrm{Arg}(z) \leq \pi$".

Quiz

Q8. Find $\mathrm{Arg}(5\mathrm{i})$, $\mathrm{Arg}(2+2\mathrm{i})$, $\mathrm{Arg}(-1-\mathrm{i})$, and $\mathrm{Arg}(1-\mathrm{i})$.

Due to the subtlety of $\mathrm{Arg}(z)$, we do not discuss $\mathrm{Arg}(z)$ in the rest of this document.

Advanced note: You can see the subtlety in the next example. Consider $z = \mathrm{e}^{2\mathrm{i}}$. Then, $\mathrm{Arg}(z) = 2$, but $\mathrm{Arg}(z^2) = \mathrm{Arg}(\mathrm{e}^{4\mathrm{i}}) \approx -2.3$ (Why?). Similarly, $\mathrm{Arg}(z^3) \approx -0.28$ and $\mathrm{Arg}(z^4) \approx 1.72$.

6.3 Euler's formula

We have seen five types of operations on complex numbers: $\bar{z}$, $|z|$, $z+w$, zw, and z^n for $n \in \mathbb{Z}$. The next step is to define e^z for $z \in \mathbb{C}$, but we want to define it nicely: we want to keep the key properties of e^x such as $\mathrm{d}\mathrm{e}^{ax}/\mathrm{d}x = a\mathrm{e}^{ax}$ and $\mathrm{e}^{a+b} = \mathrm{e}^a\mathrm{e}^b$ even for $a, b \in \mathbb{C}$. So,

Definition 6.9

We **define**, for a complex number $z = a + b\mathrm{i}$ ($a, b \in \mathbb{R}$),

$$\mathrm{e}^{a+b\mathrm{i}} := \mathrm{e}^a\big(\cos b + \mathrm{i}\sin b\big). \tag{6.13}$$

In particular, for $\theta \in \mathbb{R}$, $\quad \mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta \quad$ (**Euler's formula**). (6.14)

Advanced note: Let $a, b, x \in \mathbb{R}$. Then, we want to keep $\mathrm{e}^{(a+b\mathrm{i})x} = \mathrm{e}^{ax}\mathrm{e}^{b\mathrm{i}x}$ and thus we just have to define $\mathrm{e}^{\mathrm{i}\theta}$ for a real number $\theta = bx$. Letting $\mathrm{e}^{\mathrm{i}\theta} = f(\theta) + \mathrm{i}g(\theta)$, the requirement of derivative is given by $\mathrm{d}\mathrm{e}^{\mathrm{i}bx}/\mathrm{d}x = \mathrm{i}b\mathrm{e}^{\mathrm{i}bx}$, where the LHS is $b \cdot \mathrm{d}\mathrm{e}^{\mathrm{i}\theta}/\mathrm{d}\theta = bf'(\theta) + \mathrm{i}bg'(\theta)$ and the RHS is $\mathrm{i}b(f(\theta) + \mathrm{i}g(\theta)))$. It means $f'(\theta) = -g(\theta)$ and $g'(\theta) = f(\theta)$. Together with $f(0) = 1$ and $g(0) = 0$, we get $f(\theta) = \cos\theta$ and $g(\theta) = \sin\theta$ and now we have understand our wanted definition is $\mathrm{e}^{a+\mathrm{i}b} = \mathrm{e}^a(\cos b + \mathrm{i}\sin b)$.

Accordingly, the polar form of complex numbers is given by

$$z = a + b\mathrm{i} = r(\cos\theta + \mathrm{i}\sin\theta) = r\exp(\mathrm{i}\theta) = |z|\exp\Big(\mathrm{i}\,\mathrm{Arg}(z)\Big). \tag{6.15}$$

As we defined e^z so that the key properties are kept, the following formulae are valid. (Compare with Theorem 4.9).

Theorem 6.10 (Properties of Exponential Functions (2))

For $z, w \in \mathbb{C}$, $\quad \dfrac{1}{e^z} = e^{-z}, \quad e^z e^w = e^{z+w}, \quad \dfrac{e^z}{e^w} = e^{z-w}, \quad (e^z)^w = e^{zw}.$ (6.16)

For $z \in \mathbb{C}$ and $a > 0$, $\quad a^z = \left(e^{\ln a}\right)^z = e^{z\ln a}$. (6.17)

Currently, we know $\ln a$ only for $a > 0$. So, the second statement is only for $a > 0$ (see Table 2). Discussions on a^z for $a \le 0$ and $a \notin \mathbb{R}$ will be given in ♣TODO♣later?.

Quiz

Q9. Express the following number in the form $a + bi$:

(1) e^{i} **(2)** e^{2i} **(3)** e^{-i} **(4)** e^{1+i} **(5)** $(e^{i})^4$ **(6)** e^{i^4}

(7) $e^{i \ln 3}$ **(8)** 2^{i} **(9)** 2^{2+i} **(10)** 2^{2+2i} **(11)** 4^{1+i} **(12)** 5^{i^2}

The next theorem is extremely important:

Theorem 6.11

$$\text{For } \theta \in \mathbb{R}, \quad \cos\theta = \mathrm{Re}(e^{i\theta}) = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin\theta = \mathrm{Im}(e^{i\theta}) = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \tag{6.18}$$

Then for $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$, de Moivre's theorem $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$ holds, but we do not need to memorize it. Just use the equation $e^{in\theta} = (e^{i\theta})^n$, e.g.,

Example 6.12

Express $\cos 4\theta$ in terms of $\cos\theta$.

Solution

Write $c = \cos\theta$ and $s = \sin\theta$ for brevity. Then.

$$\cos 4\theta = \mathrm{Re}(e^{4i\theta}) = \mathrm{Re}(e^{i\theta})^4 = \mathrm{Re}(c + is)^4 = \mathrm{Re}(c^4 + 4c^3 is + 6c^2 i^2 s^2 + 4ci^3 s^3 + i^4 s^4).$$

As $s^2 = 1 - c^2$, we obtain $\cos 4\theta = c^4 - 6c^2(1 - c^2) + (1 - c^2)^2 = 8\cos^4\theta - 8\cos^2\theta + 1$.

Quiz

Q10. Express $\cos 3\theta$, $\sin 3\theta$, $\cos(\theta + \varphi)$, and $\sin(\theta + \varphi)$ in terms of $\cos\theta$, $\sin\theta$, $\cos\varphi$, and $\sin\varphi$. [Hint: Expand both sides of $e^{i(\theta+\varphi)} = e^{i\theta} e^{i\varphi}$.]

Problems

💪**6.11** Compute the following and express in the form $a + bi$.

(1) $e^{i\pi/6}$ **(2)** $e^{i\pi}$ **(3)** $e^{2i\pi}$ **(4)** $2e^{-i\pi/3}$

(5) $\sqrt{2}e^{-i\pi/4}$ **(6)** $e^{i\pi/2} + e^{-i\pi/2}$ **(7)** $(1 + i)^4$ **(8)** $\left(\sqrt{3} + i\right)^6$

★**6.12** Convert to polar form $re^{i\theta}$, choosing $\mathrm{Arg}(z) \in (-\pi, \pi]$.

(1) $1 + \sqrt{3}i$ **(2)** -2 **(3)** $-i$ **(4)** $3 - 3i$

(5) $-1 + i$ **(6)** $-\sqrt{3} - i$ **(7)** $4i$ **(8)** $1 - \sqrt{3}i$

*****6.13** Using the polar form, compute the following. Express in the form $a + bi$.

(1) $(1 + i)^8$ **(2)** $(\left(\sqrt{3} + i\right)/2)^{12}$ **(3)** $(1 - i)^{10}$

(4) $\left(1 + \sqrt{3}i\right)^{-3}$ **(5)** $((1 + i)/(1 - i))^{20}$ **(6)** $\left(\left(\sqrt{3} + i\right)/(1 + i)\right)^6$

*****6.14** Use de Moivre's theorem to express $\cos(4\theta)$ and $\sin(4\theta)$ only with $\cos\theta$ (without $\sin\theta$).

**6.15 Express $\cos^3\theta$ and $\sin^3\theta$ as a linear combination of $\cos(n\theta)$ and $\sin(n\theta)$ for various n.

**6.16 Using the polar form $z = re^{i\theta}$, find all solutions of the following equations in $\mathbb{C}$.

(1) $z^3 - 8 = 0$ (2) $z^4 + 16 = 0$ (3) $z^6 - 1 = 0$

(4) $z^3 + i = 0$ (5) $z^4 - 4z^2 + 3 = 0$ (6) $z^6 + z^3 - 2 = 0$

**6.17 For $n \in \mathbb{N}^+$ the equation $z^n = 1$ has n solutions. They are called the n-th roots of unity.

(1) Let $\omega = e^{2\pi i/n}$. Show that ω^k with $k = 0, 1, \ldots, n-1$ are the solutions of $z^n = 1$.

(2) Show the following equations for $n \geq 2$:

(a) $\displaystyle\sum_{k=0}^{n-1} \omega^k = 0$ (b) $\displaystyle\prod_{k=0}^{n-1} \omega^k = (-1)^{n+1}$ (c) $\displaystyle\sum_{k=0}^{n-1} \cos\left(\frac{2\pi k}{n}\right) = \sum_{k=0}^{n-1} \sin\left(\frac{2\pi k}{n}\right) = 0$

6.4 Trigonometric and Hyperbolic functions

For real numbers, we have defined the trigonometric and hyperbolic functions (see page 42) by

$$\cos x = \operatorname{Re} e^{ix} = \frac{e^{ix} + e^{-ix}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2},$$

$$\sin x = \operatorname{Im} e^{ix} = \frac{e^{ix} - e^{-ix}}{2i}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}.$$

We naturally extend this definition to complex numbers:

$$\cos z := \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z := \frac{e^{iz} - e^{-iz}}{2i}, \quad \cosh z := \frac{e^z + e^{-z}}{2}, \quad \sinh z := \frac{e^z - e^{-z}}{2}.$$

Then, almost obviously,

Theorem 6.13

For $x \in \mathbb{R}$,

$$\cos ix = \cosh x, \quad \sin ix = i \sinh x, \quad \cosh ix = \cos x, \quad \sinh ix = i \sin x. \tag{6.19}$$

Other functions, such as $\tan z$, $\cot z$, $\tanh z$, … are defined similarly.

6.5 Complex vectors

The next step is "complex vectors". Let us recall three interpretations of vectors, given in Chapter 5:

1. an arrow in n-dimensional space ($n \in \mathbb{N}^+$, but usually $n = 3$),
2. a list of n numbers arranged vertically ($n \in \mathbb{N}^+$), and
3. an element of a vector space (such as a Hilbert space).

There we started from the first interpretation (Definition 5.1) and reached the second interpretation (Definition 5.17). For complex vectors, the first interpretation seems not nice, but Definition 5.17 looks nice to *define* complex vectors: we just extend real numbers to complex numbers.

Definition 6.14 (Inner product of Complex vectors)

We define n-dimensional **complex vectors** by a list of n complex number: (cf. Definition 5.17)

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}; \quad v_k \in \mathbb{C}, \quad n \in \mathbb{N}^+. \tag{6.20}$$

With this definition, we can analyze complex vectors similarly as real vectors, except for one caveat. We would like to use Eq. (5.11) to define inner products, but then it would break nice properties such as Eq. (5.4) and **(C)** of Theorem 5.9.

Quiz

Q11. Confirm that we cannot use Eq. (5.11) for complex vectors since it would break Eq. (5.4) and **(C)** of Theorem 5.9 in some cases such as $\binom{1}{i}$ or $\binom{0}{i}$.

So, we define inner products for complex vectors by (compare with Eq. (5.11))

Definition 6.15 (Complex vectors)

Consider n-dimensional complex vectors $\vec{a}$ and $\vec{b}$. We define the **inner product** by

$$\langle\, \vec{a} \,|\, \vec{b} \,\rangle = \overline{a_1}\, b_1 + \overline{a_2}\, b_2 + \ldots + \overline{a_n}\, b_n = \sum_{k=1}^{n} \overline{a_k}\, b_k \qquad \text{(for complex vectors).} \tag{6.21}$$

We use a different notation $\langle\, \vec{a} \,|\, \vec{b} \,\rangle$ to indicate it is *complex* inner product.

Quiz

Be careful: Important quizzes!

Q12. Check that $\langle\, \vec{a} \,|\, \vec{b} \,\rangle = \langle\, \vec{b} \,|\, \vec{a} \,\rangle$ is false (incorrect). [Hint: Find a counterexample.]

Q13. Let $\vec{v} = k\vec{a}$ with $k \in \mathbb{C}$. Check that $\langle\, \vec{v} \,|\, \vec{b} \,\rangle = k\langle\, \vec{a} \,|\, \vec{b} \,\rangle$ is false.

Remark: Cross products are not defined for complex vectors.

For completeness, we give the definition of addition and scalar multiplication of complex vectors:

Definition 6.16 (Addition and Scalar multiplication of complex vectors)

For complex vectors $\vec{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$, $\quad \vec{a} + \vec{b} = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix}$, $\quad k\vec{a} = \begin{pmatrix} ka_1 \\ \vdots \\ ka_n \end{pmatrix}$, (6.22)

where $k \in \mathbb{C}$. (Compare with Eq. (5.10): there we only considered $k \in \mathbb{R}$, but here $k \in \mathbb{C}$.)

With these definitions, Theorem 5.7 holds **as is** for complex vectors (compare!):

Theorem 6.17

For n-dimensional complex vectors $\vec{a}$ and $\vec{b}$ and $p, q \in \mathbb{C}$,

(A) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$, **(E)** $p\vec{a} + p\vec{b} = p(\vec{a} + \vec{b})$,

(B) $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$, **(F)** $p\vec{a} + q\vec{a} = (p + q)\vec{a}$,

(C) $\vec{a} + \vec{0} = \vec{a}$, **(G)** $(pq)\vec{a} = p(q\vec{a})$,

(D) $\vec{a} + (-\vec{a}) = \vec{0}$, **(H)** $1\vec{a} = \vec{a}$.

Properties in Theorem 5.9 are also valid for complex vectors *with slight modifications*.

Theorem 6.18 (Complex-vector inner product)

For n-dimensional complex vectors $\vec{a}$ and $\vec{b}$ and a constant $k \in \mathbb{C}$,

(A) $\langle \vec{a} \,|\, \vec{b} \rangle = \overline{\langle \vec{b} \,|\, \vec{a} \rangle}$, **(E)** $\langle \vec{a} + \vec{b} \,|\, \vec{c} \rangle = \langle \vec{a} \,|\, \vec{c} \rangle + \langle \vec{b} \,|\, \vec{c} \rangle$,

(B) $\langle \vec{a} \,|\, \vec{a} \rangle \geq 0$ for any vector $\vec{a}$, **(F)** $\langle \vec{a} \,|\, \vec{b} + \vec{c} \rangle = \langle \vec{a} \,|\, \vec{b} \rangle + \langle \vec{a} \,|\, \vec{c} \rangle$,

(C) $\langle \vec{a} \,|\, \vec{a} \rangle > 0$ for any vector $\vec{a} \neq \vec{0}$, **(G)** $\langle k\vec{a} \,|\, \vec{b} \rangle = \bar{k} \langle \vec{a} \,|\, \vec{b} \rangle$,

(D) $\langle \vec{a} \,|\, \vec{a} \rangle = 0 \iff \vec{a} = \vec{0}$, **(H)** $\langle \vec{a} \,|\, k\vec{b} \rangle = k \langle \vec{a} \,|\, \vec{b} \rangle$.

Using **(B)** of the above, we can define the **magnitude** of complex vectors by

Theorem 6.19 (Magnitude of complex vectors)

For a complex vector $\vec{a}$, we define its **magnitude** by $|\vec{a}| := \sqrt{\langle \vec{a} \,|\, \vec{a} \rangle}$. (6.23)

Theorem 6.20 (Properties of complex-vector magnitude)

For complex vectors $\vec{a}$ and $\vec{b}$ and a constant $k \in \mathbb{C}$,

$$|\vec{a}| \geq 0, \qquad |\vec{a}| = 0 \iff \vec{a} = \vec{0}, \qquad |k\vec{a}| = |k|\,|\vec{a}|, \qquad |\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|. \tag{6.24}$$

Quiz

Q14. In Eq. (6.24), what do $|k|$ and $|\vec{a}|$ mean, respectively? What are their definitions?

Q15. Compare Theorem 6.18 with Theorem 5.9 and find all the differences.

Q16. Prove **(A)** and **(G)** of Theorem 6.18 using component-wise notation.

Q17. Prove **(B)** and **(H)** of Theorem 6.18. [Hint: (A) and (G) might be useful.]

Problems

Compare these problems with Theorem 5.10.

★**6.18** **(1)** Verify $|\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + 2\,\mathrm{Re}\langle\,\vec{a}\,|\,\vec{b}\,\rangle + |\vec{b}|^2$.

(2) Simplify $|(3+4\mathrm{i})\vec{v}| + |5\vec{v}|$.

(3) Expand $|\vec{v}-\vec{w}|^2$, $\left|\vec{v}+\mathrm{i}\vec{w}\right|^2$, and $\left|\vec{v}+k\vec{w}\right|^2$ with $k \in \mathbb{C}$.

*****6.19** Prove **(C)**, **(D)**, **(E)**, **(F)** of Theorem 6.18.

****6.20** Prove Theorem 6.20. Also, prove **Cauchy-Schwarz inequality**, $\left|\langle\,\vec{a}\,|\,\vec{b}\,\rangle\right| \le |\vec{a}||\vec{b}|$.

Recall that the component-wise notation was introduced in Section 5.7 with respect to a *specific* orthonormal basis vectors. Our discussion in this section is also built over a *specific* orthonormal basis vectors,

$$\vec{e}_1 = \begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix},\quad \vec{e}_2 = \begin{pmatrix}0\\1\\\vdots\\0\end{pmatrix},\ldots,\quad \vec{e}_n = \begin{pmatrix}0\\0\\\vdots\\1\end{pmatrix},\quad \text{with which } \vec{v} = \begin{pmatrix}v_1\\v_2\\\vdots\\v_n\end{pmatrix} = \sum_{k=1}^{n} v_k\vec{e}_k. \tag{6.25}$$

Definition 6.21 (Orthonormal basis vectors for complex vectors)

Assume that we are considering n-dimensional *complex* vectors. If n *complex* vectors $\vec{e}_1, \ldots, \vec{e}_n$ satisfy the following properties, we call them **an orthonormal basis** (for complex vectors):

- All of them are unit vectors, i.e., $|\vec{e}_i| = 1$ for all i.
- Any of them are perpendicular, i.e., $i \ne j \implies \langle\,\vec{e}_i\,|\,\vec{e}_j\,\rangle = 0$.
- We cannot add any more vectors without violating the above two rules.

Quiz

Q18. Compare this definition with Definition 5.14 and find the differences.

Q19. Check that $\{\vec{e}_1, \ldots, \vec{e}_n\}$ in Eq. (6.25) is a basis.

Advanced note: In mathematical literature, you may find different notations for the inner product. Most physicists write $\langle\,\vec{a}\,|\,\vec{b}\,\rangle = \sum \overline{a_k} b_k$, which we use in this document. Meanwhile, mathematicians tend to denote inner products by $(\vec{a}, \vec{b})$ and define it by $(\vec{a}, \vec{b}) = \sum a_k \overline{b_k}$, or even $\langle\,\vec{a}\,|\,\vec{b}\,\rangle = \sum a_k \overline{b_k}$.

Chapter 7

Matrices

7.1 Review: Vectors as Lists of Numbers

We discuss matrices[#7] in this chapter. A matrix is deeply tied to vectors. Since a matrix is described as a two-dimensional list of numbers,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1+\mathrm{i} \\ 1-\mathrm{i} & 0 \end{pmatrix}, \quad \text{etc.,}$$

we in this chapter interpret vectors as (one-dimensional) lists of numbers[#8], not arrows, until ♣TODO♣a section.

> **Advanced note:** Our discussion up to this chapter (Chapter 6 to Chapter 7) also assumes that the dimension of the space is finite.

Let us introduce some notation for vectors. $\boxed{\mathbb{R}^n}$ is the set of all real n-dimensional vectors. i.e., $\boxed{\vec{a} \in \mathbb{R}^n}$ means "$\vec{a}$ is a real n-dimensional vector." Similarly, $\boxed{\mathbb{C}^n}$ is the set of all complex n-dimensional vectors. i.e., $\boxed{\vec{a} \in \mathbb{C}^n}$ means "$\vec{a}$ is a complex n-dimensional vector." This notation is consistent with $\boxed{\mathbb{R}^{m,n}}$ and $\boxed{\mathbb{C}^{m,n}}$ (the set of all $m \times n$ real or complex matrices). The k-th component of a vector $\vec{a}$ should be denoted by $\boxed{(\vec{a})_k}$, but we may write it as $\boxed{a_k}$ if not confusing.

We will also use a shorthanded notation, $\boxed{\mathbb{K}}$, to express either $\mathbb{R}$ or $\mathbb{C}$, and you need to identify its meaning from the context. For example, $\boxed{\text{If } \vec{v} \in \mathbb{K}^3, \text{ then } v_k \in \mathbb{K} \text{ but } |\vec{v}| \in \mathbb{R}}$ is to mean

> If $\vec{v}$ is a (real / complex) 3-dimensional vector, then its k-th component is a (real / complex) number and its magnitude $|\vec{v}|$ is always a real number.

First, we summarize what we have discussed about vectors. A n-dimensional (real / complex) vector, $\vec{v} \in \mathbb{K}^n$, is a column array of n (real / complex) numbers $v_k \in \mathbb{K}$:

$$\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}; \qquad \vec{v} + \vec{w} = \begin{pmatrix} v_1 + w_1 \\ \vdots \\ v_n + w_n \end{pmatrix}, \qquad k\vec{v} = \begin{pmatrix} kv_1 \\ \vdots \\ kv_n \end{pmatrix} \quad (k \in \mathbb{K}). \tag{7.1}$$

Notice that $k \in \mathbb{K}$ means k can only be a real number if we are considering real vectors, while k is complex if we are considering complex vectors.

[#7] One dog, two dogs. One matrix, two matrices. One vertex, two vertices. One index, two indices.
[#8] ⟶ The beginning of Chapter 5.

The inner product and magnitude are given by

$$\begin{aligned} &\text{for } \vec{v}, \vec{w} \in \mathbb{R}^n, \quad \vec{v} \cdot \vec{w} = \sum_{k=1}^{n} v_k w_k, \quad |\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}} = \sum_{k=1}^{n} v_k^2; \\ &\text{for } \vec{v}, \vec{w} \in \mathbb{C}^n, \quad \langle\, \vec{v} \,|\, \vec{w} \,\rangle = \sum_{k=1}^{n} \overline{v_k} w_k, \quad |\vec{v}| = \sqrt{\langle\, \vec{v} \,|\, \vec{v} \,\rangle} = \sum_{k=1}^{n} |v_k|^2. \end{aligned} \tag{7.2}$$

> **Remark:** We **must** use the notation $\langle\, \vec{v} \,|\, \vec{w} \,\rangle$ for complex vectors. Meanwhile, for real vectors, we usually use $\vec{v} \cdot \vec{w}$ but also we may write $\langle\, \vec{v} \,|\, \vec{w} \,\rangle$.

For their properties, review Chapter 5 and Chapter 6. Cross products are only defined for $\mathbb{R}^3$:

$$\text{for } \vec{v}, \vec{w} \in \mathbb{R}^3, \quad \vec{v} \times \vec{w} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}. \tag{7.3}$$

For complex vectors or real vectors other than 3d, the cross product is not defined.

We also discussed orthogonal bases of $\mathbb{K}^n$. *An* orthogonal basis is made of n vectors $\vec{e}_1, \dots, \vec{e}_n$, where $\vec{e}_k \in \mathbb{K}^n$, that satisfy

$$\big(\, |\vec{e}_k| = 1 \,\big) \wedge \big(\, j \neq k \Longrightarrow \langle\, \vec{e}_j \,|\, \vec{e}_k \,\rangle = 0 \,\big) \quad \text{for all } j, k = 1, \dots, n, \tag{7.4}$$

or more simply,

$$\langle\, \vec{e}_j \,|\, \vec{e}_k \,\rangle = \delta_{jk} \quad \text{for all } j, k = 1, \dots, n, \tag{7.5}$$

where we introduced the **Kronecker delta** δ_{jk}, which is defined as

$$\delta_{jk} := \begin{cases} 1 \text{ if } j = k, \\ 0 \text{ if } j \neq k. \end{cases} \tag{7.6}$$

Problems

★7.1 Prove Eq. (7.5) and Eq. (7.6) are equivalent.

***7.2** This discussion assumes that an orthonormal basis of $\mathbb{K}^n$ always have n vectors, which we have not proved yet. So, prove that the number of basis vectors in an orthonormal basis of $\mathbb{K}^n$ is always n.

7.2 Change of basis

For vectors $\mathbb{K}^n$, we have many (actually infinite) orthonormal bases. For example, for $\mathbb{C}^2$,

$$\{\vec{e}_1, \vec{e}_2\} \text{ with } \vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \{\vec{f}_1, \vec{f}_2\} \text{ with } \vec{f}_1 = \begin{pmatrix} 0 \\ \mathrm{i} \end{pmatrix}, \quad \vec{f}_2 = \begin{pmatrix} -\mathrm{i} \\ 0 \end{pmatrix} \tag{7.7}$$

are both orthonormal bases.

Quiz

Q1. Show $\langle \vec{e}_j \,|\, \vec{e}_k \rangle = \delta_{jk}$ and $\langle \vec{f}_j \,|\, \vec{f}_k \rangle = \delta_{jk}$.

Recall that the component-wise notation is based on a specific orthonormal basis. If we used $\{\vec{f}_1, \vec{f}_2\}$ instead of $\{\vec{e}_1, \vec{e}_2\}$, the component-based notation would be different. For example, $\vec{v} = \left(\begin{smallmatrix} 3 \\ 2 \end{smallmatrix}\right)$ actually means

$$\vec{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 3\vec{e}_1 + 2\vec{e}_2, \quad \text{which is equal to} \quad -2\mathrm{i}\begin{pmatrix} 0 \\ \mathrm{i} \end{pmatrix} + 3\mathrm{i}\begin{pmatrix} -\mathrm{i} \\ 0 \end{pmatrix} = -2\mathrm{i}\vec{f}_1 + 3\mathrm{i}\vec{f}_2. \tag{7.8}$$

So, if we had chosen $\{\vec{f}_1, \vec{f}_2\}$ as our favorite basis, we would write

$$\vec{v} = \begin{pmatrix} -2\mathrm{i} \\ 3\mathrm{i} \end{pmatrix}_f, \tag{7.9}$$

where the subscript f emphasizes that the components are under the basis $\{\vec{f}_1, \vec{f}_2\}$.

In physics, we often switch from one basis to another. Choosing a basis means choosing the axes of the space[#9]. We should choose the basis so that we can analyze the problem easily. The axes belong to the whole space, not to a single vector. Therefore, we cannot change the basis of just one vector. We have to rewrite all the vectors in the new basis.

Suppose we have vectors $\vec{a}, \vec{b}, \ldots$, written under the basis $\{\vec{e}_1, \ldots, \vec{e}_n\}$: (For some reason, it is useful to write $\vec{e}_1 a_1$ instead of $a_1 \vec{e}_1$ when we discuss basis transformation.)

$$\vec{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}_e = \vec{e}_1 a_1 + \ldots + \vec{e}_n a_n, \quad \text{etc.} \tag{7.10}$$

How can we change the basis of all of them at once, to the new basis $\{\vec{f}_1, \ldots, \vec{f}_n\}$? The key idea is that the old basis vectors $\vec{e}_k$ can be expressed in terms of the new basis vectors $\vec{f}_k$:

$$\vec{e}_k = \vec{f}_1 u_{1k} + \vec{f}_2 u_{2k} + \ldots + \vec{f}_n u_{nk} = \sum_{j=1}^{n} \vec{f}_j u_{jk} = \begin{pmatrix} u_{1k} \\ \vdots \\ u_{nk} \end{pmatrix}_f \quad \text{for } k = 1, \ldots, n, \tag{7.11}$$

where, again, we use the subscript f to emphasize that the components are under the new basis. So,

[#9] $\longrightarrow$page 55

$$\vec{a} = \sum_{k=1}^{n} \vec{e}_k a_k = \sum_{k=1}^{n} \left(\sum_{j=1}^{n} \vec{f}_j u_{jk} \right) a_k = \sum_{j=1}^{n} \vec{f}_j \sum_{k=1}^{n} u_{jk} a_k = \sum_{j=1}^{n} \vec{f}_j a'_j \quad \text{with } a'_j := \sum_{k=1}^{n} u_{jk} a_k \tag{7.12}$$

and this equation already shows the expression of $\vec{a}$ under the new basis: we have got $\vec{a} = \begin{pmatrix} a'_1 \\ \vdots \\ a'_n \end{pmatrix}_f$.

Quiz

Q2. Explain each equal symbol "=" in Eq. (7.12). Also, explain why the equation means $\vec{a} = \begin{pmatrix} a'_1 \\ \vdots \\ a'_n \end{pmatrix}_f$.

Q3. The coefficient u_{jk} in Eq. (7.11) is given by $u_{jk} = \langle \vec{f}_j \,|\, \vec{e}_k \rangle$. Explain why. [Hint: Calculate the inner product, using Theorem 6.18.]

Theorem 7.1 (Change of basis)

Consider $\mathbb{K}^n$ and two bases on it: $\{\vec{e}_1, \dots, \vec{e}_n\}$ ("old" basis) and $\{\vec{f}_1, \dots, \vec{f}_n\}$ ("new" basis). If a vector $\vec{a}$ is written under the old basis as

$$\vec{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}_e = \sum_{k=1}^{n} \vec{e}_k a_k, \tag{7.13}$$

then it can be rewritten under the new basis as

$$\vec{a} = \begin{pmatrix} a'_1 \\ \vdots \\ a'_n \end{pmatrix}_f = \sum_{k=1}^{n} \vec{e}_k a'_k, \tag{7.14}$$

where the new components a'_j are given by

$$a'_j := \sum_{k=1}^{n} u_{jk} a_k, \qquad u_{jk} = \langle \vec{f}_j \,|\, \vec{e}_k \rangle. \tag{7.15}$$

We will come back to this discussion later.

7.3 Matrices

Remark: This is a quick review; you should already be familiar with matrix arithmetic.

We introduce matrices as a 2d extension of vectors.

Definition 7.2 (Matrix)

A (complex / real) $m \times n$ **matrix** is defined as a 2d array of (complex / real) numbers,

$$A = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{pmatrix}; \quad a_{jk} \in \mathbb{K}. \tag{7.16}$$

In particular; we name a **rows**, a **column** and a **component** as follows:

$$j\text{-th row: } (A_{j1} \ \ A_{j2} \ \ \dots \ \ A_{jn}); \quad k\text{-th column: } \begin{pmatrix} A_{1k} \\ \vdots \\ A_{mk} \end{pmatrix}; \quad (j,k)\text{-component: } A_{jk}; \tag{7.17}$$

"$n \times m$" is called the **size** of the matrix. We write the set of all $n \times m$ matrix as $\mathbb{K}^{m,n}$.

For example, we can write $\boxed{M \text{ is a complex } 3 \times 2 \text{ matrix}}$ by $\boxed{M \in \mathbb{C}^{3,2}}$.

Remark:
- The last component of a $m \times n$ matrix is A_{mn}.
- Columns are tall because "**l**" is tall; rows are flat because "r-o-w" is flat.

Quiz

Q4. Write a 5×2 matrix. Answer the number of its components. Answer how many rows and columns does it have. Answer the number of the components in its first row. Answer the number of the components in its first column.

Q5. Consider a $a \times b$ matrix. Answer the number of its components. Answer how many rows and columns does it have. Answer the number of the components in its first row. Answer the number of the components in its first column.

Also, vectors $\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ are called **column vectors**. We will later see **row vectors** $\vec{v}^{\mathrm{T}} = (v_1 \ \ \dots \ \ v_n)$.

Quiz

Q6. $1 \times n$ matrices and $n \times 1$ matrices can be identified as column vectors or row vectors. Which is which?

Be careful: $\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ and $(v_1 \ \ \dots \ \ v_n)$ are different objects.

You will consider complex matrices in quantum mechanics courses, while in other physics we usually use real matrices. A real matrix is (of course) a complex matrix, so it is almost enough if you learn about complex matrices.

7.4 Basic operations on matrices

You need to learn following operations for matrices:

- addition and scalar multiplication
- transposition, complex conjugate, and Hermitian conjugate
- matrix multiplication (Section 7.5)
- trace and determinant (Section 7.6)

Addition and **scalar multiplication** are defined just like vectors:

Definition 7.3 (Addition and scalar multiplication)

Addition and **scalar multiplication** of $m \times n$ matrices are defined by

$$A + B := \begin{pmatrix} A_{11}+B_{11} & A_{12}+B_{12} & \dots & A_{1n}+B_{1n} \\ A_{21}+B_{21} & A_{22}+B_{22} & \dots & A_{2n}+B_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1}+B_{m1} & A_{m2}+B_{m2} & \dots & A_{mn}+B_{mn} \end{pmatrix}, \quad kA := \begin{pmatrix} kA_{11} & \dots & kA_{1n} \\ kA_{21} & \dots & kA_{2n} \\ \vdots & \ddots & \vdots \\ kA_{m1} & \dots & kA_{mn} \end{pmatrix} \tag{7.18}$$

for the matrices A and B with same size.

Notice that $A + B$ is *not defined* if A and B are in different size.

Transposition, **complex conjugate**, and **Hermitian conjugate** are defined as follows:

Definition 7.4 (Transposition, complex conjugate, and Hermitian conjugate)

Let A be a $m \times n$ matrix. Its **transposition** A^{T} is defined by a $n \times m$ matrix with $\left(A^{\mathrm{T}}\right)_{ij} = A_{ji}$. The **complex conjugate** $\overline{A}$ of a matrix A is defined by $\left(\overline{A}\right)_{ij} := \overline{A_{ij}}$. The **Hermitian conjugate** $A^\dagger$ of a matrix A is defined by $A^\dagger := \overline{A}^T = \overline{A^T}$.

Be careful:

Example 7.5

Let $A = \begin{pmatrix} 1 & 2+\mathrm{i} & 3\mathrm{i} \\ -4\mathrm{i} & 5+6\mathrm{i} & -7\mathrm{i} \end{pmatrix}$. Then its transposition is $A^T = \begin{pmatrix} 1 & -4\mathrm{i} \\ 2+\mathrm{i} & 5+6\mathrm{i} \\ 3\mathrm{i} & -7\mathrm{i} \end{pmatrix}$, complex conjugate is $\overline{A} = \begin{pmatrix} 1 & 2-\mathrm{i} & -3\mathrm{i} \\ 4\mathrm{i} & 5-6\mathrm{i} & 7\mathrm{i} \end{pmatrix}$, and Hermitian conjugate is $A^\dagger = \begin{pmatrix} 1 & 4\mathrm{i} \\ 2-\mathrm{i} & 5-6\mathrm{i} \\ -3\mathrm{i} & 7\mathrm{i} \end{pmatrix}$.

Quiz

Q7. Let $A = \begin{pmatrix} 1 & 2+\mathrm{i} \\ -3\mathrm{i} & 4 \end{pmatrix}$. Compute A^T, $\overline{A}$, and $A^\dagger$.

7.4.1 Matrix-vector multiplication

We now define how a matrix acts on a vector, which corresponds to applying the linear transformation.

Definition 7.6 (Matrix-vector multiplication)

Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n$. The **product** $A\vec{v}$ is the vector in $\mathbb{R}^m$ defined by

$$A\vec{v} = \begin{pmatrix} a_{11}v_1 + a_{12}v_2 + \dots + a_{1n}v_n \\ a_{21}v_1 + a_{22}v_2 + \dots + a_{2n}v_n \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \dots + a_{mn}v_n \end{pmatrix}. \tag{7.19}$$

In other words, the i-th entry of $A\vec{v}$ is $\sum_{j=1}^{n} a_{ij}v_j$.

Theorem 7.7 (Matrix as a linear transformation)

The map $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T(\vec{v}) = A\vec{v}$ is a linear transformation. Conversely, every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ corresponds to a unique matrix $A \in \mathbb{R}^{m\times n}$, where the j-th column of A is $T(\vec{e}_j)$.

So matrices and linear transformations are in one-to-one correspondence (once a basis is fixed).

Example 7.8 (Reading a linear transformation as a matrix)

The linear transformation that sends $\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+2y \\ 3x-y \end{pmatrix}$ corresponds to which matrix?

Solution

Check: $T(\vec{e}_x) = T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $T(\vec{e}_y) = T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. These become the columns, so $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$. Indeed, $\begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+2y \\ 3x-y \end{pmatrix}$.

Quiz

Q8. Compute the following matrix-vector products.

(1) $\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ **(2)** $\begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ **(3)** $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

Q9. For each of the following linear transformations $T : \mathbb{R}^2 \to \mathbb{R}^2$, find the corresponding matrix.

(1) $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 3x \\ y \end{pmatrix}$ **(2)** $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} y \\ x \end{pmatrix}$

(3) $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x-y \\ x+y \end{pmatrix}$ **(4)** $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ (zero map)

Problems

7.3 Compute the following.

(1) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ **(2)** $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ **(3)** $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

(4) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ **(5)** $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ **(6)** $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$

(7) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ **(8)** $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$ **(9)** $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

★**7.4** For each T below, find the matrix A such that $T(\vec{v}) = A\vec{v}$.

(1) Reflection across the x-axis: $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ -y \end{pmatrix}$.

(2) Projection to the line $y = x$: $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} \frac{x+y}{2} \\ \frac{x+y}{2} \end{pmatrix}$.

(3) The map $T : \mathbb{R}^3 \to \mathbb{R}^2$ given by $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} x+y \\ y-z \end{pmatrix}$.Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

(4) The map $T : \mathbb{R}^2 \to \mathbb{R}^3$ given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ x-y \\ 2y \end{pmatrix}$.

7.5 Matrix Multiplication

If $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^l$ are linear transformations, their composition $T_2 \circ T_1 : \mathbb{R}^n \to \mathbb{R}^l$ is also linear. Matrix multiplication is defined to correspond to composition of linear transformations.

Definition 7.9 (Matrix multiplication)

Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ and $B = (b_{jk}) \in \mathbb{R}^{n\times l}$. The **product** $AB \in \mathbb{R}^{m\times l}$ is defined by

$$(AB)_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}. \tag{7.20}$$

The (i, k)-entry of AB is the inner product of the i-th *row* of A with the k-th *column* of B.

Be careful: AB is only defined when the number of columns of A equals the number of rows of B. In general, $AB \neq BA$; matrix multiplication is **not commutative**.

Example 7.10 (Matrix multiplication)

Compute $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}$.

Solution

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1\cdot 0 + 2\cdot 1 & 1 \cdot {-1} + 2\cdot 2 \\ 3\cdot 0 + 4\cdot 1 & 3 \cdot {-1} + 4\cdot 2 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}.$$

Theorem 7.11 (Properties of matrix multiplication)

For matrices of compatible sizes and scalars k:

- **(A)** $(AB)C = A(BC)$ (associativity)
- **(D)** $k(AB) = (kA)B = A(kB)$
- **(B)** $A(B + C) = AB + AC$ (distributivity)
- **(E)** $AB \neq BA$ in general
- **(C)** $(A + B)C = AC + BC$
- **(F)** $AB = AC$ does not imply $B = C$

The **identity matrix** I_n (or just I) satisfies $AI = IA = A$ for square A.

Definition 7.12 (Identity matrix and zero matrix)

The $n \times n$ **identity matrix** is $I_n = (\delta_{ij})$, where $\delta_{ij} = 1$ if $i = j$ and 0 if $i \neq j$:

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{7.21}$$

The **zero matrix** O has all entries 0: $AO = OA = O$.

Quiz

Q10. Compute the following.

(1) $\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}3 & 5\\-2 & 4\end{pmatrix}$ **(2)** $\begin{pmatrix}3 & 5\\-2 & 4\end{pmatrix}\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$ **(3)** $\begin{pmatrix}2 & 1\\0 & 3\end{pmatrix}\begin{pmatrix}1 & -1\\0 & 1\end{pmatrix}$

(4) $\begin{pmatrix}1 & -1\\0 & 1\end{pmatrix}\begin{pmatrix}2 & 1\\0 & 3\end{pmatrix}$ **(5)** $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}^2$ **(6)** $\begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}^4$

Q11. Show that $AI = A$ for any 2×2 matrix A and $I = I_2$.

Problems

7.5 Compute the following.

(1) $\begin{pmatrix}2 & 1\\-1 & 3\end{pmatrix}\begin{pmatrix}1 & 2\\0 & -1\end{pmatrix}$ **(2)** $\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}\begin{pmatrix}a & b\\c & d\end{pmatrix}$ **(3)** $\begin{pmatrix}a & b\\c & d\end{pmatrix}\begin{pmatrix}0 & 1\\1 & 0\end{pmatrix}$

(4) $\begin{pmatrix}1 & 0 & -1\\2 & 1 & 0\end{pmatrix}\begin{pmatrix}1 & 2\\-1 & 0\\3 & 1\end{pmatrix}$ **(5)** $\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}\begin{pmatrix}a & b\\c & d\end{pmatrix} - \begin{pmatrix}a & b\\c & d\end{pmatrix}\begin{pmatrix}1 & 2\\3 & 4\end{pmatrix}$ **(6)** $\begin{pmatrix}\cos\theta & -\sin\theta\\\sin\theta & \cos\theta\end{pmatrix}^2$

★7.6 Prove the associativity $(AB)C = A(BC)$ for $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times l}$, $C \in \mathbb{R}^{l\times p}$. [Hint: Show both sides have the same (i, k)-entry.]

*****7.7** A matrix A is called **idempotent** if $A^2 = A$. Show that $P = \begin{pmatrix}\frac{1}{2} & \frac{1}{2}\\\frac{1}{2} & \frac{1}{2}\end{pmatrix}$ is idempotent.

Interpret P geometrically.

****7.8** Show that if $AB = BA$ for *all* matrices B (of compatible size), then A must be a scalar multiple of the identity matrix.

7.6 Determinant and Trace

Two important numbers attached to a square matrix are the **determinant** and the **trace**.

7.6.1 Trace

Definition 7.13 (Trace)

The **trace** of an $n \times n$ matrix $A = (a_{ij})$ is the sum of diagonal entries:

$$\operatorname{tr}(A) := a_{11} + a_{22} + \ldots + a_{nn} = \sum_{i=1}^{n} a_{ii}. \tag{7.22}$$

Theorem 7.14

For $n \times n$ matrices A and B and scalar k:

$$\operatorname{tr}(A+B) = \operatorname{tr}(A) + \operatorname{tr}(B), \quad \operatorname{tr}(kA) = k\operatorname{tr}(A), \quad \operatorname{tr}(AB) = \operatorname{tr}(BA). \tag{7.23}$$

7.6.2 Determinant

The determinant of a square matrix A, written $\det(A)$ or $|A|$, is a single number that encodes important information about the linear transformation. Its most important meaning: $\det(A) = 0$ if and only if the transformation is not invertible (it "squashes" space).

Definition 7.15 (Determinant: 1×1 and 2×2)

For a 1×1 matrix (a): $\det(a) := a.$

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\det(A) = \det\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} := ad - bc. \tag{7.24}$$

Definition 7.16 (Determinant: 3×3 (cofactor expansion along first row))

For a 3×3 matrix $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$:

$$\det(A) = a_{11}\det\left(\begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix}\right) - a_{12}\det\left(\begin{pmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{pmatrix}\right) + a_{13}\det\left(\begin{pmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix}\right). \tag{7.25}$$

Each 2×2 determinant is called a **minor** the pattern of signs $(+, -, +)$ is important.

Be careful: The pattern of signs in the 3×3 determinant is $+, -, +$ for the first row. For the second row it is $-, +, -$, and for the third row $+, -, +$.

Example 7.17 (2×2 and 3×3 determinants)

Compute $\det\begin{pmatrix} 3 & -1 \\ 2 & 5 \end{pmatrix}$ and $\det\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$.

Solution

$$\det\begin{pmatrix}3 & -1\\ 2 & 5\end{pmatrix} = 3\cdot 5 - (-1)\cdot 2 = 15 + 2 = 17.$$

$$\det\begin{pmatrix}1 & 2 & 3\\ 0 & 4 & 5\\ 1 & 0 & 6\end{pmatrix} = 1\cdot\det\begin{pmatrix}4 & 5\\ 0 & 6\end{pmatrix} - 2\cdot\det\begin{pmatrix}0 & 5\\ 1 & 6\end{pmatrix} + 3\cdot\det\begin{pmatrix}0 & 4\\ 1 & 0\end{pmatrix} = 1(24) - 2(-5) + 3(-4) = 24 + 10 - 12 = 22.$$

Theorem 7.18 (Properties of the determinant)

For $n \times n$ matrices A and B and scalar k:

(A) $\det(AB) = \det(A)\det(B)$

(B) $\det(kA) = k^n \det(A)$

(C) $\det(A^T) = \det(A)$

(D) $\det(A) \neq 0 \iff A$ is invertible

(E) $\det(I_n) = 1$

(F) If two rows of A are swapped, det changes sign.

(G) If a row of A is multiplied by k, det is multiplied by k.

(H) If a multiple of one row is added to another, det is unchanged.

Quiz

Q12. Compute the following determinants.

(1) $\begin{vmatrix}2 & 3\\ 1 & 4\end{vmatrix}$ **(2)** $\begin{vmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{vmatrix}$ **(3)** $\begin{vmatrix}a & b\\ -b & a\end{vmatrix}$

(4) $\begin{vmatrix}1 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3\end{vmatrix}$ **(5)** $\begin{vmatrix}1 & 2 & 0\\ 0 & 1 & 3\\ 0 & 0 & 1\end{vmatrix}$ **(6)** $\begin{vmatrix}1 & 1 & 1\\ 1 & 2 & 3\\ 1 & 4 & 9\end{vmatrix}$

Q13. What does $\det(A) = 0$ tell you geometrically for a 2×2 matrix A?

Fail safe note: Think about what happens to the area of the unit square under the transformation A.

Problems

7.9 Compute $\det(A)$ and $\operatorname{tr}(A)$ for each.

(1) $A = \begin{pmatrix}3 & -2\\ 1 & 4\end{pmatrix}$ **(2)** $A = \begin{pmatrix}-1 & 5\\ 0 & 2\end{pmatrix}$

(3) $A = \begin{pmatrix}2 & 1 & 0\\ -1 & 3 & 2\\ 0 & 1 & -1\end{pmatrix}$ **(4)** $A = \begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{pmatrix}$

★7.10 Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for $A, B \in \mathbb{R}^{n\times n}$. [Hint: Write out $(AB)_{ii} = \sum_j a_{ij}b_{ji}$ and sum over i.]

★7.11 For $A = \begin{pmatrix}a & b\\ c & d\end{pmatrix}$, prove $A(ad-bc)I = (ad-bc)A$ and hence $A^{-1} = \dfrac{1}{ad-bc}\begin{pmatrix}d & -b\\ -c & a\end{pmatrix}$ if $\det A \neq 0$.

*****7.12** Show that $\det(A^2) = \det(A)^2$ and $\det(A^n) = \det(A)^n$ for any $n \in \mathbb{N}$.

**7.13 Prove: if A is an $n \times n$ matrix and $\det(A) \neq 0$, then the equation $A\vec{v} = \vec{0}$ has only the trivial solution $\vec{v} = \vec{0}$.

7.7 Special Matrices

Several special classes of matrices arise frequently in physics.

7.7.1 Transpose and conjugate transpose

Definition 7.19 (Transpose and conjugate transpose)

For a matrix $A = (a_{ij}) \in \mathbb{C}^{m\times n}$:

- The **transpose** $A^T = (a_{ji}) \in \mathbb{C}^{n\times m}$: swap rows and columns.
- The **conjugate transpose** (or **Hermitian conjugate**) $A^\dagger = \overline{A}^T = (\overline{a_{ji}}) \in \mathbb{C}^{n\times m}$: transpose *and* take complex conjugates.

Remark: Some textbooks write A^* instead of $A^\dagger$ for the conjugate transpose. In physics, $A^\dagger$ (read: "A dagger") is the standard notation. For a real matrix, $A^\dagger = A^T$.

Quiz

Q14. Let $A = \begin{pmatrix} 1+\mathrm{i} & 2 \\ -\mathrm{i} & 3-\mathrm{i} \end{pmatrix}$. Find A^T and $A^\dagger$.

Q15. Show that $(AB)^T = B^T A^T$ and $(AB)^\dagger = B^\dagger A^\dagger$.

Fail safe note: Compute the (i, j)-entry of both sides.

7.7.2 Special classes

Definition 7.20 (Special square matrices)

Let $A \in \mathbb{C}^{n\times n}$. We say A is:

Name	Condition	Example
symmetric	$A^T = A$	$\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$
anti-symmetric	$A^T = -A$	$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$
Hermitian	$A^\dagger = A$	$\begin{pmatrix} 1 & \mathrm{i} \\ -\mathrm{i} & 2 \end{pmatrix}$
anti-Hermitian	$A^\dagger = -A$	$\begin{pmatrix} \mathrm{i} & 1 \\ -1 & \mathrm{i} \end{pmatrix}$
orthogonal	$A^T A = I, A \in \mathbb{R}^{n\times n}$	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$
unitary	$A^\dagger A = I$	
diagonal	$a_{ij} = 0$ for $i \neq j$	$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

Remark: A symmetric matrix is a special case of Hermitian (when all entries are real). An orthogonal matrix is a special case of unitary. In physics, Hermitian and unitary matrices are especially important (quantum mechanics!).

Quiz

Q16. For $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$, verify $A^T = A$.

Q17. For $A = \begin{pmatrix} 1 & \mathrm{i} \\ -\mathrm{i} & 2 \end{pmatrix}$, verify $A^\dagger = A$.

Q18. For $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, verify $A^T A = I$.

Problems

★**7.14** Classify each of the following matrices.

(1) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ **(2)** $\begin{pmatrix} 1 & 1+\mathrm{i} \\ 1-\mathrm{i} & -1 \end{pmatrix}$

(3) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ **(4)** $\dfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$

(5) $\begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix}$ **(6)** $\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$

*****7.15** Show that for any matrix $A \in \mathbb{C}^{n\times n}$:

(1) $A + A^\dagger$ is Hermitian.

(2) $A - A^\dagger$ is anti-Hermitian.

(3) Any matrix can be written as a sum of a Hermitian and an anti-Hermitian matrix.

*****7.16** Show that if A is unitary, then $|\det A| = 1$.

****7.17** Show that the eigenvalues of a Hermitian matrix are real. [Hint: Let $A\vec{v} = \lambda\vec{v}$ with $\vec{v} \neq \vec{0}$. Compute $\langle \vec{v} \,|\, \overrightarrow{Av} \rangle$ in two ways.]

7.8 Rotation, Reflection, and Scaling in 2d

We now apply matrices to concrete geometric transformations in $\mathbb{R}^2$. Write $\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ and use the standard basis $\{\vec{e}_x, \vec{e}_y\}$.

7.8.1 Scaling

Scaling by (s_x, s_y) means multiplying the x-component by s_x and the y-component by s_y:

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} s_x x \\ s_y y \end{pmatrix} = \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \tag{7.26}$$

Scaling by a single constant s in all directions (dilation) uses sI.

7.8.2 Reflection

Reflection across the x-axis sends $(x, y) \mapsto (x, -y)$:

$$R_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{7.27}$$

Reflection across the y-axis: $R_y = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

What about reflection across the line $y = x \tan\alpha$ (a line through the origin at angle α)? The general formula is

$$R_\alpha = \begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}. \tag{7.28}$$

Quiz

Q19. Verify Eq. (7.28) for $\alpha = 0$ (reflection across the x-axis) and $\alpha = \pi/4$ (reflection across $y = x$).

Q20. Show that $(R_\alpha)^2 = I$ for any α. Explain this geometrically.

7.8.3 Rotation

Counterclockwise rotation by angle θ sends $\vec{e}_x \mapsto \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$ and $\vec{e}_y \mapsto \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$. These become the columns:

Definition 7.21 (Rotation matrix)

The counterclockwise rotation by θ in $\mathbb{R}^2$ is represented by

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{7.29}$$

Theorem 7.22

$$R(\theta)R(\varphi) = R(\theta + \varphi), \quad R(\theta)^{-1} = R(-\theta), \quad \det R(\theta) = 1, \quad R(\theta)^T R(\theta) = I. \tag{7.30}$$

Example 7.23 (Rotation by $\pi/4$)

The vector $\begin{pmatrix}1\\0\end{pmatrix}$ rotated counterclockwise by $\pi/4$ gives $R(\pi/4)\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}\cos(\frac{\pi}{4}) & -\sin(\frac{\pi}{4})\\ \sin(\frac{\pi}{4}) & \cos(\frac{\pi}{4})\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}$.

Quiz

Q21. Verify $R(\theta)R(\varphi) = R(\theta+\varphi)$ by direct computation. [Hint: Use the angle-addition formulas for sin and cos.]

Q22. For what θ does $R(\theta) = I$? For what θ does $R(\theta)$ equal reflection across the x -axis?

Problems

7.18 Compute $R(\theta)\vec{v}$ for:

(1) $\theta = \pi/2, \vec{v} = \begin{pmatrix}1\\0\end{pmatrix}$ **(2)** $\theta = \pi, \vec{v} = \begin{pmatrix}0\\1\end{pmatrix}$ **(3)** $\theta = \pi/3, \vec{v} = \begin{pmatrix}1\\1\end{pmatrix}$

(4) $\theta = -\pi/4, \vec{v} = \begin{pmatrix}\sqrt{2}\\0\end{pmatrix}$ **(5)** $\theta = \pi/6, \vec{v} = \begin{pmatrix}\sqrt{3}\\1\end{pmatrix}$ **(6)** $\theta = 2\pi/3, \vec{v} = \begin{pmatrix}1\\-1\end{pmatrix}$

★7.19 Show the following.

(1) $R(\theta)$ is orthogonal for all θ.

(2) $\det R(\theta) = 1$ for all θ.

(3) $R(\theta)^{-1} = R(-\theta)$.

(4) Verify the identity $R(\theta)R(\varphi) = R(\theta+\varphi)$.

***__7.20__ The composition of two reflections is a rotation. Specifically, show that $R_{\pi/2}R_0 = R(\pi)$, where R_0 is reflection across the x-axis and $R_{\pi/2}$ is reflection across the y-axis. Then, more generally, show that $R_\beta R_\alpha = R(2(\beta-\alpha))$, where R_α and R_β are reflections from Eq. (7.28).

**__7.21__ A rotation in $\mathbb{R}^3$ about the z-axis by angle θ is represented by

$$R_{z(\theta)} = \begin{pmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{pmatrix}. \tag{7.31}$$

Write down analogous matrices $R_{x(\theta)}$ and $R_{y(\theta)}$ for rotations about the x- and y-axes. Show that $R_{x(\theta)}$, $R_{y(\theta)}$, and $R_{z(\theta)}$ are all orthogonal with determinant 1.

(1) Show that in general $R_{x(\alpha)}R_{y(\beta)} \neq R_{y(\beta)}R_{x(\alpha)}$ (rotations in 3d do not commute).

7.9 Linear Transformations

Definition 7.24 (Linear transformation)

A map $T : \mathbb{R}^n \to \mathbb{R}^m$ (or $T : \mathbb{C}^n \to \mathbb{C}^m$) is called a **linear transformation** (or **linear map**) if, for any vectors $\vec{v}, \vec{w}$ in the domain and any scalar k,

$$T(\vec{v} + \vec{w}) = T(\vec{v}) + T(\vec{w}), \quad T(k\vec{v}) = kT(\vec{v}). \tag{7.32}$$

Equivalently, $T(p\vec{v} + q\vec{w}) = pT(\vec{v}) + qT(\vec{w})$ for any scalars p, q.

Be careful: Not every map is linear. For example, the map $T : \mathbb{R}^1 \to \mathbb{R}^1$ defined by $T(x) = x^2$ is *not* linear, because $T(2) = 4$ but $2T(1) = 2$.

Example 7.25 (Examples of linear transformations in 2d)

1. **Scaling** by a constant $c > 0$: $T(\vec{v}) = c\vec{v}$.
2. **Rotation** by angle θ counterclockwise: $T(\vec{v})$ is $\vec{v}$ rotated by θ.
3. **Projection** onto the x-axis: $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ 0 \end{pmatrix}$.
4. **Zero map**: $T(\vec{v}) = \vec{0}$ for all $\vec{v}$.
5. **Identity**: $T(\vec{v}) = \vec{v}$ for all $\vec{v}$.

Quiz

Q23. Verify that scaling (Eq. (7.32) is satisfied for $T(\vec{v}) = c\vec{v}$.

Q24. Check that the map $T(\vec{v}) = \vec{v} + \vec{a}$ (translation by a fixed $\vec{a} \neq \vec{0}$) is **not** linear.

Q25. For a linear transformation T, show that $T(\vec{0}) = \vec{0}$. [Hint: Apply the definition with $k = 0$.]

7.9.1 A linear transformation is determined by its values on basis vectors

A key fact is that once you know what T does to each basis vector, you know T completely.

Theorem 7.26 (A linear transformation is determined by basis images)

Let $\{\vec{e}_1, \ldots, \vec{e}_n\}$ be an orthonormal basis of $\mathbb{R}^n$ (or $\mathbb{C}^n$). A linear transformation T is completely determined by the m vectors $T(\vec{e}_1), T(\vec{e}_2), \ldots, T(\vec{e}_n) \in \mathbb{R}^m$. Specifically, for any $\vec{v} = v_1\vec{e}_1 + \ldots + v_n\vec{e}_n$,

$$T(\vec{v}) = v_1 T(\vec{e}_1) + v_2 T(\vec{e}_2) + \ldots + v_n T(\vec{e}_n). \tag{7.33}$$

Proof: By linearity, $T(\vec{v}) = T\left(\sum_k v_k \vec{e}_k\right) = \sum_k v_k T(\vec{e}_k)$.

Quiz

Q26. A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ satisfies $T\left(\begin{pmatrix}1\\0\end{pmatrix}\right) = \begin{pmatrix}1\\2\end{pmatrix}$ and $T\left(\begin{pmatrix}0\\1\end{pmatrix}\right) = \begin{pmatrix}-1\\3\end{pmatrix}$. Find $T\left(\begin{pmatrix}3\\-2\end{pmatrix}\right)$ and $T\left(\begin{pmatrix}a\\b\end{pmatrix}\right)$.